TECHNICAL REPORT

FACE-ATLAS

A Transdiagnostic Latent-axis Stratification of Severe Mental Illness

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ABSTRACT

Psychiatric diagnosis aggregates biologically and prognostically heterogeneous patients into single categories. **FACE-ATLAS** reorganizes the harmonized three-cohort FACE baseline (bipolar disorder, schizophrenia, major depression; $N = 9{,}013$) around *continuous, diagnosis-agnostic axes of variation*, and asks how far such a map can be pushed — through five dependent questions: does it *exist*, *organize*, *persist*, *predict*, and *guide treatment*?

The map is estimated by a single global, missingness-aware **Bayesian sparse bifactor / ESEM** model with **mixed likelihoods**, fit to each patient's *observed cells only* (no imputation, full sample), with diagnosis held out as *metadata* and the axes discovered on the baseline visit. From ten candidate constructs the data resolve a **nine-dimension map** — a general functional-burden factor G plus eight weakly-correlated axes (cognition, metabolic, inflammatory, sleep, developmental-risk, suicidality, mania, substance) — with depression/anxiety instruments placed as *cross-loading windows* and anhedonia *rejected*. The headline result: **biology — metabolic and inflammatory load — is the least severity-entangled domain** (correlations with G of 0.12 and 0.07, versus 0.39 for cognition and 0.42 for sleep), so biological risk rides largely on axes that overall clinical severity does not capture.

On these coordinates the space is a graded **continuum, not discrete biotypes** (five structure-discovery methods agree, and a single-Gaussian falsification null reproduces the apparent clustering — the best silhouette is 0.140 against a null of 0.141 ± 0.002; adjusted Rand index ≈ 0 against the DSM-5 subtypes), described by **four extreme phenotypes (an A=4 archetype simplex)** with every patient a soft blend and severity as its spine. The soft tessellation is exported as a nested K-family with **no privileged K** — the operative granularity is an outcome question, settled in the prognosis chapter: none. Scored forward onto the follow-up visits the geometry *persists* — *durable biology, moving symptoms*. It also *predicts*, but modestly and for *functioning* only: the four archetypes — biology read as the worst-prognosis corner of the simplex — add *group-level* signal beyond diagnosis, severity, and the baseline outcome (course-dependent; two-year functional-remission AUC +0.011), not a large individual-decision gain. And on observational treatment-as-usual the map does *not* reliably *moderate treatment* response — lithium in bipolar disorder, the cleanest contrast, is a well-powered *null*.

Taken together, FACE-ATLAS demonstrates a transdiagnostic biological map that is real, stable, and a *continuum* rather than a set of biotypes, carrying a small-but-genuine group-level prognostic signal for functioning and *no* demonstrable treatment moderation in observational data — a deliberately *calibrated* account in which the modest and negative results are reported as such. Establishing individual-level clinical utility or treatment guidance would require incident-event outcomes, randomized treatment data, and external validation beyond this baseline observational cohort.

Contents

1 Introduction and scientific framing

1.1 The problem: heterogeneity inside the diagnostic label

Bipolar disorder (BP), schizophrenia (SZ), and major depressive disorder (DR) are defined by polythetic, consensus criteria: a patient qualifies by meeting some threshold number of symptoms drawn from a longer list. Two patients with the *same* diagnosis can therefore share almost no symptoms, while a given patient's biology, cognition, and course can resemble someone with a *different* diagnosis more than a same-diagnosis peer. For prognosis and treatment selection this categorical resolution is too coarse: the label predicts the *average* patient in a class, not the individual in front of the clinician.

Two research programs have grown up in response, and they bet on opposite shapes for psychiatric variation. The *dimensional* program — the NIMH Research Domain Criteria (RDoC) and the Hierarchical Taxonomy of Psychopathology (HiTOP) [1, 2] — replaces categories with *continuous axes* of dysfunction that cut across diagnoses and units of analysis (behaviour, cognition, physiology). The *biotype* program instead searches for *new discrete subgroups*: data-driven clusters meant to carve patients more sharply than DSM does [27]. The two make incompatible claims — smooth gradients versus latent kinds — and FACE-ATLAS treats that disagreement as an empirical question rather than a premise; we return to how the FACE-ATLAS results sit within both programs, and the wider biomedical literature, in Chapter 13.

1.2 Measurement before stratification

Asking the question cleanly requires separating two problems that are routinely conflated.

- **Measurement** — where does each patient sit on a small set of continuous, diagnosis-agnostic axes of clinical and biological variation? This *builds the map.*
- **Stratification** — do those coordinates fall into decision-relevant regions, and are those regions *discrete or graded*? This *asks whether the coordinates fall into discrete regions or form a continuum, and how to read a patient's position either way.*

Collapsing the two is the classic error. Clustering raw instruments and naming the clusters is a *theory-imposed scoring system*, not a discovery: it bakes the answer into the choice of features and the choice of distance. FACE-ATLAS instead estimates the measurement model first and lets stratification act on the estimated coordinates — carrying their uncertainty — only afterwards. Whether the map harbours biotypes or a continuum is then something the data *answer* (Chapter 8), not something the method assumes.

1.3 What this report establishes

On the harmonized three-cohort FACE baseline ($N = 9{,}013$), the program reaches one calibrated result, stated here in advance so the rest of the report can be read as its defence:

> *The FACE cohorts hide a real, stable, transdiagnostic map — nine dimensions — on which biological load is nearly independent of clinical severity; the map is a graded* **continuum** *rather than a set of biotypes; it forecasts two-year functioning at the* group *level beyond diagnosis and severity; and, on observational data, it does* not *reliably guide treatment.*

The load-bearing finding is the first. Metabolic and inflammatory burden are the *least severity-entangled* domains in the map — they correlate with the general functional-burden factor at only 0.12 and 0.07, against 0.39 for cognition and 0.42 for sleep — so biological risk rides on axes that a clinician's global impression of severity does not see. Two consequences follow and organize the later chapters: a stratification can exploit that largely independent signal (Chapter 8), and a prognostic model can read future functioning from it (Chapter 10).

The shape of that map is the second framing decision, and it is settled by evidence, not assumed. When we project patients onto the nine coordinates and ask whether they fall into clean clusters, the honest answer is *no*: the space is a graded continuum. The decisive check is a **single-Gaussian falsification null** — we ask how clustered a deliberately structureless cloud (one smooth Gaussian blob with the same shape as our data) would *look* to the same clustering tools, and the real data are no more clustered than that straw man (the best silhouette is 0.140 in the data against 0.141 ± 0.002 in the null). What organizes the continuum instead is a small set of **four extreme phenotypes** — an $A{=}4$ archetype simplex, with every patient a soft blend of the corners rather than a member of one box (Chapter 8). When a clinician needs hard groups anyway, we export a nested family of soft tessellations with **no privileged** K: the right number of regions is not a property of the geometry but a question about what the regions are *for*, and that question is answered downstream in prognosis (the operative answer there is: none — the continuous and archetype encodings dominate any hard cut).

A word on stance. The positive claims are deliberately bounded, and the *negative and modest* results — a continuum rather than clean biotypes, a group-level rather than individual-level prognostic gain, and *no* demonstrable treatment moderation in observational data — are reported as the contribution, not buried beneath the positive ones. Scientific validity (a real, stable map carrying genuine signal) is within reach of this baseline cohort; strong individual-level clinical utility is not (the two-year functional-remission gain over diagnosis, severity, and the baseline outcome is only AUC $+\,0.011$), and the report says so wherever it applies.

1.4 The method in one paragraph: hybrid discovery

The map is estimated by a *single* global, missingness-aware Bayesian sparse bifactor / ESEM model — a factor model with one general axis touching every indicator (a general dimension of the kind reported across psychopathology [19]), plus specific axes for each domain, which additionally permits small data-driven cross-loadings (Chapter 3) — with mixed likelihoods, fit to the baseline visit. The candidate ontology is an *eligibility map plus soft priors*, not a fixed set of labels. It first says which constructs are actually measurable in the FACE common variables; constructs with no valid indicators are declared `not_testable` before fitting. For the eligible constructs, theory says where each instrument *should* load, while the likelihood is free to confirm the axis, split it, downgrade it to a proxy, or reject it. The soft priors are the formal mechanism of "theory proposes, data decides": most cross-loadings are shrunk toward zero, but the likelihood can pull any of them away if the data demand it.

Three objects kept distinct throughout

Candidate construct — a clinical idea, screened for indicators and, if eligible, encoded as soft loading priors. **Empirical dimension** — a stable, data-supported axis of covariance in the measurement map. **Patient stratum** — a downstream representation of the fixed coordinates: archetype mixtures and a coarse soft tessellation (Chapter 8). A candidate is a hypothesis; a dimension is a measurement result; a stratum is a validated description of the map's geometry; on this map it is a *reading lens* on a continuum, not a discrete kind.

The discovery pipeline therefore has two gates (Figure 1). **Ten constructs were originally proposed.** Three of them — impulsivity, negative symptoms, and sensory abnormalities — carry no FACE common-variable indicator and are reported `not_testable`; stating that is itself a result, not a failure. The eligible constructs then enter *one* global fit. That fit placed overall severity as the general factor G, confirmed cognition, sleep, and suicidality, *split* the single metabolism/immunometabolism construct into two weakly-correlated axes (metabolic and inflammatory), and downgraded neurodevelopment to a *developmental-risk proxy*; anhedonia, carried by a single thin indicator, proved inseparable from G and the depression windows and was *rejected*. Two further constructs — **mania** and **substance use** — were added once their indicators were ingested into the harmonized dataset, re-entered through the same global model, and both were confirmed. The measurement map is therefore **nine dimensions**: G plus cognition, metabolic, inflammatory, sleep, developmental-risk, suicidality, mania, and substance — discovered axes, not strata.

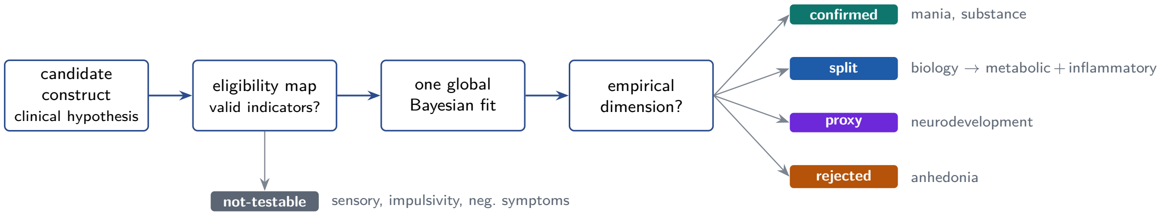
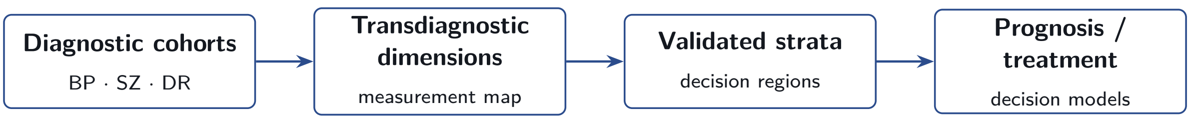


Figure 1. Hybrid discovery as eligibility plus empirical adjudication. Theory proposes; FACE coverage first declares unmeasured constructs *not-testable*; one global fit then adjudicates the eligible constructs as *confirmed*, *split*, relabelled a *proxy*, or *rejected*. Mania and substance were added after the original ten, once their indicators entered the dataset. This is the formal sense in which the data are allowed to overturn the theory.

1.5 One argument, five questions

The report is a single argument, built as five questions that depend on one another — each can be asked only once the previous one is answered *yes*, so the order is logical, not administrative. The conceptual pipeline runs in four layers (cohorts → dimensions → strata → prognosis/treatment); the five questions walk through it, and each is a chapter.



- **Exist? Transdiagnostic dimensions** — the measurement map, built on the baseline visit (Chapters 2–7).
- **Organize? How the coordinates are organized** — discrete types, or a graded continuum? (the answer reshapes M2 into a coordinate system with a reading guide) (Chapter 8).
- **Persist? Temporal coherence** — whether the coordinates and strata hold across the follow-up visits (Chapter 9).
- **Predict? Prognosis** — incremental predictive value beyond diagnosis and severity (Chapter 10).
- **Guide treatment? Treatment** — whether a stratum moderates treatment response (Chapter 11).

This document is the methods-of-record and the as-built record, written at a level of detail so that it can both seed the eventual manuscript and serve as the project's standing technical reference.

1.6 Load-bearing invariants

Four commitments are enforced at every layer of the system, from the data loader to the likelihood. They are the difference between a defensible map and an artefactual one.

Load-bearing invariant

(I1) No naive imputation. Structure is estimated from each patient's *observed* cells via an observed-data likelihood; no completed patient×variable matrix is ever created. Naive imputation manufactures covariance and fabricates strata when missingness is cohort-, site-, or severity-dependent — as it is in FACE.

(I2) Diagnosis is metadata. BP/SZ/DR labels are never indicators. They enter only as a post-hoc validation layer.

(I3) Baseline defines; follow-up validates. Dimensions are estimated on the baseline visit V0 only; visits V1–V4 are reserved for temporal coherence and prognosis, used *after* the map is fixed.

(I4) One global structure. There is *one* global factor structure on one harmonized matrix; there are no cohort-specific "modules." An instrument absent for a cohort is simply a missing cell under the observed-data likelihood, and a candidate with no valid indicators is reported `not_testable`, never back-filled with an invented proxy.

2 The data foundation

2.1 The FACE cohorts and the baseline visit

FACE-ATLAS is built on the harmonized baseline (visit `V0`) of three FACE expert-centre cohorts — bipolar disorder, schizophrenia, and major depressive disorder — each a multi-site French network in which patients receive a standardized, deep phenotyping battery. After harmonization the modelling sample is

$$N = 9{,}013 \;=\; \underbrace{6{,}252}_{\text{BP}} + \underbrace{2{,}209}_{\text{SZ}} + \underbrace{552}_{\text{DR}},$$

fit on the *full* sample with no completeness selection (Chapter 4 explains why this is a stated acceptance criterion rather than a convenience). The cohort imbalance — BP is roughly 11× DR — is handled by testing measurement invariance and by an optional inverse-frequency-weighted sensitivity arm, never by discarding data.

Table 1. Composition of the harmonized V0 modelling sample.

cohort	N	share	role
Bipolar disorder (BP)	6,252	69.4%	entry metadata / validation label
Schizophrenia (SZ)	2,209	24.5%	entry metadata / validation label
Major depressive disorder (DR)	552	6.1%	entry metadata / validation label
Total	**9,013**	100%	one global factor structure

2.2 What the harmonized variables measure

The dictionary spans the breadth of an expert-centre assessment, which is what makes a *multi-unit* transdiagnostic model possible. The largest blocks are biological: routine blood panels (BILAN BIOLOGIQUE, 45 usable variables — lipids, glycaemia, blood counts, inflammatory markers) and vital signs / ECG (14). Clinical phenotype is carried by clinician- and self-rated questionnaires (HETERO- and AUTO-QUESTIONNAIRES, 43 together — global severity, functioning, mood, mania, sleep, anxiety), a dedicated suicidality module (SUICIDE, 34), and neuropsychological testing (NEUROPSYCHOLOGIE, 12 — executive function, processing speed, memory). Liability and history are captured by psychiatric/medical antecedents (20) and perinatal items (10); substance use has its own block (6). Demographic, social, and care/work variables serve as covariates. Biology, cognition, behaviour, and history thus sit in one matrix — the precondition for a map that crosses both diagnoses and units of analysis.

2.3 The harmonization dictionary

A single curated dictionary (`data/face-common-vars.xlsx`; 225 rows, 201 usable) is the contract between the raw cohorts and the model. One row per harmonized variable records: the human label and clinical rationale; the per-cohort source column (BP/SZ/DR, blank where absent); the target dtype and value set; per-variable *sanity bounds*; a *readiness* tier (READY for 3-cohort, PARTIAL for 2-cohort, NOT USABLE excluded); and the canonical harmonized name — the modelled feature. The data layer loads this contract; nothing is hand-merged.

The harmonization problem is real: the three cohorts often encode the same variable differently — BP/SZ store French text labels where DR stores integer codes; the same column mixes units (milliseconds and seconds for ECG intervals; g/L and g/dL for haemoglobin concentration); sentinel and ceiling tokens (`>10`) contaminate counts. Each such variable has a registered transformer that reconciles the encodings and records its rule and provenance.

2.4 The pipeline: harmonize, bound, decode — never impute

The data layer (`src/face/data/`) exposes three composable entry points used by the build script. The flow is shown in Figure 2.

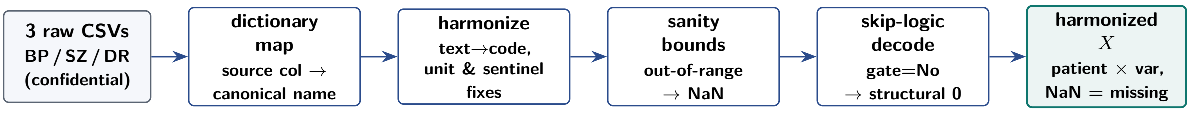


Figure 2. The no-imputation data pipeline. `build_unified_dataframe` reads each per-cohort source column, applies the harmonization rule and the sanity bounds, and stacks the cohorts; `to_harmonized_dataset` restricts to the baseline visit and applies skip-logic structural-zero decoding. Missing cells remain `NaN` at every stage.

Sanity bounds. Each variable carries plausibility bounds tuned to catch gross scale/unit errors (a glucose of 488 from a mg/dL slip), *not* to trim genuine clinical extremes. Out-of-range cells are set to `NaN` — never winsorized to the boundary, never filled. Formally, with bounds $[\ell_j, h_j]$,

$$X_{ij} \leftarrow \mathrm{NaN} \quad \text{iff} \quad X_{ij} < \ell_j \vee X_{ij} > h_j, \qquad \text{(observed cells only; already-missing cells untouched).}$$

Skip-logic structural-zero decoding. Some blanks are not missing data — they are structural zeros implied by an instrument's own gate. In the suicidality module the item "ever attempted?" (`isf05`) gates the attempt sub-battery — "how many times?" (`isf07`) and the attempt detail items (`isf08`, `isf08a`). When the gate is *No*, the downstream items are not asked and arrive blank, but the correct value is 0. The decoder fills such cells, and only such cells:

Structural zero vs. unknown — the decision that recovers data without imputing

For a gate g with no-value 0 and dependent d, the decoded value is

$$d \leftarrow 0 \quad \text{iff} \quad (g = 0) \;\wedge\; (d = \mathrm{NaN}).$$

A gate that is *Yes* leaves d missing (genuinely unobserved); a gate that is *unknown* leaves d missing (NaN $= 0$ is false, so nothing fires); an existing value is *never* overwritten. The rule cascades down the whole sub-battery (`isf05`=No $\Rightarrow$ `isf07`=0, `isf08`=0, `isf08a`=0). On FACE this lifts `isf07` coverage from $\approx 30\%$ to $\approx 90\%$ *without inventing a single value* — the recovered zeros are observed values, logged with rule and provenance.

The invariant, enforced everywhere. "No imputation" is not one check but a property maintained at each stage: sanity bounds null but never fill; every text→code recode maps unknown text to missing rather than a guessed code; skip-logic fills only gate-determined zeros; numeric coercion turns un-parseable values to `NaN`; and the output contract is typed as `NaN = missing`. The model-ready table is persisted to Parquet (`data/processed/baseline_v0.parquet`, one row per patient $\times$ modelled indicator) while raw data stay confidential CSV.

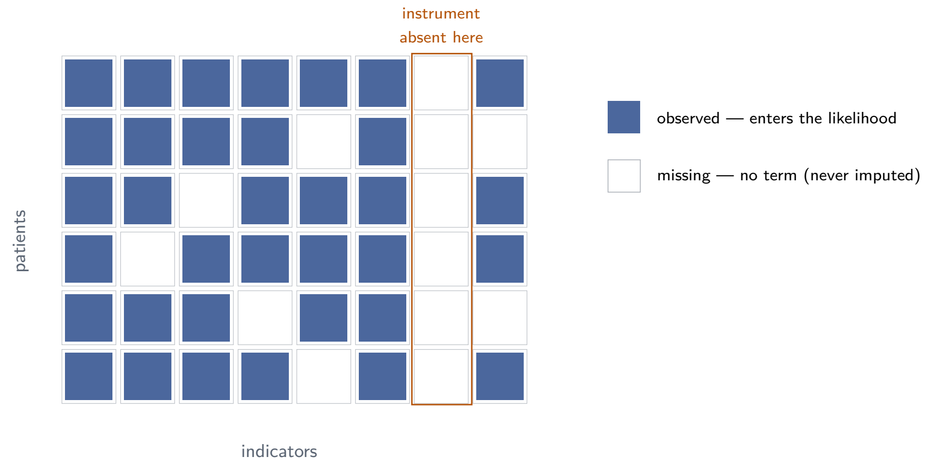


Figure 3. No imputation, made literal. Each patient contributes only the cells actually observed (filled); a missing cell contributes *no term* to the likelihood and is never filled. Whole columns can be empty for a cohort (an instrument absent by design) without distorting anyone's coordinates — the model learns that factor from the cohorts that measure it. Filling these holes would manufacture covariance and bias the map toward the best-measured patients.

2.5 Diagnosis and identifiers as metadata

The cohort label and DSM arm are carried as covariates and validation targets, never as indicators or clustering inputs (invariant I2). Identifiers — patient id, cohort, arm, visit, and the recruitment-site code `siteid_city` — are never modelled on; the site code is persisted as a side table for the later site bootstrap but excluded from the feature matrix. A strict schema rejects, at construction time, any feature id that carries a longitudinal suffix, encoding invariant I3 at the type level.

2.6 Coverage, missingness, and why the observed-data likelihood is licensed

Mean cell missingness across the modelled indicators is substantial and *structured* — by cohort design (instruments present in some cohorts only), by site practice, by clinical routing, and by skip-logic. This structure is the reason the observed-data likelihood (invariant I1) is not merely convenient but *principled*.

The missingness mechanism. The observed-data likelihood — equivalently, full-information maximum likelihood — returns consistent loadings and factor correlations under *Missing At Random* (MAR): missingness may depend on *observed* quantities but not, given those, on the unobserved value itself [6]. In FACE the dominant missingness is structural and by design: an instrument is administered in some cohorts or sites and not others, or it is gated by skip-logic. That is missingness driven by *design* variables (cohort, site, the gate) — MAR, indeed MCAR-conditional-on-design — not by the latent quantity being measured. Filling such holes is therefore not just unnecessary but harmful: a completed matrix would describe the best-measured sub-population and inject covariance the model would read as a spurious factor. Where missingness might instead be *informative* (MNAR — e.g. a lab skipped *because* the patient looked well), a selection-model arm is available as a sensitivity analysis (Chapter 4); it is not the primary fit, precisely because the structural-by-design component dominates here.

Coverage is information, not noise. Of the 201 usable variables, 109 are present in all three cohorts. Of the 92 two-cohort variables, 78 are present in BP and DR but *absent in SZ* — chiefly the affective and functioning self-reports (FAST, MADRS, QIDS, the Altman mania self-rating, C-SSRS) — while 13 are present in BP and SZ but absent in DR (notably the alcohol/cannabis substance-use items), and one is SZ/DR only. So **SZ, not DR, is the cohort most often missing an instrument**, and the gaps are systematic rather than random. That asymmetry shapes later findings rather than corrupting them: because SZ lacks the FAST functioning anchor, the general factor G re-anchors there on CGI-S/EGF/EQ-5D (a documented invariance partial, Chapter 6); because the substance-use items are BP/SZ-only, substance is reported as a two-cohort axis and not claimed for DR.

Caveat

The DR cohort ($N = 552$) is small, and the SZ cohort lacks the affective and functioning self-report battery (FAST, MADRS, QIDS, Altman, C-SSRS). Factors whose anchors are cohort-specific are tested for measurement invariance (Chapter 6) and their per-patient reliability is flagged by observed-indicator count, rather than assuming a uniform map across cohorts.

3 The measurement model

This chapter is written as a tutorial rather than as a compressed methods note. The central object is still the same: a Bayesian sparse bifactor/ESEM measurement model for the FACE transdiagnostic space. The practical aim is that the reader can move from a single clinical measurement, such as BMI or CRP, to the full observed-data likelihood used by the sampler, without treating any step as a black box.

The chapter follows the operational order used in the implementation. We first discuss the optional *Gaussian-copula* data layer, because it explains what numerical object is actually sent into the Gaussian engine in the acceleration variant (§3.2). We then write the bifactor/ESEM measurement equation and interpret the patient coordinates, loadings, and measurement slopes (§3.3). After that we separate the native mixed likelihood from the Gaussian-copula acceleration path (§3.4), derive the marginal Gaussian likelihood and the role of the factor-correlation matrix Φ (§3.5), write the dense observed-data log-likelihood with and without missing cells (§3.6), and only then introduce Woodbury and Cholesky as computational devices, not as new statistical assumptions (§3.7). The chapter ends with the interpretation of the general factor G, the identification problem, the priors, and the Bayesian learning procedure used to estimate the model parameters.

3.1 The candidate ontology and the eligibility map

Theory proposes a set of candidate dimensions. Mapping them against the FACE *common-variable* instrument coverage fixes the eligible model set (Table 2) — and this map is itself a primary result, because stating which candidates have no measurable indicators is a finding, not a failure. At this stage each candidate is judged only *testable* (it has indicators) or *not-testable*; the empirical verdict (confirmed / split / proxy / rejected) is adjudicated from the fitted model in Chapter 5.

Table 2. The candidate ontology and eligibility. Three constructs (impulsivity, negative symptoms, sensory) have no FACE common indicator and are reported `not_testable`; the rest enter the fit. *Mania* and *substance* were added to the original ten once their indicators were ingested. The hypothesised *split* of biology and *proxy* reading of neurodevelopment are tested, not assumed; the empirical verdicts are adjudicated in Chapter 5.

candidate	role in model	anchors (FACE common)	cohorts
Overall severity	G (functional burden)	CGI-S, EGF, EQ-5D-VAS, FAST, employment	3-cohort
Cognitive flexibility	cognition	TMT-B, WAIS coding/digit-span, fluency, CVLT	3-cohort
Metabolism / immuno	metabolic + inflammatory (split?)	BMI, waist, glucose, lipids / CRP, WBC, neutrophils	3-cohort
Sleep / circadian	sleep	PSQI total + sub-scores	3-cohort
Suicidality	suicidality (mixed)	ISF ideation/attempt; C-SSRS lethality	3-cohort ISF
Neurodevelopment	developmental-risk (proxy?)	perinatal, CTQ, age-of-onset, family history	3-cohort
Mania / activation	mania	YMRS, Altman self-report	3-cohort YMRS
Substance use	substance (mixed)	alcohol/cannabis lifetime SUD, nicotine	BP/SZ
Anhedonia	anhedonia (thin; rejected)	QIDS anhedonia item	BP/DR
Impulsivity	`not_testable`	no common scale	—
Negative symptoms	`not_testable`	no PANSS / SANS	—
Sensory abnormalities	`not_testable`	none	—

The composite depression/anxiety instruments (MADRS, QIDS, STAI) are deliberately *not* a dimension: each is seeded as a cross-loading *window* on the axes it clinically touches, and the data place it. There is no separate affective factor and no extra “depression” dimension unless one emerges under model comparison (Chapter 6). The general factor G does one job — transdiagnostic functional burden, anchored by functioning and global severity *only* — so symptom severity informs the axes it belongs to rather than contaminating the general factor.

3.2 The Gaussian-copula data layer: when the item scale is not the modelling scale

Before writing the measurement equation, it is important to be explicit about the scale of the variables entering that equation. In the native model, continuous indicators are oriented, transformed if needed (for example a log transform for strongly skewed positive laboratories), residualized for covariates when that mode is enabled, and then standardized. Binary, ordinal, and count variables keep their native likelihoods. In the optional `gaussian_copula` mode, however, a broader set of indicators is first mapped to a standard-normal rank scale, and the same Gaussian factor machinery is applied to those rank-normalized values.

A copula separates two questions that are otherwise entangled. The first is the *marginal* question: what is the distribution of each indicator by itself? CRP may be skewed, BMI may be closer to continuous, and a count variable may be integer-valued with a heavy right tail. The second is the *dependence* question: do patients who are high on one indicator also tend to be high on another, after translating all indicators onto a comparable rank scale? The Gaussian copula keeps the marginal distributions empirical and models the dependence geometry through a Gaussian latent representation.

For an oriented raw indicator y_{ij}, let $\widehat{F}_j$ be its empirical cumulative distribution function. The Gaussian-copula transform is

$$z_{ij} = T_j(y_{ij}) = \Phi_N^{-1}\left(\widehat{F}_j(y_{ij})\right), \tag{1}$$

where Φ_N^{-1} is the inverse CDF of the standard normal distribution. In the implementation this is the rank-based inverse-normal transform

$$u_{ij} = \frac{\mathrm{rank}(y_{ij})}{n_j + 1}, \qquad z_{ij} = \Phi_N^{-1}(u_{ij}), \tag{2}$$

with average ranks for ties. The transformation is monotone: a patient with a higher oriented CRP than another patient keeps a higher Gaussianized CRP score. What changes is the unit of measurement. A loading in copula mode is a slope on the Gaussian rank-score scale, not on the raw biochemical or clinical scale.

The common language analogy

The original indicators speak different measurement languages. CRP speaks a skewed biochemical language, BMI a continuous anthropometric language, and a high-cardinality count variable an integer language. The Gaussian copula translates each variable into the same question: "where is this patient in the population ranking for this item?" The percentile is then converted into a standard-normal score. The downstream measurement model can therefore model dependence on a common scale without pretending that all original variables had Gaussian marginal distributions.

The copula path is not the certified faithful default. It is an acceleration and sensitivity vertical. In this report, however, the copula fit is the *reported* object: once it was shown to reproduce the native-likelihood factor structure, the copula vertical was promoted to the measurement vehicle for the entire M1→M5 chain documented here (it is better calibrated for the heavy-tailed and count indicators; §3.2). Continuous families are always Gaussianized in this mode. Ordinal or count indicators are promoted into the Gaussianized marginalized block only if they have enough distinct values and are not dominated by one modal category; binary and low-cardinality ordinal variables remain in the native discrete likelihood. This tiering avoids a category with too little measurement resolution being treated as if it were a genuine continuous ruler.

Once transformed, the same Gaussian factor model is applied to z_i rather than to the native x_i:

$$z_i \mid \Theta \sim \mathcal{N}\left(0,\ \Lambda\Phi\Lambda^\top + \Psi\right). \tag{3}$$

Thus the copula mode changes the observation scale before the likelihood. It does not change the meaning of Φ as the latent-factor correlation matrix, nor does it change the role of Λ as the item-by-factor measurement map. It changes the interpretation of the item loadings: they become sensitivities of normalized ranks rather than sensitivities of native-scale measurements.

Because the empirical marginal map is stored, the copula model can also be inverted for synthetic generation. A generated Gaussian score z_{ij} is mapped back to the oriented original scale by

$$y_{ij} = \widehat{F}_j^{-1}(\Phi_N(z_{ij})), \tag{4}$$

usually by monotone interpolation on the stored empirical map. A synthetic patient can therefore be generated in three conceptual steps: draw latent coordinates f_i, draw Gaussianized item scores from the fitted measurement model, and invert each item through its empirical marginal distribution.

3.3 The bifactor/ESEM measurement equation

We now write the core measurement equation. Behind the many measured quantities we posit a smaller number of unobserved causes: latent factors. Each patient i has a general-factor coordinate and a vector of specific coordinates,

$$G_i \sim \mathcal{N}(0,1), \qquad D_i = (D_{i1}, \dots, D_{iK})^\top, \tag{5}$$

where G_i is the patient's position on the general severity / functional-burden dimension, and D_{ik} is the patient's position on specific dimension k — for example cognition, metabolic burden, inflammation, sleep, developmental risk, suicidality, mania/activation, or substance use.

For one observed indicator j in patient i, the linear predictor is

$$\eta_{ij} = \alpha_j + \lambda_{jG} G_i + \sum_{k=1}^{K} \lambda_{jk} D_{ik} + \beta_j^\top c_i. \tag{6}$$

Read Eq. (6) as the recipe for the expected reading of instrument j. It contains: a baseline item level α_j; a contribution from general burden, $\lambda_{jG} G_i$; contributions from the specific factors, $\sum_k \lambda_{jk} D_{ik}$; and a covariate calibration term $\beta_j^\top c_i$. The covariates adjust an item's mean. They are not psychiatric dimensions. Correcting an item for age, sex, education, site, or cohort is like correcting a thermometer for altitude before reading the temperature; altitude changes the reading, but it is not itself the temperature factor.

For a Gaussian indicator, the observed value is the expected value plus item-specific noise,

$$x_{ij} = \eta_{ij} + \varepsilon_{ij}, \qquad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_j^2). \tag{7}$$

In copula mode the same equation is written for z_{ij} rather than for native x_{ij}.

Coordinates and measurement slopes. A frequent source of confusion is the difference between a patient coordinate and a loading. The coordinate $D_{i,\text{metabolic}}$ says where patient i sits on the metabolic axis. The loading $\lambda_{\text{BMI},\text{metabolic}}$ says how sensitive the BMI indicator is to movement along that axis. In derivative form,

$$\frac{\partial \mathbb{E}[x_{i,\text{BMI}} \mid f_i]}{\partial D_{i,\text{metabolic}}} = \lambda_{\text{BMI},\text{metabolic}}, \qquad \frac{\partial \mathbb{E}[x_{i,\text{BMI}} \mid f_i]}{\partial G_i} = \lambda_{\text{BMI},G}. \tag{8}$$

The word *partial* matters. It means we ask how the expected BMI changes when one latent coordinate moves, while all other latent coordinates are held fixed. The loading is therefore a measurement slope: expected item change per one-unit movement on a latent axis.

The sensor analogy

The latent factor is the hidden signal; an observed item is a sensor. BMI is not the metabolic factor. BMI is a sensor that reacts to the metabolic coordinate, possibly also to general burden and weak neighbouring axes. A large $\lambda_{\text{BMI},\text{metabolic}}$ means BMI is a sensitive metabolic sensor. A small $\lambda_{\text{BMI},G}$ means BMI carries little general-severity signal after the metabolic coordinate is held fixed.

Stacking all J observed indicators of patient i gives

$$x_i = \mu_i + \Lambda f_i + \varepsilon_i, \qquad f_i = (G_i, D_i^\top)^\top, \qquad \mu_i = \alpha + B c_i. \tag{9}$$

Here $x_i \in \mathbb{R}^J$ is the observed item profile, $f_i \in \mathbb{R}^{K+1}$ is the latent coordinate profile, $\Lambda \in \mathbb{R}^{J\times(K+1)}$ is the loading matrix, and ε_i is the residual item-noise vector. The matrix product Λf_i simply applies every item row of loadings to the patient's latent coordinates. For BMI, the BMI row of Λ says how BMI reads general burden, metabolic burden, inflammation, and any other allowed factor.

This structure is **bifactor/ESEM**. It is bifactor because a specific-domain indicator may load both on the general factor G and on its home specific factor. It is ESEM because an indicator may also have small, theory-seeded cross-loadings onto other plausible dimensions, rather than being forced into exactly one box as in a rigid confirmatory model. The general factor is the hub; the specific factors are the spokes; broad symptom scales such as MADRS, QIDS, and STAI are windows onto several axes rather than standalone dimensions.

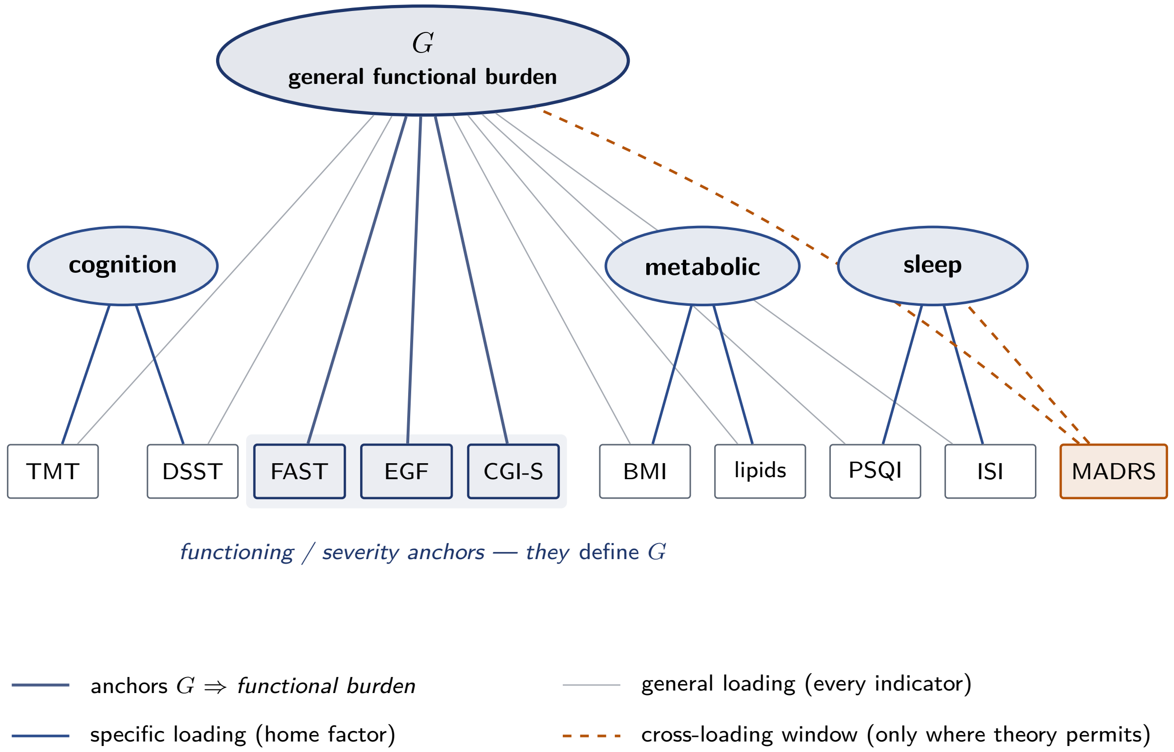


Figure 4. The bifactor / ESEM structure (schematic; 3 of the 8 specific factors shown, with example indicators labelled). The general factor G links to *all* indicators (thin grey); each specific factor links to its own indicator pool (blue). A cross-loading "window" such as MADRS (dashed amber) loads only where theory permits. Crucially, G is *anchored* by a pool of functioning / global-severity indicators (FAST, EGF, CGI-S; solid dark) that carry no competing specific factor — a construct becomes "general" only when many indicators share its signal, and here that signal is functioning content, so G reads as *functional burden* rather than as a generic symptom factor.

3.4 Native mixed likelihood and the Gaussian-copula acceleration branch

The per-item likelihood $p(x_{ij} \mid f_i, \Theta)$ should match the measurement type. This is generalized linear-model logic: the same latent predictor η_{ij} can be translated through different observation models, depending on whether the item is continuous, binary, ordinal, or a count,

$$X_{ij} \sim F_j(\eta_{ij}, \theta_j). \tag{10}$$

Use the right ruler for each quantity. You weigh flour on a scale, count eggs as whole numbers, and record whether the oven is on as yes/no. Forcing all these quantities onto the same

Table 3. The native mixed-likelihood observation model. The continuous block can be marginalized analytically; the non-Gaussian block carries explicit patient latents unless a high-resolution item is promoted by the optional Gaussian-copula acceleration branch.

indicator type	native likelihood	form
continuous / z-scored neuropsych	Gaussian	$X_{ij} \sim \mathcal{N}(\eta_{ij}, \sigma_j^2)$
skewed lab (CRP, triglycerides)	log-Gaussian	$\log X_{ij} \sim \mathcal{N}(\eta_{ij}, \sigma_j^2)$
ordinal item (C_j categories)	ordered logistic	$P(X_{ij} \le c) = \operatorname{logit}^{-1}(\tau_{jc} - \eta_{ij})$
binary (ISF ideation/attempt; SUD)	Bernoulli-logit	$X_{ij} \sim \operatorname{Bernoulli}(\operatorname{logit}^{-1}(\eta_{ij}))$
count (attempts, cigarettes)	negative binomial	$X_{ij} \sim \operatorname{NegBin}(e^{\eta_{ij}}, \phi_j)$

continuous ruler can manufacture fake covariance. The native mixed-likelihood design therefore keeps binary suicidality, ordinal severity items, counts, and continuous biology on appropriate likelihood families while allowing them to live in one latent factor space through Φ.

The Gaussian-copula branch is the controlled exception. It asks whether a non-Gaussian item has enough resolution to be treated as a monotone measurement of an underlying Gaussian rank score. If yes, it enters the same fast marginalized Gaussian block as the continuous indicators. If not, it remains in the native discrete likelihood. Thus the branch does not abandon the mixed-likelihood principle; it creates a tiered computational compromise: faithful native likelihood for low-resolution variables, Gaussian-copula acceleration for high-resolution variables whose rank geometry is more important than their native unit distances.

Observed cells only

With $\mathcal{O}_i \subseteq \{1, \ldots, J\}$ the indicators observed for patient i, missing cells contribute no likelihood term. The model never completes the patient-by-item matrix:

$$\log p(X_{\text{obs}} \mid \Theta) = \sum_{i=1}^{N} \sum_{j \in \mathcal{O}_i} \log F_j(X_{ij} \mid \eta_{ij}, \theta_j)$$

when the latent coordinates are explicit. In the marginalized Gaussian block the same rule appears as an observed sub-vector likelihood, developed below.

3.5 From conditional likelihood to the marginalized Gaussian likelihood

The conditional Gaussian statement follows immediately from Eq. (9) if $\varepsilon_i \sim \mathcal{N}(0, \Psi)$:

$$x_i \mid f_i, \Theta \sim \mathcal{N}(\mu_i + \Lambda f_i, \Psi). \tag{11}$$

This assumes we know the patient's latent coordinate vector f_i. In reality we do not. Therefore the model must also state how patient coordinates are distributed in the population:

$$f_i \sim \mathcal{N}(0, \Phi). \tag{12}$$

The matrix Φ is the latent-factor covariance or correlation matrix. Its diagonal fixes the scale of the latent dimensions, usually to one; its off-diagonal entries encode how dimensions co-vary across patients. A loading λ_{jk} answers how item j reacts to dimension k. A correlation Φ_{kl} answers whether patients high on dimension k tend also to be high on dimension l. These are different scientific objects.

Because f_i is hidden, the observed-data likelihood averages the conditional likelihood over all plausible latent coordinates:

$$p(x_i \mid \Theta) = \int p(x_i \mid f_i, \Theta)\, p(f_i \mid \Theta)\, df_i. \tag{13}$$

This is the probability sum rule. If a hidden cause could have taken many possible values, the probability of the observed data is the weighted average of the probability of those data under each possible hidden value.

For Gaussian factors and Gaussian residuals, the integral has a closed form. Taking the mean,

$$\mathbb{E}[x_i] = \mathbb{E}[\mu_i + \Lambda f_i + \varepsilon_i] = \mu_i, \tag{14}$$

because $\mathbb{E}[f_i] = 0$ and $\mathbb{E}[\varepsilon_i] = 0$. Taking the covariance,

$$\mathrm{Cov}(x_i) = \mathrm{Cov}(\Lambda f_i + \varepsilon_i) = \Lambda\,\mathrm{Cov}(f_i)\Lambda^\top + \mathrm{Cov}(\varepsilon_i) = \Lambda\Phi\Lambda^\top + \Psi. \tag{15}$$

Therefore

$$x_i \mid \Theta \sim \mathcal{N}(\mu_i, \Sigma), \qquad \Sigma = \Lambda\Phi\Lambda^\top + \Psi. \tag{16}$$

This is the covariance identity at the heart of factor analysis. The shared latent signal creates covariance between observed indicators; the diagonal residual matrix Ψ captures private item noise.

The item-pair view. For two observed indicators j and j', the implied covariance is

$$\mathrm{Cov}(x_{ij}, x_{ij'} \mid \Theta) = \lambda_j^\top \Phi \lambda_{j'} + \Psi_{jj'}, \tag{17}$$

where $\lambda_j^\top$ is row j of Λ. Since Ψ is diagonal, for $j \neq j'$ the residual term is zero:

$$\mathrm{Cov}(x_{ij}, x_{ij'} \mid \Theta) = \lambda_j^\top \Phi \lambda_{j'}. \tag{18}$$

That is the operational meaning of local independence: after the latent factors explain shared structure, the leftover item noise does not create cross-item covariance.

3.6 The actual Gaussian log-likelihood: dense, then missing-data

Equation (16) gives the distribution. The likelihood is obtained by evaluating the Gaussian density of the observed data under candidate parameter values. The data are fixed; the parameters vary. For one candidate

$$\Theta = \{\Lambda, \Phi, \Psi, \mu\}, \tag{19}$$

the model constructs

$$\Sigma(\Theta) = \Lambda\Phi\Lambda^\top + \Psi \tag{20}$$

and scores how probable the observed patient profiles are under that covariance structure.

Assume first that every patient has every item observed. For one patient define the residual

$$r_i = x_i - \mu_i. \tag{21}$$

The multivariate Gaussian density is

$$p(x_i \mid \Theta) = \frac{1}{(2\pi)^{J/2}|\Sigma|^{1/2}} \exp\left[-\frac{1}{2} r_i^\top \Sigma^{-1} r_i\right]. \tag{22}$$

Taking logs gives

$$\log p(x_i \mid \Theta) = -\frac{1}{2}\left[J\log(2\pi) + \log|\Sigma| + r_i^\top \Sigma^{-1} r_i\right]. \tag{23}$$

The three pieces have distinct roles. The term $J\log(2\pi)$ is the Gaussian normalizing constant. The log-determinant $\log|\Sigma|$ penalizes overly diffuse covariance models: a model that says "almost anything is plausible" spreads probability mass too widely. The quadratic form $r_i^\top \Sigma^{-1} r_i$ is the covariance-adjusted distance between the observed profile and the expected profile. A residual in a direction where the model expects high variability is less surprising; a residual in a direction where the model expects low variability is heavily penalized.

For N independent patients,

$$\log L(\Theta) = \sum_{i=1}^{N} \log p(x_i \mid \Theta). \tag{24}$$

This is not yet a Bayesian posterior; it is only the likelihood score for a candidate value of Θ.

Real FACE data contain missing cells, so each patient contributes the density of their observed sub-vector only. Let $\mathcal{O}_i$ be the observed item set and $k_i = |\mathcal{O}_i|$. The marginal property of the multivariate normal gives

$$x_{i,\mathcal{O}_i} \mid \Theta \sim \mathcal{N}(\mu_{i,\mathcal{O}_i}, \Sigma_{\mathcal{O}_i\mathcal{O}_i}), \tag{25}$$

where the same index appears twice because Σ is a matrix: we keep the rows corresponding to observed items and the columns corresponding to those same observed items. The observed residual is

$$r_{i,\mathcal{O}_i} = x_{i,\mathcal{O}_i} - \mu_{i,\mathcal{O}_i}. \tag{26}$$

The missing-data Gaussian log-likelihood for patient i is therefore

$$\log p(x_{i,\mathcal{O}_i} \mid \Theta) = -\frac{1}{2}\left[k_i \log(2\pi) + \log|\Sigma_{\mathcal{O}_i\mathcal{O}_i}| + r_{i,\mathcal{O}_i}^\top \Sigma_{\mathcal{O}_i\mathcal{O}_i}^{-1} r_{i,\mathcal{O}_i}\right]. \tag{27}$$

The full observed-data likelihood is

$$\log L(\Theta) = \sum_{i=1}^{N} \log p(x_{i,\mathcal{O}_i} \mid \Theta). \tag{28}$$

No missing value is set to zero, to the mean, or to a model prediction. A zero may appear in code as a finite placeholder for tensor arithmetic, but the mask removes that cell from every likelihood term.

Likelihood as a candidate-parameter score

At each candidate Θ, the model builds $\Sigma(\Theta)$, extracts the observed submatrix $\Sigma_{\mathcal{O}_i\mathcal{O}_i}$ for each patient, computes the residual penalty and log-determinant, and sums the result across patients. Candidate A may say BMI and waist share a metabolic factor and CRP shares an inflammatory factor; Candidate B may put those relationships elsewhere. The higher likelihood belongs to the candidate whose implied covariance makes the actually observed patient profiles less surprising.

3.7 Woodbury and Cholesky: computational devices, not new model assumptions

The dense likelihood in Eq. (27) is clear but expensive. It appears to require, for many patients and many missingness patterns, the determinant and inverse of $\Sigma_{\mathcal{O}_i\mathcal{O}_i}$. The model covariance has special structure:

$$\Sigma = \Lambda\Phi\Lambda^\top + \Psi. \tag{29}$$

The implementation rewrites the low-rank part as

$$\Lambda\Phi\Lambda^\top = (\Lambda R)(\Lambda R)^\top, \qquad RR^\top = \Phi, \tag{30}$$

where R is the Cholesky factor, or matrix square root, of Φ. This R has no separate clinical meaning. It is an internal computational representation of the factor-correlation matrix. Define

$$L_t = \Lambda R, \qquad \Sigma = L_t L_t^\top + \Psi. \tag{31}$$

Woodbury then avoids inverting the large item covariance matrix. For an observed item set $O = \mathcal{O}_i$, write L_O for the observed rows of L_t and Ψ_O for the diagonal residual covariance of the observed items. The inverse identity is

$$(\Psi_O + L_O L_O^\top)^{-1} = \Psi_O^{-1} - \Psi_O^{-1} L_O \left(I + L_O^\top \Psi_O^{-1} L_O\right)^{-1} L_O^\top \Psi_O^{-1}. \tag{32}$$

The determinant identity is

$$\log |\Psi_O + L_O L_O^\top| = \log |\Psi_O| + \log \left|I + L_O^\top \Psi_O^{-1} L_O\right|. \tag{33}$$

The expensive matrix on the left is $k_i \times k_i$, where k_i is the number of observed items for patient i. The smaller matrix on the right is $F \times F$, where F is the number of factors. In this application F is far smaller than the number of indicators, so the computation is much cheaper.

The matrix

$$A_O = I + L_O^\top \Psi_O^{-1} L_O \tag{34}$$

is then Cholesky-decomposed for numerical stability. This is the second place Cholesky appears. The first Cholesky, $R = \text{chol}(\Phi)$, is used to express the low-rank covariance. The second Cholesky is used to compute the log-determinant and solve linear systems involving A_O without materializing a large inverse. Neither Cholesky changes the model; both are safe ways to evaluate the same likelihood.

The implementation also groups patients by missingness pattern. The determinant term depends on which columns are observed, not on the observed values themselves. Therefore all patients sharing the same observed-item mask can reuse the same small matrix work, while each patient's residual vector supplies its own quadratic penalty.

3.8 The general factor G in depth

The general factor is what makes this a bifactor model rather than an ordinary correlated-factors model. Three facts distinguish G: unit variance, anchoring by functioning/global-severity indicators, and orthogonality to the specific factors in the primary bifactor parameterization. The purpose is conceptual hygiene. G should read as "how much overall functional burden," while the specific factors carry "what kind of burden."

In the primary bifactor model,

$$\Phi = \begin{pmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & \Phi_D \end{pmatrix}. \tag{35}$$

The zero off-diagonal block encodes $G \perp D_k$. The specific block Φ_D may still contain correlations among the specific dimensions. In a correlated-G sensitivity model, the zero block is relaxed:

$$\Phi = \begin{pmatrix} 1 & \phi_{Gd}^\top \\ \phi_{Gd} & \Phi_D \end{pmatrix}. \tag{36}$$

This sensitivity model asks whether severity is empirically entangled with each specific factor. The vector ϕ_{Gd} is therefore not a nuisance. It is a scientific object: it quantifies how much each domain is tied to functional burden.

Why the G row matters

The statement "biology is relatively separable from severity" is not a visual slogan; it is a statement about ϕ_{Gd} and about the direct bifactor loadings λ_{jG} of biological indicators. If metabolic and inflammatory entries in the G row of Φ remain small in the correlated-G sensitivity model, while sleep or cognition are more severity-linked, the model has quantified separability rather than assumed it.

3.9 Soft-prior loading roles and identification

Theory enters through the prior on each loading. Every item-factor cell receives a role: a primary/home loading, a general-factor loading, a plausible cross-loading, an unlikely cross-loading, or a near-zero cell keeping a G anchor off the specifics. The home loadings are positive, which fixes the sign of each factor so that higher coordinates mean more burden after item-level sign orientation.

Table 4. Soft-prior roles and the induced prior on the loading λ_{jk}. Hard-zero primary runs fix the unlikely cells exactly at zero; the soft-unlikely arm frees them with tight shrinkage as a sensitivity analysis.

role	meaning	prior on λ_{jk}
`primary` / `g_anchor`	item should define this factor	$\mathcal{N}_{+}(0.60,\, 0.30^2)$, truncated > 0
`bifactor_G` / `plausible_cross`	item may carry this signal	$\mathcal{N}(0,\, 0.25^2)$, possibly tighter by block
`unlikely_cross`	should be near zero	exact zero in primary; $\mathcal{N}(0,\, 0.05^2)$ in soft arm
`g_anchor_on_specific`	G anchor held off specifics	exact zero or near-zero sensitivity prior
covariate / outcome	adjust for / reserve for validation	not in Λ

The prior is more than decoration. A Gaussian factor model has a rotation problem: for suitable invertible matrices M,

$$\Lambda^\star = \Lambda M, \qquad \Phi^\star = M^{-1}\Phi M^{-\top} \quad \Longrightarrow \quad \Lambda^\star\Phi^\star(\Lambda^\star)^\top = \Lambda\Phi\Lambda^\top. \tag{37}$$

The likelihood alone sees the same Σ after such a rotation. The loading matrix is therefore not identified by the likelihood alone. The sparse prior breaks this symmetry because a rotated loading matrix no longer aligns with the theory map: home loadings may no longer be positive and substantial, and cells intended to be near zero may become large. The posterior concentrates on the orientation where the empirical covariance is explained while the loading pattern remains interpretable.

The recipe analogy

Two different-looking loading matrices can cook the same observable covariance dish if one is a rotation of the other. The likelihood tastes only the final dish. The prior checks whether the recipe respects the intended ontology: CRP should define inflammation, BMI should define metabolism, FAST should anchor general burden, and unlikely cross-loadings should stay small unless the data strongly justify them.

3.10 Bayesian inference and model-parameter learning

Bayesian inference combines the likelihood with the priors:

$$p(\Theta \mid X_{\text{obs}}) \propto p(X_{\text{obs}} \mid \Theta)\, p(\Theta). \tag{38}$$

In log form,

$$\log p(\Theta \mid X_{\text{obs}}) = \log L(\Theta) + \log p(\Theta) + C, \tag{39}$$

where C does not depend on Θ. The data are fixed. The posterior is a landscape over candidate parameter values. At one candidate value, the model proposes a loading matrix Λ, a factor-correlation matrix Φ, residual scales σ_j, and any active intercept/covariate parameters.

It builds $\Sigma = \Lambda\Phi\Lambda^\top + \Psi$, evaluates the observed-cell likelihood, adds the prior score, and obtains a posterior log density. At another candidate value, this entire score changes.

The continuous marginalized model samples structural parameters but not patient coordinates. The Gaussian patient scores f_i have been integrated out. Therefore the sampler targets posterior draws of the continuous parameter block

$$\Theta_{\text{cont}} = \{\Lambda, \Phi, \sigma\} \tag{40}$$

and, in the in-likelihood covariate mode, also α and B. This is why marginalization is computationally valuable: instead of sampling thousands of patient-level latent coordinates, the continuous stages sample the population-level measurement map.

Parameter learning is performed by MCMC, specifically NUTS/Hamiltonian Monte Carlo in the implementation. It is better to think of this as posterior exploration than as ordinary deterministic optimization. A maximum-likelihood or MAP optimizer would try to find a single best point. MCMC instead produces draws from the posterior, quantifying uncertainty in every loading, correlation, and residual scale. The optimization-like part is the sampler's repeated use of gradients to move toward high posterior-density regions; the inferential output is a cloud of plausible parameter values, not only one point estimate.

The staged fitting strategy is a continuation method. It first estimates a stable continuous backbone, then adds correlations among specific factors and symptom windows, then adds additional continuous factors, and finally runs the full mixed stage. Each rung starts from a simpler geometry and hands information forward. This is not a change in scientific estimand; it is risk management for a high-dimensional posterior with sparse loadings, missingness, and mixed measurement families.

Mixed-stage inference. In the native mixed model, not all variables can be marginalized. Binary, low-cardinality ordinal, and count likelihoods do not collapse into the same Gaussian covariance identity. Factors carrying such variables require explicit patient latent coordinates in the sampler. The mixed model therefore partitions the factors into explicit and marginalized blocks: the explicit block is sampled per patient because discrete likelihoods attach to it, while the remaining Gaussian block is still marginalized conditionally. In the Gaussian-copula acceleration path, some high-resolution ordinal/count indicators are promoted into the Gaussianized block, reducing the explicit burden.

Diagnostics. The posterior draws must be checked. $\widehat{R}$ asks whether independent chains agree; ESS asks how many independent samples the autocorrelated chains are worth; divergences flag failures of the Hamiltonian trajectories to follow the posterior geometry. These diagnostics are not administrative details. They tell us whether the reported loading atlas and factor correlations are numerically credible.

3.11 Patient scoring after fitting

Fitting estimates the population-level measurement map. Patient scoring asks a different question: after fixing Λ, Φ, and Ψ at fitted values, where does patient i sit in latent space given their observed items? For the Gaussian block this is another conditional Gaussian calculation:

$$f_i \mid x_{i,O} \sim \mathcal{N}\Big(\Phi\Lambda_O^\top\Sigma_{OO}^{-1}(x_{i,O} - \mu_{i,O}),\ \Phi - \Phi\Lambda_O^\top\Sigma_{OO}^{-1}\Lambda_O\Phi\Big), \tag{41}$$

where $O = \mathcal{O}_i$. The posterior mean is the patient's coordinate estimate; the posterior covariance gives score uncertainty. A patient with many observed home indicators has a tighter coordinate estimate. A patient with few or no observed home indicators is partly or fully prior-dominated.

This separation is essential. The marginalized likelihood learns Λ, Φ, and Ψ without sampling patient scores. Scoring then projects patients into the learned map. The two operations are connected by the same Gaussian algebra, but they answer different operational questions.

3.12 The prior atlas: the theory half of the deliverable

The soft-prior loading matrix is itself an artifact — the *prior atlas* (Figure 34, Appendix B), an indicator×factor heatmap of where each instrument is *expected* to load, generated from configuration alone with no patient data. It encodes a block-diagonal primary structure (each factor anchored by its own pool), a faint G column over every specific item, theory-permitted windows, and hypothesised biological bridges. The scientific deliverable of the measurement map is the *prior* $\rightarrow$ *posterior* comparison: this theory map placed beside the fitted empirical atlas, with a per-candidate verdict. As built, 143 indicators across the candidate factors enter as modelled cells.

End-to-end summary

The measurement pipeline is: orient indicators so higher means more burden; optionally Gaussianize suitable marginals through the copula transform; write the bifactor/ESEM measurement equation; integrate Gaussian patient latents to obtain $x_i \mid \Theta \sim \mathcal{N}(\mu_i, \Lambda\Phi\Lambda^\top + \Psi)$ or its copula-scale equivalent; evaluate the observed-cell log-likelihood with missing-data submatrices; compute it efficiently with Woodbury and Cholesky; combine it with sparse identifying priors; sample the posterior with NUTS; and finally project patients back into the learned latent map with uncertainty.

4 Estimation: staged continuation to the global fit

4.1 The target and the three difficulties

The scientific object is the single global posterior $p(\Theta \mid X_{\text{obs}})$ with $\Theta = \{\Lambda, \alpha, \theta, \Phi, \text{latent}\}$ for all eligible dimensions, all indicators, all likelihoods, on the full V0 sample. Fitting it cold is hard for three concrete reasons, each a failure mode:

1. **Funnel geometry at scale.** $\sim$9,000 patients $\times$ $\sim$8 latent dimensions $\approx$ 70,000 per-patient latent parameters with hierarchical variance produce Neal's funnel and NUTS divergences.
2. **Weak identification.** G + correlated specifics + cross-loadings is identified only up to rotation/sign/label; on sparse, cohort-structured missingness the posterior can be multimodal.
3. **Mixed-likelihood stiffness.** Gaussian + ordinal + Bernoulli + negative-binomial across cohort-specific missingness at full N is a large, stiff joint model.

4.2 Continuation (homotopy) staging

Staging is *not* a scientific sequence of results — it is a continuation method. We solve an easy, well-conditioned sub-model, then deform toward the full target one source of difficulty at a time, carrying each converged posterior forward as the warm-start for the next. Formally we fit a nested sequence $\mathcal{M}_1 \subset \mathcal{M}_2 \subset \cdots \subset \mathcal{M}_5 =$ global and initialise $\theta^{(s+1)}_{\text{init}} = \mathbb{E}[\theta \mid \mathcal{M}_s]$.

What staging buys: *fault isolation* (a failure at stage s implicates only what s added); *warm-starts* (the hardest fit begins in a good basin); *risk quarantine* (each failure mode is introduced at a known stage with its mitigation); *stability-under-elaboration* as robustness evidence; and a

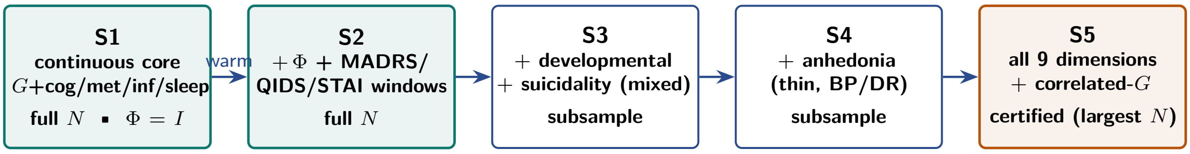


Figure 5. Staged continuation. Each stage adds exactly one source of difficulty; the converged posterior warm-starts the next. Green = certified at full N; amber = the reported nine-dimension global fit, documented at the largest N that mixes (cross-seed Tucker φ 0.993). *Only $\mathcal{M}_5$ is interpreted.* S3–S4 also adjudicated developmental-risk, suicidality, and anhedonia (rejected).

natural QC/discussion gate after each step. After the seven-dimension backbone first stabilised, two candidates that had been *deferred* (mania, substance) were folded in and the *joint* nine-dimension model re-certified — not bolted on — so the reported map is one internally consistent object (Chapter 5).

The interpretation rule

Only $\mathcal{M}_5$ (the global fit) is interpreted or reported. Stages S1–S4 are convergence/identification checkpoints with *provisional* reads. Publishing an intermediate stage is exactly the error to avoid — a prior iteration concluded "no general factor" from a sub-model that omitted the severity anchors. Each stage emits a report and is followed by a discussion gate before advancing.

A stage advances only through the acceptance gate: $\widehat{R} \le 1.01$; ESS ≥ 400 (bulk and tail); 0 divergences; no Heywood case ($\max_j |\lambda_j|$ within a cap); and posterior-predictive checks not grossly violated.

4.3 The Gaussian-block marginal, and its identity with FIML

For continuous indicators the latent factors can be integrated out (*marginalized*) analytically — averaged over rather than estimated one patient at a time, which is what removes the per-patient "funnel" that otherwise makes these models hard to sample. Stacking the continuous loadings into Λ, the full factor covariance into Φ, and the residual variances into $\Psi = \operatorname{diag}(\sigma_j^2)$, the marginal over patient i's observed continuous sub-vector is Gaussian with

$$X_{i,\mathcal{O}_i} \sim \mathcal{N}(\mu_{i,\mathcal{O}_i},\, \Sigma_{\mathcal{O}_i\mathcal{O}_i})\,, \qquad \Sigma = \Lambda\,\Phi\,\Lambda^\top + \Psi, \quad \mu_i = \alpha + Bc_i. \tag{42}$$

Summing $\log \mathcal{N}(\cdot)$ over each patient's observed sub-vector is *exactly the FIML objective.*

Why this makes a separate FIML confirmation redundant

The marginalized Bayesian model and a full-information maximum-likelihood SEM optimize the *same* observed-data likelihood; they differ only by the presence of priors. A flat-prior MAP therefore equals the MLE equals the FIML solution. Hence (i) FIML adds no independent estimator evidence beyond a prior-free refit (Chapter 6), and (ii) this marginal is also the computational route that removes the per-patient latent funnel for the Gaussian block while keeping the full sample.

4.4 The low-rank Woodbury likelihood

Evaluating Eq. (42) naively costs a determinant and a solve on $\Sigma_{\mathcal{O}_i\mathcal{O}_i}$, which lives in indicator space (up to 71×71 for a well-measured patient) — and NUTS needs this, with its gradient, many times per draw. But $\Sigma = \Psi + \Lambda\Lambda^\top$ has a diagonal-plus-low-rank structure (Ψ diagonal; $\Lambda\Lambda^\top$ rank $F = K{+}1$) — per-indicator noise plus a handful of shared factor directions. The

Woodbury identity (with the matrix-determinant lemma) is the standard algebraic shortcut for exactly this shape: it lets the costly determinant and solve be carried out in the small *factor* space rather than the full indicator space — the work moves into a room a few-by-a-few wide instead of seventy-by-seventy. With $W = \Psi^{-1}$ (the observed-cell precision; $W_{ij} = 0$ where indicator j is missing for patient i),

$$A_i \;=\; I_F + \Lambda_{\mathcal{O}_i}^\top W_i \Lambda_{\mathcal{O}_i} \in \mathbb{R}^{F\times F}, \qquad b_i = \Lambda_{\mathcal{O}_i}^\top W_i x_{i,\mathcal{O}_i}, \tag{43}$$

$$\log|\Sigma_i| = \log|\Psi_i| + \log|A_i|, \qquad x^\top \Sigma_i^{-1} x = x^\top W_i x - b_i^\top A_i^{-1} b_i, \tag{44}$$

so each patient's marginal log-likelihood is

$$\ell_i = -\tfrac{1}{2}\Big[k_i \log 2\pi + \log|\Psi_i| + \log|A_i| + x_i^\top W_i x_i - b_i^\top A_i^{-1} b_i\Big], \quad k_i = |\mathcal{O}_i|. \tag{45}$$

The only matrix factorized is A_i, of size $F \times F$ (e.g. 5×5 at S1) rather than up to 71×71. Correlated factors fold in through the reparameterization $\widetilde{\Lambda} = \Lambda\,\mathrm{chol}(\Phi)$, so $\Lambda\Phi\Lambda^\top + \Psi = \widetilde{\Lambda}\widetilde{\Lambda}^\top + \Psi$ and the same kernel runs unchanged. The likelihood is added to the model as a single potential $\sum_i \ell_i$; NUTS then samples only the loading/residual parameters — the ~70,000 patient latents are gone.

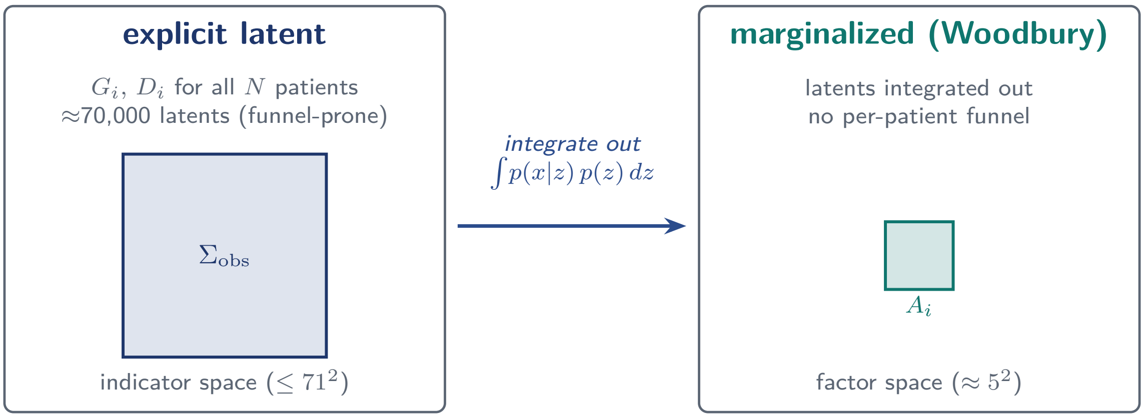


Figure 6. The mechanism behind full-sample tractability. *Left:* the explicit model instantiates a latent vector for every patient (≈70,000 parameters) and evaluates the Gaussian likelihood with a determinant and solve on the observed sub-covariance Σ_{obs} — a dense matrix in *indicator* space, up to 71×71. *Right:* integrating the Gaussian latents out and applying the Woodbury identity moves the only matrix that must be factorized, A_i, into *factor* space ($\approx 5 \times 5$), and removes the per-patient funnel entirely. Same likelihood, vastly cheaper geometry.

Full-sample inference without imputation is affordable

The marginalized Woodbury parameterization fits the S1 continuous core on *all* $N = 9{,}013$ patients ($J = 68$, 415,531 observed cells), with no imputation and no subsampling, passing every convergence gate: $\max \widehat{R} = 1.010$, $\min \mathrm{ESS} = 1{,}939$, 0 divergences. The point is not an absolute runtime but that the full-sample, no-imputation invariant is reached by a sequence of *exact* reparameterizations (§4.6) rather than by approximation or by selecting an easier sub-population — a cold explicit-latent fit of the same model does not converge at this scale.

Recovering patient scores. Marginalization does not discard the coordinates: for a fixed posterior draw the conditional $z_i \mid x_{i,\mathcal{O}_i}$ is Gaussian with precision A_i and mean $A_i^{-1} b_i$ — the *same* A_i, b_i used in Eq. (45). Fitting and scoring thus separate cleanly: fit the map once on the full sample, then project any patient's observed cells onto it (with uncertainty), so a downstream cohort can be scored without driving the measurement fit.

4.5 The mixed-likelihood frontier (S3–S5)

Binary/ordinal/count indicators cannot be Gaussian-marginalized. The engine uses a *partial collapse*: the factors touching non-Gaussian indicators ($f_e = \{G, \text{suicidality}, \text{developmental}\}$) carry explicit non-centered latents, while the pure-continuous specifics (f_m = {cognition, metabolic, inflammatory, sleep}) stay marginalized and are coupled to f_e through the shared Φ by the Gaussian conditional

$$f_m \mid f_e \sim \mathcal{N}(M f_e, S), \qquad M = \Phi_{me}\Phi_{ee}^{-1}, \quad S = \Phi_{mm} - \Phi_{me}\Phi_{ee}^{-1}\Phi_{em}. \tag{46}$$

The continuous block becomes a per-patient-mean Woodbury (mean Bf_e, residual covariance $\Lambda_m S \Lambda_m^\top + \Psi$); the non-Gaussian indicators get Bernoulli/negative-binomial/ordered-logistic likelihoods on f_e. This is exactly the "partial collapse" the methods plan flagged as the hardest step, and it works: the mixed fit attains 0 divergences.

A reparameterization ladder, not hardware, resolved the mixing frontier

With 0 divergences but slow mixing, the mixed-likelihood stages were weak-identification *ridges*, not funnels — so the fix is reparameterization, not more compute. Two moves did it: (i) tightening every explicit specific's $\to G$ cross-loading toward zero (the dimensions that touch non-Gaussian items are $\approx\perp G$ anyway, so this removes a ridge without touching the biology$\perp G$ estimand), and (ii) building Φ from a *unit-row* Cholesky factor so it stays a valid correlation matrix (the same fix as the $\Phi = LL^\top$ bug, Chapter 14). The explicit-latent block remains the documented mixing limit, so the joint map is documented at the *largest N that mixes* with a cross-seed resample-stability guard.

4.6 What makes the fit expensive — and the optimizations that remove it

It is worth being explicit about where the time goes. The cost of one NUTS draw factorizes as

$$\text{cost per draw} \sim \underbrace{(\#\text{ sampled parameters})}_{\text{geometry}} \times \underbrace{(\text{leapfrog steps per draw})}_{\text{step size}} \times \underbrace{(\text{cost of one gradient})}_{\text{linear algebra}},$$

and a cold, explicit-latent fit is large on all three. The *geometry* factor is enormous: instantiating G_i and D_{ik} for every patient adds ∼45,000–70,000 latent parameters, and their hierarchical variance creates Neal's funnel, which forces tiny steps and produces divergences (so the *step-size* factor is bad too). The *linear-algebra* factor is also large: every gradient evaluates the observed-data Gaussian likelihood, which naively requires a determinant and a solve on each patient's observed sub-covariance $\Sigma_{\mathcal{O}_i\mathcal{O}_i}$ — a dense matrix in indicator space, up to 71×71, across all ∼9,000 patients — and NUTS performs this many times per retained draw, so the three factors multiply.

The engine attacks each factor with a transformation that leaves the inferential target unchanged (Table 5); the one approximation in the stack is confined to the heavier checkpoint stages and is labelled as such.

Two entries deserve a note. First, the tuning is counter-intuitive: because marginalization removes the funnel, the posterior is benign (its typical set is only ∼21 leapfrog steps deep), so the right move is to *lower* the target acceptance and cap the tree depth — bounding the expensive deep warmup trajectories — rather than the usual reflex of *raising* target acceptance to chase divergences, which here would only shrink the step size and slow a posterior that has no pathology to avoid. Second, the model is written in PyMC for a readable specification and sampled through the NumPyro/JAX backend, which compiles and batches the $F \times F$ linear algebra across patients. Together these turn an intractable cold fit into one that converges at full N, within a standard workstation's memory, in about an hour.

Table 5. The optimization stack. Every row preserves the inferential target (the same posterior, or the same observed-data likelihood) except the last — an explicitly labelled approximation used only where exact marginalization is unavailable.

factor attacked	optimization	effect
geometry	marginalize the Gaussian latents (Eq. 42)	removes ∼45–70k parameters; eliminates the funnel; the MCMC state drops to a few hundred structural parameters
linear algebra	Woodbury identity + determinant lemma (Eq. 45)	the per-patient solve moves from indicator space ($\leq 71^2$) to factor space (F^2, e.g. 5^2)
linear algebra	pattern-grouped, batched assembly	A_i as one matrix multiply; its Cholesky computed once per *unique missingness pattern* ($\approx 2.75\times$, log-likelihood-identical)
step size	tuning matched to the (now benign) geometry	lower target acceptance (0.85–0.90) + tree-depth cap 8 bound the deep warmup trajectories ($\approx 2.7\times$)
warmup length	staged warm-starts (continuation, Chapter 4)	each stage starts in the previous basin; shorter tuning, fewer failed trajectories
implementation	vectorization + cached preprocessing	the $F \times F$ work is batched over patients; harmonization is not recomputed per fit
checkpoints only	balanced random sub-sample (*labelled approximation*)	the mixed stages, whose non-Gaussian block has no closed-form marginal, run at $N = 4$–5k

Why the mixed-likelihood stages cost more, and the sub-sample fallback

The whole stack hinges on marginalizing the Gaussian latents. The non-Gaussian indicators (binary/ordinal/count) have *no closed-form latent integral*, so the suicidality and developmental factors must keep explicit per-patient latents — which restores the geometry cost the continuous stages had removed. The S3–S5 checkpoints therefore run on a *random, cohort-balanced* sub-sample ($N = 4{,}000$–$5{,}000$; realistic missingness, with a resample-stability check) — never completeness- or attrition-selected, so the missingness structure is preserved. This is the one approximation in the stack, and it is labelled as such: the reported map targets the full sample or the largest N that certifies, and a full-N certified S5 is a matter of additional sampling effort, not a change of method.

5 Results: the nine-dimension measurement map

This chapter reports what the measurement model found. It is the empirical companion to Chapter 3: there we wrote the bifactor/ESEM equation and the Gaussian-copula data layer that feeds it; here we read off the fitted map. One point of vocabulary carries through everything below. The map is estimated on the *Gaussian-copula scale*: every indicator is first translated into a standard-normal rank score (Eq. (1)), and the same Gaussian factor machinery ($z_i \sim \mathcal{N}(0, \Lambda\Phi\Lambda^\top + \Psi)$) is applied to those rank scores. The copula is therefore the **estimation vehicle** for the map — it puts a skewed laboratory value, a near-continuous anthropometric reading, and a high-cardinality count onto one common ruler so they can share a factor space honestly. A loading below is a sensitivity of normalized ranks; a factor coordinate is a position on a rank-score axis. The structure that comes out — which axes exist, how they relate, how entangled each is with severity — is the scientific object, and it is what the rest of this report builds on.

On the harmonized FACE V0 baseline (N = 9,013 = BP 6,252 · SZ 2,209 · DR 552), the copula-scale bifactor/ESEM yields a **nine-dimension transdiagnostic map**: a general factor G (functional burden) plus eight specific axes — cognition, metabolic, inflammatory, sleep, developmental-risk, suicidality, mania, and substance (Table 6). Every coordinate is estimated from each patient's observed cells only, with no imputation; missing cells contribute no likelihood term.

Table 6. The reported transdiagnostic map. Loadings are posterior means on the Gaussian-copula rank-score scale (higher means more burden after item orientation). $|\lambda_G|$ is the mean absolute bifactor loading on G — the *severity entanglement* of each domain (§5.3 refines biology's value under the correlated-G identification).

dimension	anchored by	primary λ	$\lvert\lambda_G\rvert$
G — functional burden	FAST 0.90, EGF 0.73, EQ-5D, CGI-S (functioning + severity only; no symptom content)	— (general)	—
cognition	CVLT, WAIS coding/digit-span, fluency, TMT-B	0.57	0.26
metabolic	BMI, waist, BP, glucose, HbA1c, lipids	0.32	0.08
inflammatory	CRP, WBC, neutrophils, platelets, eosinophils	0.39	0.07
sleep	PSQI objective sub-scores, ESS	0.48	0.25
developmental-risk	CTQ, age-of-onset, WURS, perinatal, family history	0.42	0.12
suicidality	ISF binary ideation/attempt + count	2.2–2.7	0.3–0.6
mania	YMRS 0.57, Altman 0.76 (BP)	0.49–0.76	0.15
substance	alcohol/cannabis lifetime SUD, nicotine	0.38–0.69	0.13

5.1 The per-patient coordinate object: a point with honest error bars

The loadings above describe the *axes*; the other half of the deliverable is *where each patient sits* on them. Every one of the 9,013 patients is a single **point in this nine-dimensional space**, its coordinates oriented so higher means more burden and scaled so 0 is the population mean and ±1 one standard deviation. So "metabolic +1.5, inflammatory +1.2, suicidality −0.3" is a complete, quantitative, *transdiagnostic* description of one person — built without ever using their diagnosis.

The crucial part is that the point is **not** a point. Each coordinate carries a posterior **mean**, **SD**, and **HDI**, a **reliability tier**, and — in the faithful arm — the full per-patient covariance S_i and joint posterior *draws* (§7). Clinically this is the difference between confidence and bluff: a patient with a full laboratory panel gets a *tight* biological coordinate; a patient with two substance items and no labs gets a *fuzzy* substance coordinate — not a confident wrong one. Unobserved or structurally-absent cells (substance for the DR cohort, skip-logic zeros) are **marginalized, never imputed**: they neither place the patient nor get filled in.

These error bars are load-bearing, not decoration. If one clustered the posterior *means* as though they were exact, the precisely measured axes (metabolic, inflammatory) would dominate the geometry while the ill-measured ones injected phantom structure. Propagating the uncertainty instead is exactly what dissolves that artefact: re-run over the posterior draws, the apparent multi-component look of the means collapses to a *single* component in all twenty draws (§8.1). The same uncertainty makes patient-to-patient similarity honest — nearest neighbours are ranked by an uncertainty-weighted distance,

$$d^2(i,j) = \sum_d \frac{(\mu_{i,d} - \mu_{j,d})^2}{\sigma_{i,d}^2 + \sigma_{j,d}^2 + \varepsilon}, \tag{47}$$

so an axis a patient is uncertain about contributes little to who counts as similar, and an ill-measured patient simply gets a *fuzzy* neighbourhood (the M2 similarity tool, §8.6). That is the continuum-honest core of the whole map: "your most similar patients, across diagnoses, look like this."

5.2 The empirical atlas, beside the theory

The scientific deliverable of the measurement map is the *prior* → *posterior* comparison (Figure 7): the theory map of where instruments were *expected* to load, placed beside the fitted map of where they *actually* load. Reading the two together is the whole story — the block-diagonal primary structure is recovered, the faint G column appears over the functioning anchors only, the windows land on G, and the biology candidate resolves into two distinct blocks. The posterior atlas alone is the per-indicator loading fingerprint of the map (Figure 8).

5.3 The headline: biology is the least severity-entangled domain

CERTIFIED The load-bearing finding concerns how much each specific axis is entangled with overall illness burden G. Because the strict bifactor constraint ($G \perp$ specifics) could in principle *understate* that entanglement, it is tested under the correlated-G sensitivity arm, which *frees* G to correlate with the specifics (Figure 9).

Biology rides on axes severity does not see

Freeing G to correlate with the specifics, it correlates +0.07 with inflammatory and +0.12 with metabolic, versus +0.39 with cognition and +0.42 with sleep. So metabolic and inflammatory load carry 98.5–99.5% of their variance *independent* of how impaired a patient is overall. Two patients who look equally ill on a clinician's global impression can differ sharply in biological load — precisely the severity-independent signal a biology-aware stratification can exploit.

A short note on why these numbers are trustworthy. The copula scale removes one common way such a correlation can be inflated: a heavy right tail shared by a biochemical marker and a severity rating can manufacture apparent coupling on the raw scale. By comparing patients on their population rank rather than their raw value, the copula estimate asks the cleaner question — do patients who rank high on metabolic load also rank high on functional burden? — and the answer remains a small +0.12. The separability is a property of the rank geometry, not of any one instrument's units.

The conservative reading of G itself supports this. G is anchored *only* by functioning and global-severity items (FAST 0.90, EGF 0.73) and carries *no* symptom content — a patient's subjective-illness rating loads ≈ 0. G is therefore a transdiagnostic *impairment* axis, not a latent "p-factor" liability to psychopathology — a deliberately modest interpretation that avoids the bifactor over-claim.

5.4 Distinct axes, and depression as a window

DOCUMENTED The dimensions are *weakly* correlated — mean $|\Phi_{\text{off-diagonal}}| \approx 0.10$ (Figure 10). The map carries genuine multidimensional information, not one severity factor in

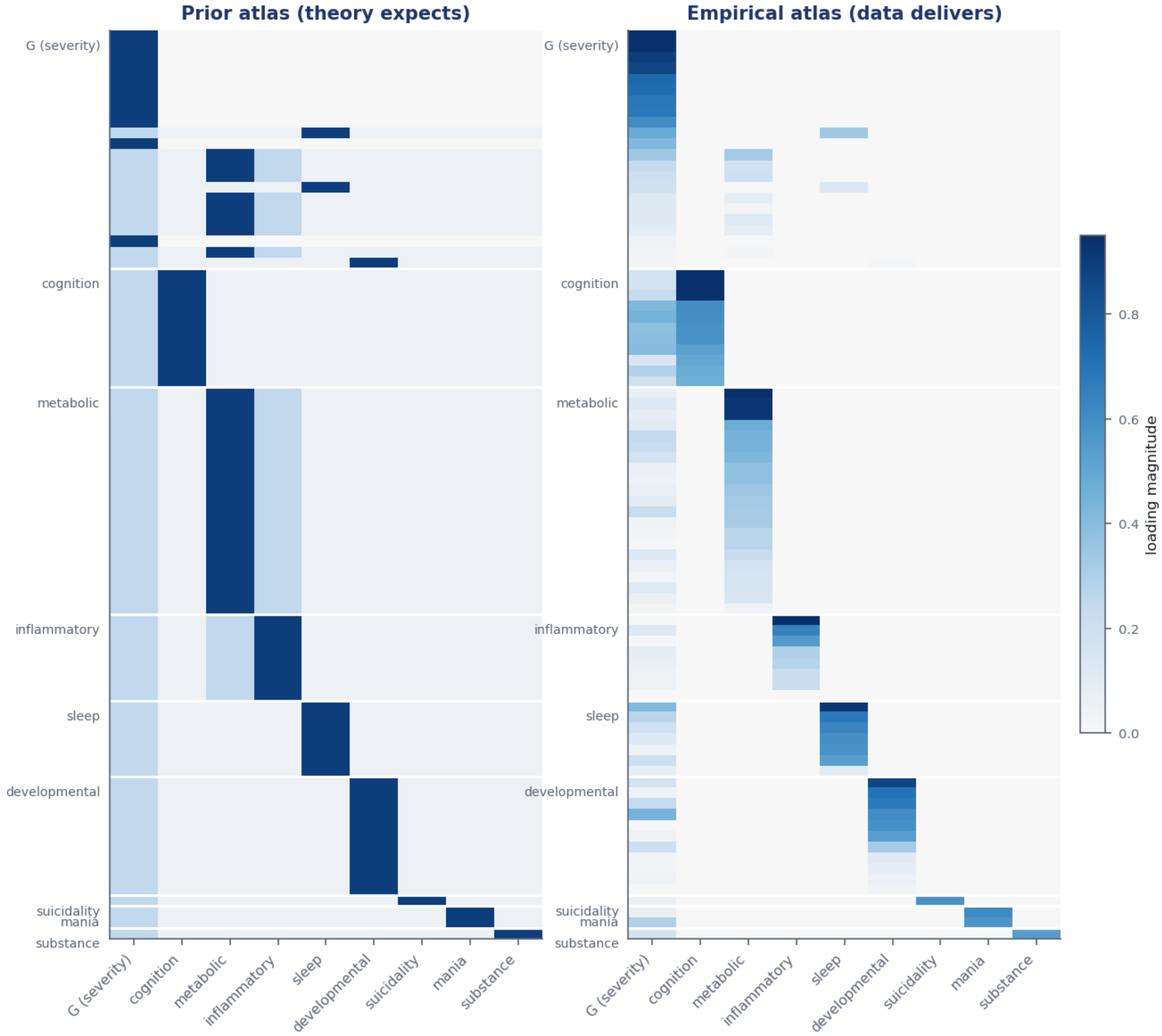


Figure 7. Prior → posterior. *Left:* the soft-prior atlas (theory — where each instrument is *expected* to load). *Right:* the fitted posterior atlas (data — where it *actually* loads). Rows are the modelled indicators grouped by home factor, in a shared order; columns are the nine dimensions. The data recover the block-diagonal primaries theory proposed, add a faint general-factor column over the functioning anchors, and — crucially — earn structure theory only hypothesised (the metabolic/inflammatory split). This side-by-side is the hybrid model adjusting theory with evidence.

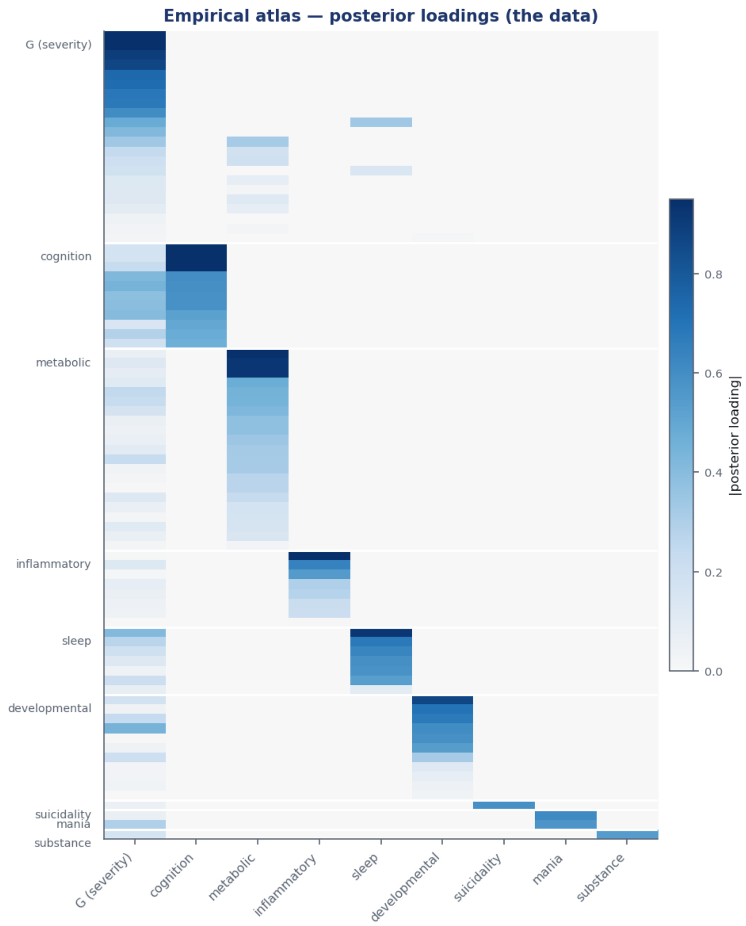


Figure 8. The empirical atlas — posterior loading magnitudes, rows grouped by home factor. Each dimension is anchored by its own block; the leftmost (G) column is faint for the specific items (biology faintest of all) and dark only for the functioning anchors at the top.

disguise. The only non-trivial couplings are clinically coherent: metabolic↔inflammatory 0.19 (an immunometabolic link — the two are related but *not* collinear, vindicating the split), suicidality↔developmental 0.23 (childhood adversity → suicidality), and sleep↔developmental 0.19.

The composite depression/anxiety instruments are *not* a separate dimension: MADRS, QIDS, and STAI load 0.80, 0.77, and 0.66 on G (with minor sleep side-loadings), behaving as cross-loading *windows* onto overall burden. Depression and anxiety severity are expressions of functional burden here, not a separable affective latent — the model was free to grow an eleventh factor and declined.

5.5 The map is larger than seven: mania and substance

Two candidates had indicators but were initially *deferred* from the staged build. Closing that gap **confirmed both as real, low-G axes** and the joint nine-dimension model was re-fit rather than bolted on. **Mania** is anchored by YMRS (0.57) and the Altman self-report (0.76 in BP), with $|\lambda_G|$ 0.15. **Substance** is anchored by lifetime alcohol/cannabis use-disorder and nicotine indicators (+0.38 to +0.69), with $|\lambda_G|$ 0.13. Both axes are carried by indicators whose native form is binary or count; on the copula scale the high-resolution members of each pool enter the Gaussianized block while the sparse binary members keep their native likelihood (the tiering of §3.2). Manic activation and substance use are thus distinct transdiagnostic axes, not reducible to severity — and worth carrying into stratification.

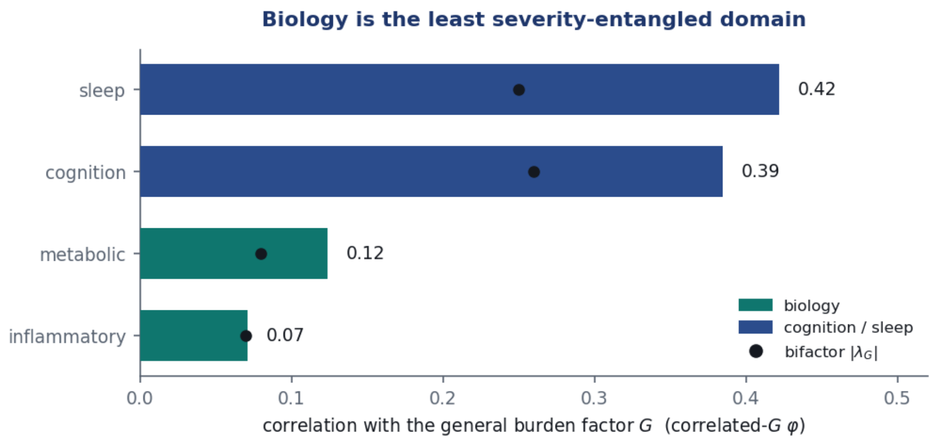


Figure 9. Severity entanglement per domain. Bars: the correlated-G factor correlation with G; dots: the stricter bifactor direct loading $|\lambda_G|$ (the two agree on the ordering). Metabolic (0.12) and inflammatory (0.07) burden are nearly independent of G, far below cognition (0.39) and sleep (0.42). Mania and substance are likewise low ($|\lambda_G|$ 0.15 and 0.13).

5.6 Suicidality as a mixed-likelihood axis

DOCUMENTED That non-Gaussian psychopathology composes with the continuous biology in one correlated-factor space is itself a result. The binary ISF ideation and attempt items are exactly the kind of low-resolution indicator the copula tiering keeps in its native Bernoulli/count likelihood rather than Gaussianizing: they load strongly on their own axis *and* modestly on G (Table 7), with 0 divergences. The proper discrete likelihoods integrate into the shared Φ alongside the Gaussianized biology without breaking identification or being crushed onto a pseudo-continuous scale.

Table 7. Suicidality indicators: home loading and bifactor G loading (logit scale; these items keep their native Bernoulli/count likelihood). Ideation and attempt load on the specific axis *and* on overall burden — clinically coherent.

ISF indicator	home (suicidality)	G (bifactor)
isf01–05 (ideation)	$+2.2$ to $+2.7$	$+0.33$ to $+0.56$
isf08 / isf09 (attempt)	$+1.8$ / $+2.0$	$+0.12$ / $+0.34$
isf08a (attempt, ordinal)	$+1.7$	$+0.12$
isf09a (attempt, count)	$+1.6$	$+0.31$

5.7 What the data rejected, and the verdict on every candidate

REJECTED **Anhedonia** does not form a stable distinct factor. With a single thin BP/DR indicator it is non-identified ($\widehat{R} = 1.54$, ESS $= 7$); when it forms, its indicator loads 0.61 on G and the QIDS-total window cross-loads 0.37 onto it (near-collinear, since the anhedonia anchor is a QIDS component). Its variance is absorbed by G + the depression windows. This is the anticipated outcome — a result, not a failure.

The full adjudication (Table 8): of the nine prior-matrix candidates, **eight are confirmed** (one a split, one a proxy), **one rejected** (anhedonia); three pre-matrix constructs are `not_testable`; depression/anxiety are windows. No candidate remains deferred.

The chapter in one object. The measurement model delivers two things at once: a set of *axes* — nine validated, transdiagnostic dimensions, with biology the least severity-entangled of them

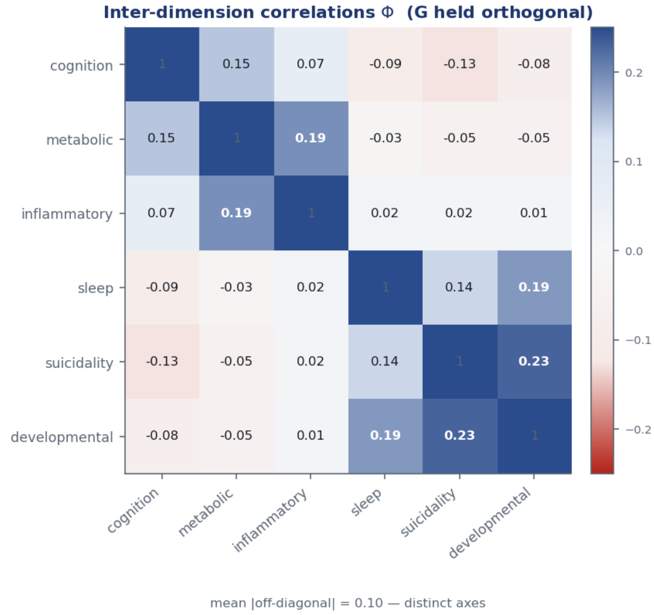


Figure 10. Inter-dimension correlations Φ (specific block; G orthogonal by construction). The off-diagonal is weak (mean 0.10); the strongest links — metabolic↔inflammatory and suicidality↔developmental — are exactly the clinically expected ones. mania and substance integrate under the same Φ with low couplings (cross-seed Tucker φ 0.993).

(§5.3) — and a per-patient *coordinate object* on those axes, every position carried with honest uncertainty rather than as a false point (§5.1). Neither half is the deliverable alone: the axes without the coordinates are a factor structure, the coordinates without the uncertainty an over-confident table. Together they are the map every later chapter reads from — a place to put any patient, and a calibrated statement of how sure that placement is. The next chapter asks the obvious question of any such map: do its positions cluster into *types*, or is it a continuum?

6 Confirmation and validation

A measurement claim is only as good as its adversarial checking. It is not enough to fit a nine-dimension map and report that it converged; we have to ask, repeatedly and in good faith, "what *else* could have produced this picture?" Every plausible alternative explanation that we cannot rule out is a crack in the foundation, because everything downstream — the strata of Chapter 8, the temporal-coherence checks of Chapter 9, the prognostic models of Chapter 10, and the treatment-moderation analysis of Chapter 11 — inherits whatever structure the measurement model invents. A map that is partly an artefact of the priors, the sampler, or the cohort mix would quietly contaminate all of it.

So we harden the map along six arms, each isolating one "could this be an artefact?" question:

- of the *prior* or the *estimator* — did the soft theory priors paint the picture rather than the data? (§6.1)
- of *confounding* — is the biology$\perp G$ separation really a medication, adiposity, or site signal in disguise? (§6.2)
- of the *sampler* — are the posterior draws numerically trustworthy, or is convergence an illusion? (§6.3)
- of *resampling and cohort imbalance* — would a different draw of patients, or dropping a cohort, give a different map? (§6.4)

Table 8. Per-candidate adjudication (the empirical dimension atlas, §6). Theory proposed; the FACE data disposed.

candidate	verdict	basis
Overall severity	confirmed ($\rightarrow G$)	clean functioning anchor; no symptom content
Cognitive flexibility	confirmed	mean primary 0.57; invariant across cohorts
Metabolism / immuno	confirmed (**split**)	metabolic + inflammatory, $\Phi \approx 0.19$
Sleep / circadian	confirmed	0.48; most invariant axis
Neurodevelopment	confirmed (**proxy**)	own axis 0.42; early-adversity/liability
Suicidality	confirmed (mixed)	binary ISF $+2.2$ to $+2.7$ logit
Mania / activation	confirmed	YMRS/Altman 0.49–0.76; $\lvert\lambda_G\rvert$ 0.15
Substance use	confirmed	SUD + nicotine, Bernoulli/NB; $\lvert\lambda_G\rvert$ 0.13
Anhedonia	**rejected**	thin; absorbed by G + depression windows
Impulsivity / Negative symptoms / Sensory	`not_testable`	no common indicators
Depression / anxiety	*windows*, not a dimension	load 0.66–0.80 on G

- of *cross-cohort non-comparability* — does each loading mean the same thing in BP, SZ, and DR, or are we averaging apples and oranges? (§6.5)
- of *absolute misfit* — can the model actually *regenerate* data that look like the real thing? (§6.6)

One simplification keeps this chapter honest and short. The map is estimated under the Gaussian-copula observed-data likelihood of Chapter 3: each indicator is carried into the model on its standard-normal rank scale, so a loading is a slope on the Gaussianized rank score, and the whole nine-dimension covariance is then read through the same marginalized Woodbury machinery. Because that marginalized likelihood and full-information maximum likelihood (FIML) optimize the *same* observed-data objective (Chapter 4), a separate FIML confirmation arm would re-ask a question we have already answered. Confirmation is therefore done *in-engine*: a prior-free refit, model comparison, and posterior-predictive checks, all on the same copula objects that produced the map. Nothing in this chapter appeals to a second pipeline; these are the results.

What "artefact" means here, concretely

An *artefact* is any feature of the reported map that would change if we changed something that should be scientifically irrelevant: the prior, the random seed, the particular patients sampled, or the cohort labels. The biology$\perp G$ separation, the split of biology into metabolic and inflammatory axes, and the mania/substance axes are scientific claims only if they *survive* those changes. Each arm below perturbs one such irrelevant thing and measures how far the map moves.

6.1 Not a prior or estimator artefact

CERTIFIED Theory entered the model softly, through the loading priors of Chapter 3: BMI is *expected* to define metabolism, CRP to define inflammation, FAST to anchor general burden. A fair sceptic asks whether those priors did the work — whether we found inflammation because the data carry it, or merely because we told the model to look there. The clean test is to delete the theory and refit.

Concretely, we run a *flat-prior* refit: the soft priors are replaced by identification constraints only (sign-oriented home cells so the factors point the right way; signed cells centered at zero with no shrinkage), a fresh seed, and no warm-start from the primary fit. If the priors were painting the picture, removing them would smear it. They do not. The prior-free fit reproduces the soft-prior loadings and the factor-correlation matrix Φ *exactly* — per-factor *Tucker congruence* $\varphi = 1.00$ and max $|\Delta\Phi_{\text{off-diag}}| = 0.00$.

Tucker congruence φ

φ is the cosine similarity between two columns of loadings — the same factor estimated two ways. It runs from 0 (orthogonal, unrelated) to 1 (identical pattern). The field's working threshold is $\varphi \geq 0.95$ for "the same factor." A value of 1.00 to two decimals means the flat-prior and soft-prior factors are, for practical purposes, the same object.

Why is this conclusive rather than merely reassuring? Because a flat-prior MAP estimate *is* the maximum likelihood estimate, which *is* FIML on the observed cells. So a flat-prior fit that lands on the same loadings tells us the structure is **earned from the data, not manufactured by the priors** — and it does so without ever running a separate FIML program. The soft priors only break the rotational tie of Chapter 3; they do not invent the inflammation axis.

That settles *whether the priors did the work*. A second, independent question is whether a general factor plus specific axes is even the right *shape* of model. We answer it by out-of-sample fit. The *WAIC* (widely-applicable information criterion — an estimate of how well a model predicts unseen data, where lower is better) decisively prefers the bifactor over both simpler alternatives (Figure 11): a single general factor is too coarse, a bag of correlated factors with no G misses the shared burden axis, and only the bifactor — a hub G plus orthogonal spokes — captures both.

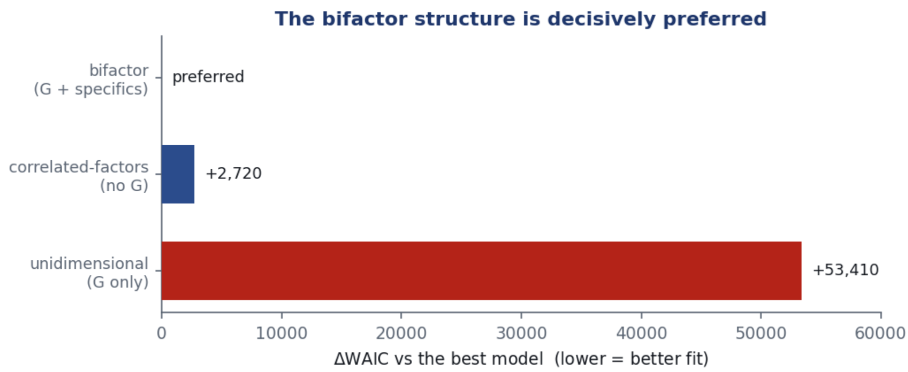


Figure 11. WAIC model comparison on a common cohort-balanced sub-sample (lower = better out-of-sample fit). The bifactor (G + specifics) is preferred over a correlated-factors model with no general factor (by $\approx 2{,}700$) and overwhelmingly over a unidimensional null (by $\approx 53{,}000$). The lesson is that both a general factor *and* the specific axes are needed: drop G and you lose the shared functional-burden signal; drop the specifics and you lose everything that makes the map transdiagnostic.

The headline separation, quantified rather than asserted

The load-bearing finding of the map — that biology is the *least* severity-entangled domain — is a statement about the G row of Φ, not a slogan about a picture. In the correlated-G sensitivity model of Chapter 3, where G is freed to correlate with each specific axis, the biological axes stay nearly orthogonal to functional burden: metabolic$\sim G \approx +0.12$ and inflammatory$\sim G \approx +0.07$, against cognition$\sim G \approx +0.39$ and sleep$\sim G \approx +0.42$. Read plainly: a patient's metabolic and immune load is almost independent of how impaired they are overall, whereas cognitive and sleep burden partly track impairment. Because G carries functioning content only and no symptom content, this is exactly the kind of signal a clinician's global impression cannot see — and exactly what a biology-aware stratification (Chapter 8) can exploit.

6.2 Is biology$\perp G$ a confound? medication, adiposity, and site

CERTIFIED The biology$\perp G$ result just above is the single most load-bearing claim in the report: everything downstream treats the biological corner as a *real* axis of illness rather than a re-description of severity. So it deserves the hardest adversarial question of all — and it is the one a journal reviewer asks first: *is the "biology axis" actually biology?* Three deflationary alternatives would make the finding far less interesting, even trivial:

- **Medication.** Antipsychotics *cause* metabolic syndrome — weight gain, dyslipidemia, glucose dysregulation. If the metabolic axis mostly tracks who is medicated, then "metabolic load independent of severity" collapses to "*drug exposure* independent of current symptoms" — a near-tautology, not a discovery.
- **Adiposity.** If the metabolic factor is dominated by body weight, "metabolic$\perp G$" is just "BMI$\perp G$."
- **Site / assay batch.** The three cohorts ran their bloodwork at different sites on different platforms. A "biological axis" that really loaded on *which laboratory ran the sample* would be a between-cohort scale artefact in a transdiagnostic costume.

Partialling out a confounder, intuitively

To "adjust for" a confounder is to ask a counterfactual: *among patients matched on age, sex, education, recruitment site, and medication, is metabolic load still independent of severity?* We answer it the direct way — by *subtraction*. For each measurement we remove the part a confounder can predict (regress, say, CRP on age/sex/site/antipsychotic and keep the leftover) and re-estimate the factor correlations on those leftovers, the *residuals*. The logic is a sieve: if the biology$\perp G$ separation were really the confounder wearing a biological mask, then subtracting the confounder would wash it out and the residuals would show nothing. If the separation is still there after the confounder's whole footprint has been removed, it was never the confounder to begin with. Each rung of the ladder below subtracts one more suspect and re-reads $\Phi(G, \cdot)$ in what remains.

We run this on the same correlated-G model that produced the headline, partialling each continuous indicator on a *growing* design of confounders before the factor model is fit.

The adjustment, and why it is conservative

For a Gaussian item, adjusting its mean by the published covariate term $\beta_j^\top c_i$ is Frisch–Waugh–Lovell-equivalent to *residualizing* the item on the covariate design c before the factor model: regress the item on c, keep the residual, then fit. The marginalized Woodbury kernel is untouched (the residual is still zero-mean Gaussian). We walk a ladder of designs, each adding the next suspect: **A0** nothing (the reported value); **A1** age(spline)+sex+education+site; **A2** A1 + antipsychotic exposure; **A3** A2 + BMI moved into the covariate block. **A2 is the headline, and it is deliberately conservative**: antipsychotics sit on the *causal path* to metabolic load (they are prescribed for severity *and* they cause metabolic change), so partialling them out removes genuine signal as well as confounding. Surviving A2 is a strong test, not a lenient one.

The antipsychotic exposure is the map-independent harmonized drug-class flag of Chapter 11 (coverage $\approx 54\%$; BP lifetime, SZ/DR current; missing values mean-imputed in the *design* only, never in the data). The model is the same correlated-G marginalized fit, on an $N \approx$ 2,000 diagnosis-balanced sub-sample, two seeds; arms A0–A2 converge cleanly ($\widehat{R} \leq 1.02$, 0 divergences).

Table 9. Biology$\perp G$ under a growing confounder design. Each cell is $\Phi(G, \text{domain})$ — the domain's correlation with functional burden — so *smaller is more severity-independent.* A0 reproduces the published map ($0.13/0.06$). Through medication and site adjustment (A2), metabolic and inflammatory stay the *two least G*-entangled domains, a factor of ~ 4 to 6 below cognition and sleep. (A3 is non-estimable — itself a result; see below.)

domain	A0 unadj.	A1 +demo+site	A2 +antipsychotic	A3 +BMI
metabolic	0.127	0.066	**0.069**	*n.e.*
inflammatory	0.064	0.051	**0.060**	*n.e.*
cognition	0.402	0.261	0.265	*n.e.*
sleep	0.371	0.358	0.361	*n.e.*

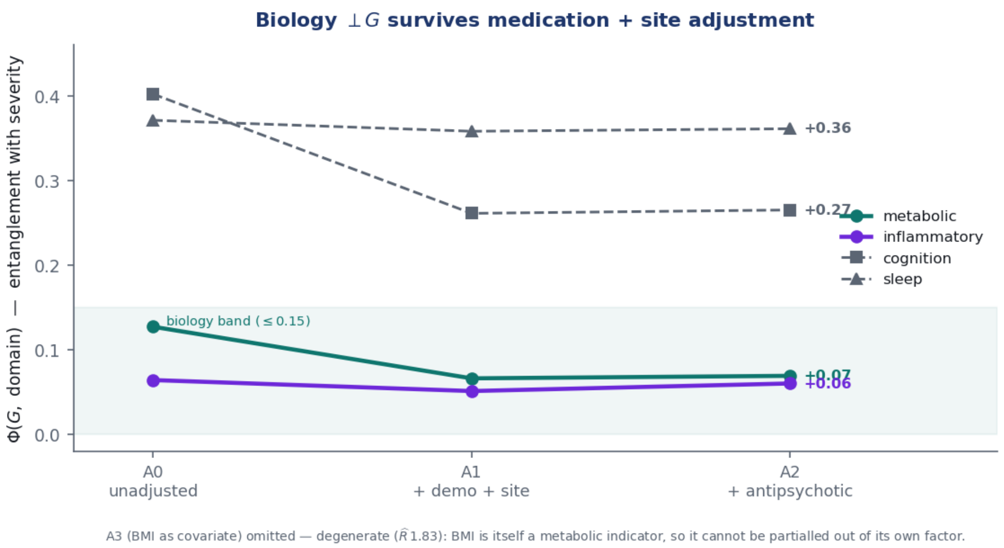


Figure 12. $\Phi(G, \text{domain})$ across the confounder ladder. The two biological axes stay inside a low "biology band" (≤ 0.15) as medication and site are partialled out — metabolic actually *halves* from A0 to A1 — while cognition and sleep remain three to six times higher. The biology axis is not a medication or site signal. (A3/BMI omitted: degenerate, see text.)

Biology $\perp G$ is not a medication, adiposity, or site artefact

After adjusting every indicator for age, sex, education, *site, and antipsychotic exposure*, the metabolic (0.069) and inflammatory (0.060) axes remain the *least* severity-entangled domains in the map — below cognition (0.265) and sleep (0.361) by roughly a factor of four to six, and essentially unchanged from their unadjusted values. Metabolic$\sim G$ does not inflate under adjustment; it *falls*. The deflationary reading — "the biology axis is just who is medicated, or how heavy, or which lab" — is not supported: the severity-independence of biological load is a property of the patients, not of their treatment, their weight, or their recruitment site.

The BMI arm is non-estimable — and that is itself the finding

Arm A3 moves BMI out of the metabolic factor and into the covariate block, asking "is metabolic$\perp G$ just adiposity?" The fit *degenerates* ($\widehat{R}\,1.83$, ESS ≈ 2): BMI is one of the metabolic factor's own primary indicators, so partialling it out partly dissolves the factor it is meant to test — you cannot cleanly remove a thing from its own definition. We report this as non-estimable rather than quote a number from a non-converged chain. The honest content is that metabolic load and adiposity are *not separable* by this method — which is exactly why the **conservative A2** (antipsychotic-adjusted, BMI *kept* as an indicator) is the headline arm. Two further bounds: antipsychotic adjustment is over-conservative (it can remove real, drug-induced biology); and a site *dummy* is coarser than the full cross-platform assay/batch harmonization this baseline data cannot supply. This is an internal robustness result on the measurement structure, not external validation.

6.3 Documented at the largest N that mixes

DOCUMENTED A map can be perfectly reproducible and still be numerical nonsense if the sampler never actually explored the posterior. Convergence diagnostics are the instruments that catch this. They ask three separate questions: do independent chains, started from different points, agree on the answer ($\widehat{R}$, where 1.00 is perfect agreement)? How many genuinely independent samples is the autocorrelated chain worth (effective sample size, ESS)? And did the Hamiltonian trajectories ever fail to track the posterior geometry (divergences, and the energy diagnostic BFMI that flags funnel pathologies)?

The reported nine-dimension fit passes the full battery across seeds: $\max \widehat{R} \le 1.04$, $\min \mathrm{ESS} \ge 112$, 0 divergences, and BFMI ≥ 0.41 (no funnel). More demanding still, it passes a *cross-seed* reproducibility check: refit from independent random seeds, the loading map returns with Tucker congruence 0.993 (largest single-loading shift 0.19). Independent runs converge on the same atlas.

There is one honest boundary, and we state it as a boundary rather than smooth it over. The continuous backbone — where the biology$\perp G$ estimand lives — mixes cleanly at the full sample ($\widehat{R} = 1.010$). The slowest-mixing object is one coupling in the discrete block (the suicidality$\sim$developmental Φ entry), whose latent coordinates cannot be marginalized and must be sampled per patient. We therefore document the joint map at the **largest N that mixes**, guarded by the cross-seed stability above. The practical reading: the point estimates are resample-stable; only the Φ-precision of that single coupling is provisional. Tightening it is more sampling effort, not a change of method or a change of map.

Why "documented", not "certified", here

The status pill on this arm is deliberately weaker than on the others. "Certified" means a check passed at full strength; "documented" means we are reporting an honest limit rather than claiming we cleared it. The distinction matters precisely because the rest of the report leans on these coordinates — a reader is owed the knowledge that one Φ entry is sampled to lower precision than the rest, even though it does not touch the headline biology$\perp G$ result.

6.4 Robust to resampling, cohort imbalance, and site

CERTIFIED The cohorts are wildly unbalanced — roughly 11 bipolar patients for every depression patient — and patients are clustered within recruitment sites. A reasonable worry is that the map is really a portrait of the dominant cohort, or of one large site, dressed up as a transdiagnostic structure. If so, it would shatter the moment we changed the sampling.

We change the sampling four ways and watch whether the map holds. *Leave-one-cohort-out* asks whether any single cohort is load-bearing. *Diagnosis-balanced subsampling* forces equal cohort representation, removing the 11 : 1 tilt. *Site cluster-bootstrap* resamples whole sites, testing for recruitment artefacts. A $1/n_{cohort}$*-weighted* fit down-weights the large cohorts directly. Across all four, the minimum Tucker congruence against the full-sample reference is **0.958** — comfortably above the 0.95 "same factor" bar. The structure is not driven by cohort imbalance, by any single cohort, or by site. It is the same map however we slice the patients.

6.5 Measurement invariance — largely invariant, with honest partials

CERTIFIED Invariance is the question of whether a measurement *means the same thing* in different groups. A loading is a translation rule from a latent axis to an observed item; if that rule differs by cohort, then comparing a BP patient's metabolic coordinate to an SZ patient's is comparing readings from two differently-calibrated instruments. We test it by refitting per cohort and comparing the primary loadings with Tucker congruence (Figure 13, left).

The core is **largely invariant**: cognition, metabolic, and sleep recover congruently in every cohort pair ($\varphi \geq 0.95$), with 12/15 comparisons fully invariant. The honest exceptions are *documented, not hidden* — and each one is itself a finding about where a single instrument means different things across populations:

- G **in BP–SZ** (φ0.92): SZ lacks the FAST behavioural anchor, so G re-anchors on CGI-S/EGF/EQ-5D there. The general factor still measures functional burden in SZ; it simply leans on different anchors.
- **inflammatory in DR** (φ0.71/0.75): a genuine immune re-weighting, not a fit failure — neutrophils load ≈ 0 in DR (0.07 vs 0.88 in BP/SZ) while eosinophils dominate. The immune axis exists in DR but is carried by different cells.
- **mania-Altman in DR** (φ0.764): clinician-rated YMRS holds across cohorts (0.57/0.41), but the *self-rated* Altman item floors out in DR ($0.76 \to 0.10$). Self-reported manic activation is near-floor in a depression cohort; lean on YMRS for mania in DR (Figure 13, right).
- **substance** is invariant BP–SZ (φ0.997) and is declared a **2-cohort axis**: its alcohol/cannabis lifetime-SUD indicators are BP/SZ-only, so the observed-cell likelihood simply omits DR rather than claiming an invariance that the data cannot support.

How to carry the partials downstream

The four partials are constraints on interpretation, not on the map's existence. When a later chapter compares strata across cohorts (Chapter 8) or tracks an axis over time (Chapter 9), the inflammatory and mania-Altman comparisons in DR, and the G comparison in SZ, carry their re-anchoring caveat; substance is read as a BP/SZ-only axis throughout. These are exactly the kinds of cross-cohort subtlety that an honest transdiagnostic map should surface rather than paper over.

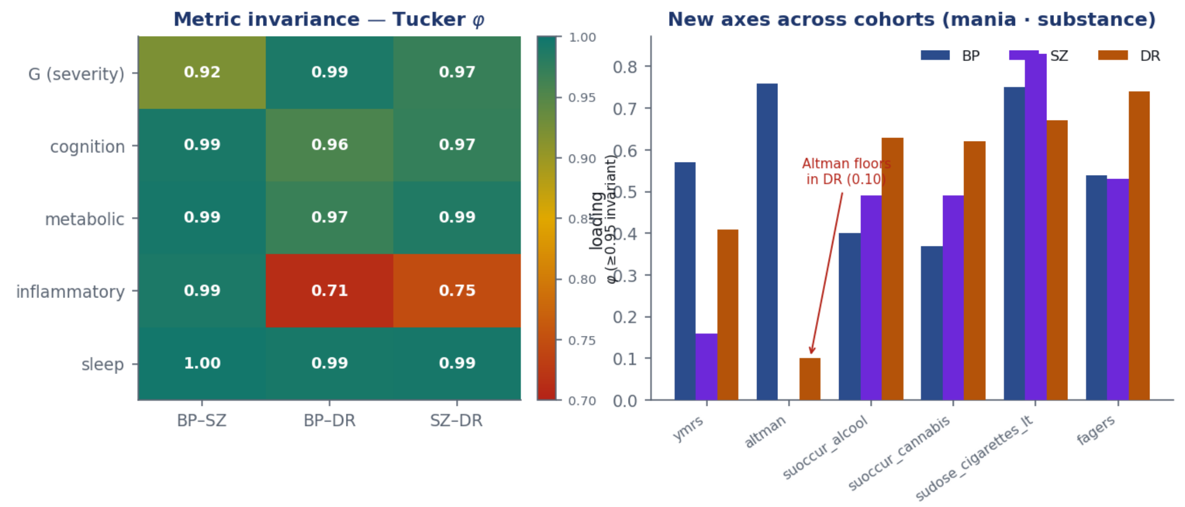


Figure 13. Measurement invariance across BP/SZ/DR. *Left:* Tucker congruence φ of the primary loadings per factor per cohort-pair (green ≥ 0.95 invariant; amber partial; red non-invariant) — the lone red block is inflammatory in DR. *Right:* per-cohort loadings of the mania and substance indicators; the self-rated Altman item does not transfer to DR, while clinician-rated YMRS does. Partials are reported as caveats, not smoothed over: each marks a place where a cross-cohort comparison on that axis must be read with the documented re-weighting in mind.

6.6 Absolute fit across both likelihood blocks

CERTIFIED The four arms above are *relative*: they show the map is reproducible, prior-free, resample-stable, and largely invariant. None of them asks the most basic question of all — can the model generate data that actually *look like* the real patients? That is the job of posterior-predictive checks (PPCs). We draw synthetic datasets from the fitted posterior and compare summary statistics of the synthetic data to the observed data. A model that cannot reproduce the data it was fit to has misspecified something, no matter how clean its diagnostics.

We run the check on both halves of the mixed likelihood. For the Gaussianized continuous block, the natural summary is the model-implied versus observed pairwise correlation structure, condensed into a *Bayesian SRMR* — a root-mean-square residual correlation, where the conventional good-fit threshold is < 0.08. The map scores SRMR ≈ 0.074. The small residual misfit is confined to *repeated-measure* clusters — the same quantity measured twice (cholesterol/LDL, supine/standing blood pressure and heart rate, AST/ALT, WAIS subtests) — which is method-factor structure (two measurements of one thing share extra variance), not a failure of the dimensional map. For the non-Gaussian block — the binary and ordinal items that keep their native likelihoods — the check is whether the model reproduces each indicator's observed endorsement rate or mean. It reproduces **21 of 22 indicators** within the 90% posterior-predictive interval (Bayesian $p \approx 0.5$, i.e. the observed value sits squarely in the middle of the predicted range; Figure 14).

One item-level mis-fit, isolated and flagged

The single exception is `isf09a`, the suicide-attempt *count* — 90.8% zeros, a hurdle a plain negative binomial over-predicts in the high-suicidality tail. The fix is the right likelihood for that one item: a hurdle / zero-inflated count, should its count precision ever be needed. Crucially this is an **item-level** caveat, not a **factor-level** one. The suicidality *dimension* is carried by its seven binary ISF ideation/attempt items, all of which reproduce cleanly (Bayesian p 0.48–0.59). The axis is sound; one count item on it is mis-specified.

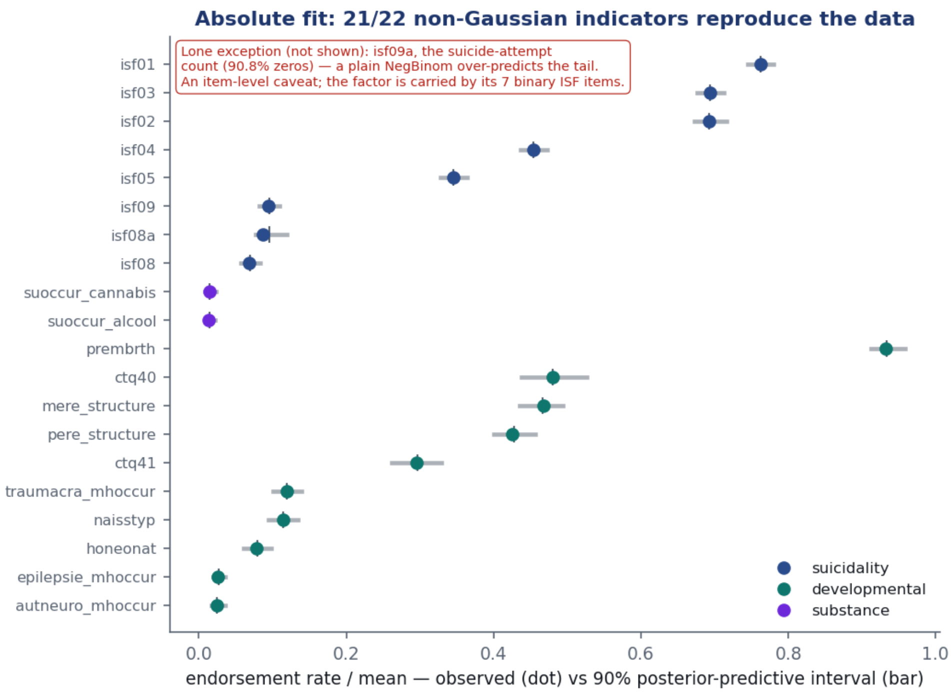


Figure 14. Absolute fit of the non-Gaussian block: observed endorsement rate (dot) vs the 90% posterior-predictive interval (bar), per binary/ordinal indicator, coloured by home factor. Every shown indicator reproduces — the observed dot lands inside its predicted bar. The lone failure (a count item, off-scale here) is the suicide-attempt count `isf09a`, discussed below.

What the six arms buy

Taken together, the arms say the nine-dimension map is **not** an artefact of the priors (flat-prior refit $\varphi = 1.00$), of *confounding* (biology$\perp G$ holds under medication and site adjustment — metabolic 0.069, inflammatory 0.060), the sampler (cross-seed $\varphi = 0.993$, clean diagnostics), the resampling (worst-case $\varphi = 0.958$ across four schemes), or cross-cohort non-comparability (largely invariant, with four documented and interpretable partials), and it **can** regenerate the observed data on both likelihood blocks (SRMR 0.074; 21/22 discrete indicators reproduce). The biology$\perp G$ separation, the metabolic/inflammatory split, and the mania and substance axes are therefore properties of the data, not of our modelling choices — a substrate solid enough to build the strata, temporal, prognosis, and treatment milestones on (Chapters 8–11).

7 Per-patient scoring

A map is only useful if every patient can be placed on it. Because fitting and scoring separate cleanly under the marginalized model (Chapter 4), the map is fit *once* and then every patient is *projected* onto it: for a fixed posterior draw, a patient's coordinate on a continuous-anchored dimension is the Gaussian conditional $z_i \mid x_{i,\mathcal{O}_i}$ with mean $A_i^{-1} b_i$ and precision A_i — the same A_i, b_i that defined the likelihood. The explicit dimensions (suicidality, developmental, substance) are read from their latent f_e on the fit sub-sample. All 9,013 patients receive, per dimension, a posterior **mean**, **SD**, and **HDI**, the **number of observed home indicators**, and a **reliability tier**, all oriented so higher = more burden.

Coordinates carry honest uncertainty — a GPS with error bars

A patient scored from six observed cognition tasks is located precisely; a patient with none is placed at the population prior with a wide error bar. Treating those two as equally measured is exactly the mistake the reliability tier prevents. A coordinate is *well-characterised* (≥ 3 observed home indicators), *partial* (1–2), or *prior-dominated* (0, i.e. informed only by the prior and by correlated dimensions).

The tiers expose coverage honestly (Figure 15). G is well-characterised for almost everyone (functioning is near-universally recorded). Cognition is *prior-dominated for* 2,506 *patients* who had no cognitive testing — their cognition coordinate is a prior-plus-correlation guess, and is flagged as such. Most starkly, **mania is *partial* for every patient**: with only two indicators (YMRS, Altman) it can never reach the ≥ 3 "well" tier, so it is never reported as well-characterised — a built-in humility about a thin axis.

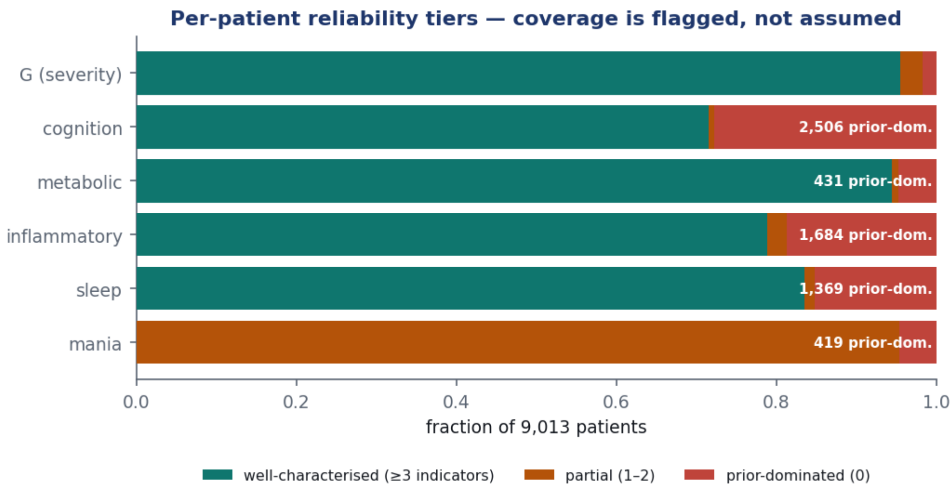


Figure 15. Per-patient reliability by continuous-anchored dimension: the fraction of the 9,013 patients that are well-characterised (≥ 3 observed home indicators), partial (1–2), or prior-dominated (0). Coverage is *flagged*, not assumed away — cognition is prior-dominated for the 2,506 untested patients, and mania (only two indicators) is partial for everyone.

Uncertainty must propagate to the stratification

The scores are a distribution, not a point. A stratification that hard-assigns patients to clusters on the posterior means would discard exactly the information these tiers encode — a prior-dominated cognition score is not evidence of average cognition, only of *absent measurement*. The per-patient SD/HDI and tier are carried forward so downstream decision regions can down-weight or abstain where a coordinate is unreliable. Two refinements remain: full-N projection of the non-Gaussian block (a logistic/count conditional, not Gaussian; the explicit scores are currently on the fit sub-sample), and a hurdle likelihood for `isf09a` if its count precision is needed.

8 Reading the map: a continuum, not biotypes

The previous chapters delivered a *map*: every one of the 9,013 patients placed on nine validated dimensions — severity, two kinds of biology (metabolic, inflammatory), cognition, sleep, mania, suicidality, developmental adversity, substance — with diagnosis deliberately held out. This chapter asks the question the whole "precision psychiatry" field chases: do patients fall into a handful of natural *types* — biotypes — that cut across the old diagnostic labels? The discipline here is to *not assume the answer*. The answer turns out to be **no**, and that negative reshapes what M2 *is*: not a typology, but a **coordinate system** for placing any patient, plus a *reading guide* for interpreting positions.

8.1 Ask whether types exist before imposing them

The classic error in clustering clinical data is to pick a number of clusters K and then "find" K groups — the algorithm always returns them, whether or not the data contain any. So before any clustering we run a **structure-discovery gate** (Figure 16): a battery of independent tests that each ask, in a different language, *is there cluster structure at all?* The decisive one is a **falsification null** — we compare the best partition of the real coordinates against the best partition of a single structureless Gaussian blob with the same mean and covariance. If the real data separate no better than the blob, there are no kinds to find.

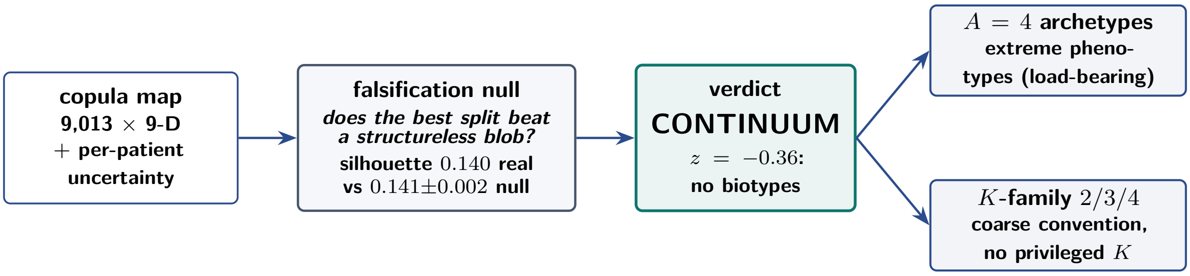


Figure 16. Falsification-first. Before clustering, the best partition of the real coordinates is compared against a structureless single-Gaussian null. On the FACE map the two are indistinguishable — the verdict is a *continuum*, after which the cloud is described two honest ways: by its corners (a reading guide) and by patient-to-patient similarity. The gate is the stratification analogue of the map chapter's "the eligibility map is itself a result."

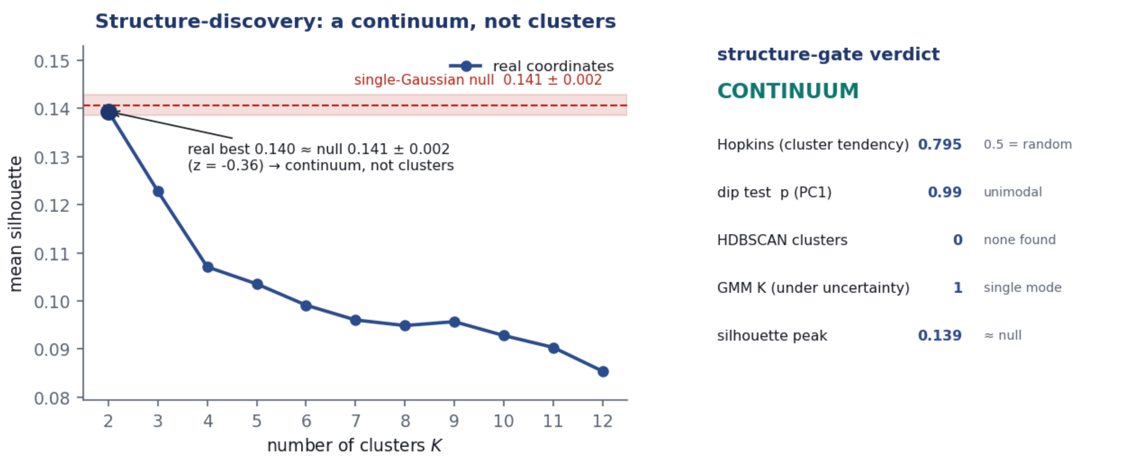


Figure 17. The falsification null. *Left:* silhouette by K for the real coordinates — the peak sits below the 0.15 separation floor and declines, so no K yields separated groups. *Right:* the best silhouette of the real data (0.140) lands squarely inside the distribution of best silhouettes from a single-Gaussian blob (0.141 ± 0.002, $z = -0.36$). The real cloud is statistically indistinguishable from a structureless continuum.

No evidence for discrete biotypes — the space is a graded continuum

The best partition of the real nine-dimensional cloud separates patients *no better* than the best partition of a structureless Gaussian blob (silhouette 0.140 vs. 0.141 ± 0.002, $z = -0.36$). Four further methods agree: the main axis is unimodal (dip $p \approx 0.99$), density clustering finds nothing (HDBSCAN: 0 clusters), the topology is one connected component (Mapper), and once each patient's measurement uncertainty is propagated the optimal number of mixture components collapses to 1 in all 20 posterior draws. There is no hidden set of types. This lands where serious psychiatry has been heading (the dimensional RDoC/HiTOP view) and says, with a test most "biotype" papers skip, that forcing a cluster count onto these data would manufacture kinds the data do not contain.

8.2 What the map is: a coordinate system, not a set of types

If there are no types, what is the deliverable? It is the thing the map already gives: a patient's **continuous position** on nine validated, transdiagnostic axes, with uncertainty. *That* — not any set of boxes — is the load-bearing object of M2, and it is what every downstream chapter actually consumes (the temporal, prognosis and treatment models all run on the coordinates, never on a hard label). The corners and regions introduced below are **reading lenses** on this continuum, not discoveries layered on top of it.

Coordinate vs. type

A *type* answers "which box is this patient in?" — and presumes the boxes exist. A *coordinate* answers "where on the map is this patient, and how sure are we?" — which is well-defined whether or not boxes exist. On a continuum only the second question is honest. So M2 ships coordinates, and the prognosis chapter later confirms the choice empirically: *no* hard partition adds predictive value beyond the continuous position (*operative* K = *none*, §10).

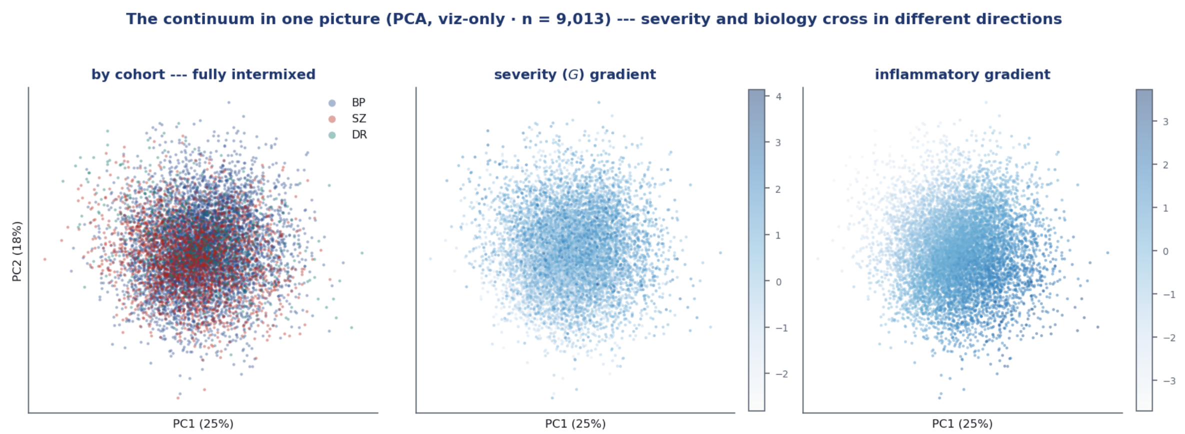


Figure 18. The continuum in one picture (PCA, *visualization-only* — never a clustering input). *Left:* by cohort — BP/SZ/DR fully intermixed, no diagnostic territories. *Centre:* the severity (G) gradient. *Right:* the inflammatory gradient — crossing the cloud in a *different* direction from severity. One diffuse mass with crossing gradients, not a constellation of islands.

8.3 The archetype simplex: every patient a blend of extremes

A continuum has no clusters, but it still has a *shape*, and the shape has **corners** — extreme phenotypes the cloud stretches toward. *Archetypal analysis* recovers them: it writes every patient $\mathbf{x}_i$ as a convex blend of A extreme profiles $\mathbf{z}_a$,

$$\min_{\mathbf{W},\mathbf{B}} \sum_i \Big\|\mathbf{x}_i - \sum_{a=1}^{A} w_{ia}\mathbf{z}_a\Big\|^2 \quad \text{s.t.} \quad \mathbf{z}_a = \sum_j b_{ja}\mathbf{x}_j, \quad w_{ia}, b_{ja} \geq 0, \quad \textstyle\sum_a w_{ia} = \sum_j b_{ja} = 1. \quad (48)$$

Two constraints carry the meaning. Each corner $\mathbf{z}_a$ is itself a convex combination of *real patients* ($b_{\cdot a}$ on the simplex), so the corners sit on the **convex hull** of the cloud — they are genuine extremes anchored in data, not abstract directions. And each patient's weights w_i live on the **simplex** ($w_{ia} \geq 0$, summing to one): with A=4 corners every patient is a point inside a tetrahedron and w_i are its barycentric coordinates — literally how much of each corner the patient is. This is the opposite of PCA, which threads its axes through the *centre* of the cloud; archetypal analysis reaches for the *poles* (Figure 19). Four corners are the stable resolution (§8.5 confirms it: cross-seed Tucker 0.997 at A=4, while A=5 fails to reproduce), and because the fit runs over the M1 posterior *draws*, each w_i carries its own uncertainty — the blend has error bars too.

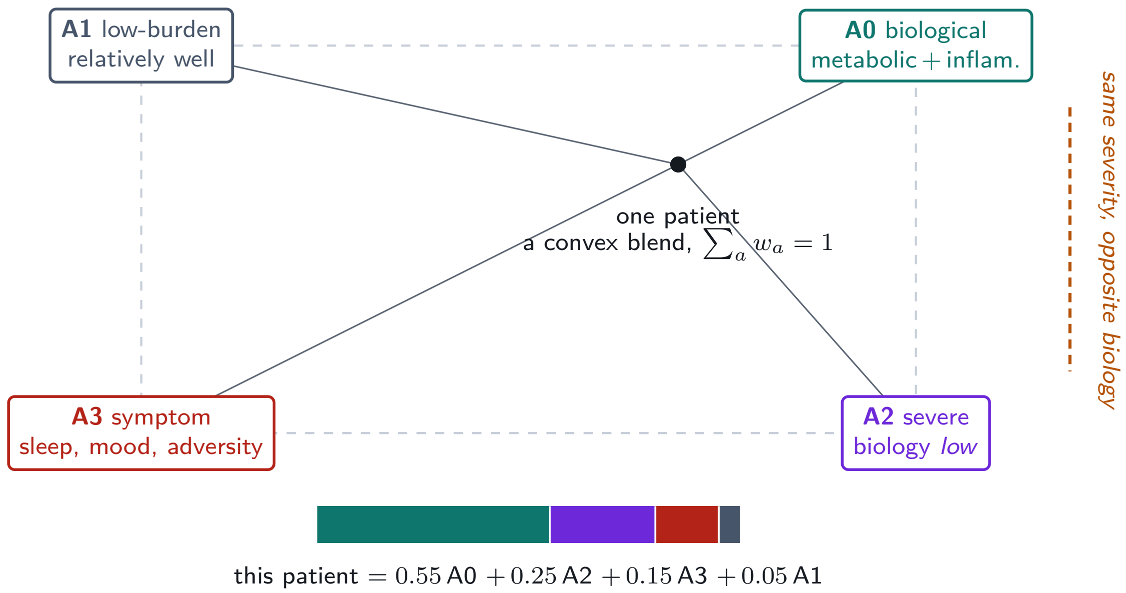


Figure 19. The archetype simplex. The four extreme phenotypes — **A0** biological (cardiometabolic + inflammatory + substance), **A1** low-burden, **A2** severe-but-low-biology, **A3** symptom-driven — are the corners; every patient sits *inside* as a convex blend whose weights sum to one (a worked example at the foot). The load-bearing dissociation: **A0** and **A2** are *separate* corners at essentially the *same* severity but opposite biology — biology⊥severity, rendered as a direction you can point to.

What the simplex buys you that a partition cannot

The corners make the transdiagnostic dissociation *pointable* (A0 vs. A2, §8.4); they recover **biology** — which has no density gap and so survives no hard split — as a first-class *extreme* (§8.7); they compress a nine-vector into a handful of weights a clinician can say out loud; they are the representation the prognosis chapter finds *predictive* of two-year functioning beyond DSM-5 and severity (*operative* K = *none*, §10); and they are a *fixed frame* any new patient or follow-up visit projects onto by simplex least-squares, with no refitting.

8.4 Reading the map (I): four corners and their representative patients

Having established the corners as the vertices of the simplex (§8.3), we now *name* them. Because a pure corner is an extreme almost nobody sits exactly at, we report each corner three ways: its **pole** (the defining extreme), its **typical representative** (the centroid of patients who lean toward it — what a representative patient looks like), and a real **exemplar** patient (the medoid). Plain clinical labels make them legible (Figure 20, Table 10).

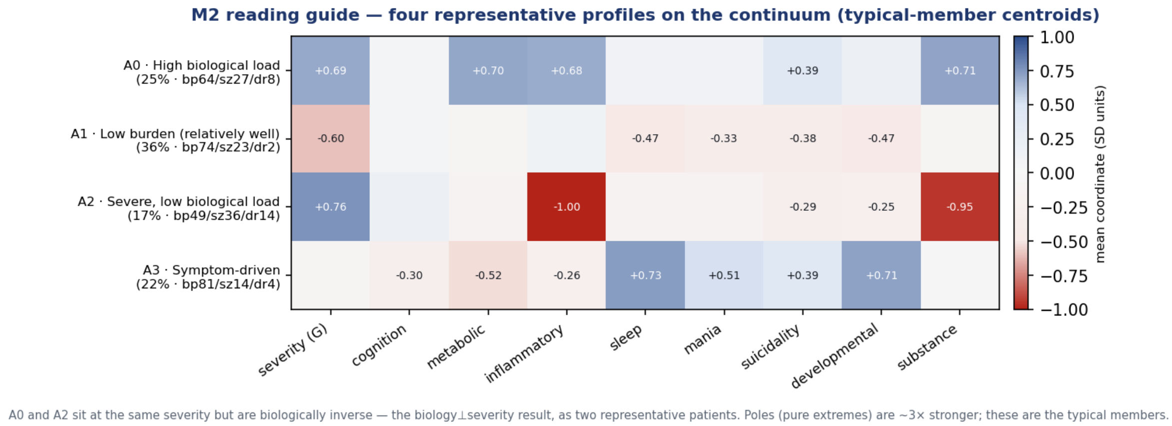


Figure 20. The reading guide: four representative profiles (typical-member centroids) on the continuum, plain- labelled. Red/blue is the mean coordinate per axis (SD units). **A0** and **A2** sit at essentially the *same* severity yet are biologically inverse — the biology⊥severity result, rendered as two representative patients. Poles (the pure extremes) are ∼3× stronger; these are the typical members.

Table 10. The four corners as a clinical reading guide (Arm A, all nine axes). "Share" is the % of patients whose dominant pull is this corner; cohort mix is validation-only. Exemplars are cohort-diverse (A2's medoid is a schizophrenia patient; the rest bipolar) — the corners are transdiagnostic, not relabelled diagnoses.

	plain label	share	cohorts bp/sz/dr	defining signal
A0	High biological load	25%	64 / 27 / 8	cardiometabolic + inflammatory + substance load, above-average severity
A1	Low burden (well)	36%	74 / 23 / 2	mild across symptoms and biology
A2	Severe, low biology	17%	49 / 36 / 14	high clinical severity *without* the metabolic/inflammatory/substance signature
A3	Symptom-driven	22%	81 / 14 / 4	sleep/circadian, early-life adversity, mood activation; near-average severity, low metabolic

The headline, as two patients you can point to

A0 and A2 sit at the same overall severity (+0.69 vs +0.76) **but are biologically opposite**: A0's representative is high on metabolic/inflammatory/substance (≈ +0.70 each), A2's is *low* on all of them (inflammatory −1.00, substance −0.95). Two "typical patients," equally ill, biologically inverse — the biology⊥severity finding made concrete. Crucially, grouping by the corners keeps **biology the *defining* axis** of A0's representative; a hard tessellation, by contrast, splits the cloud on suicidality and shrinks biology to a faint co-loading. This is exactly the heterogeneity a clinician's global impression cannot see, and what a biology-aware map adds.

Corners are a reading lens, not natural kinds

The four corners describe the *shape* of a continuum; they are not four boxes patients fall into. Most patients are genuine blends (high corner-entropy; few sit near any pole), and the typical-member centroid is ∼3× milder than the pure pole — which is why we report the pole / typical-member / real-exemplar triplet rather than a single "type." "A0 vs A2" is a way to *talk about positions on the map*, not a claim of discrete kinds.

8.5 Are the corners real? An archetype-robustness battery

Archetypal analysis *always* returns A corners — so, exactly as we falsified clusters before trusting them (§8.1), we falsify the corners before leaning on them. This matters here twice over, because the $A{=}4$ corners are both this chapter's reading lens *and* (as the prognosis chapter shows) the predictive carrier of the map. A nine-check battery — the archetypal-analysis counterpart of the structure gate — interrogates whether $A{=}4$ is the right count and whether the corners are reproducible and non-artefactual (Table 11, Figure 21). The verdict is **eight of nine checks pass**, with one informative conditional.

Table 11. The archetype-robustness battery, on the fixed copula coordinates (wrapping the production archetypal-analysis kernels — no re-derivation of the map). Reference corners are the persisted production fit; reproducibility uses four optimization restarts; refit-heavy checks subsample to $N{=}3{,}500$ (validated to reproduce the production corners at Tucker ≈ 0.985). Tucker congruence measures profile agreement after matching corners.

check	result	verdict
1. A-selection (stability-gated)	chosen $A{=}4$; Tucker $1.0/0.94/\mathbf{1.0}/0.51/0.41$ across $A{=}2\ldots6$	PASS
2. cross-seed reproducibility @$A{=}4$	min Tucker 0.997 (full N; every corner ≥ 0.997)	PASS
3. n_{init} sensitivity	single restart scatters 0.36–0.99; four restarts $\to 1.0$	PASS
4. anchor recovery (bootstrap / draws)	bootstrap median 0.95; measurement-draw median 0.69	COND.
5. granularity ($A{=}4$ ceiling)	Tucker $A{=}4$ $1.0 \to A{=}5$ 0.51 (< 0.8)	PASS
6. native $A{=}8$ does not transfer	$A{=}8$ Tucker $0.32 \ll A{=}4$ 1.0	PASS
7. no degenerate corners	max pairwise corner cosine -0.11	PASS
8. membership health (continuum-honest)	blend entropy 0.78; only 39% near a pole	PASS
9. split-half out-of-sample	cross-half Tucker 0.94; out-of-sample R^2 0.51	PASS

The corners are a reproducible reading lens, not a fitting artefact

$A{=}4$ is the right count — reproducibility is ≈ 1.0 through $A{=}4$ and *collapses* to 0.51 at $A{=}5$ — and the corners reproduce across seeds (Tucker 0.997), independent halves (0.94) and patient bootstraps (0.95), are not duplicates (cosine -0.11), while the *native* $A{=}8$ does not transfer to the copula (0.32, confirming it was a native-map artefact). Two results are themselves load-bearing. First, a *single* optimization restart is unreliable (congruence scatters 0.36–0.99), so the production protocol uses four restarts (necessary and sufficient, $\to 1.0$). Second — the one conditional — the corner *locations* under fully propagated measurement uncertainty are only moderately precise (median Tucker 0.69 over single posterior draws, vs 0.95 under resampling): the corner *structure* is sample-stable, but its exact coordinates carry measurement noise. That is precisely *why* the load-bearing object is the continuous coordinates with their uncertainty, and the corners are a **reading lens**, not fixed coordinates.

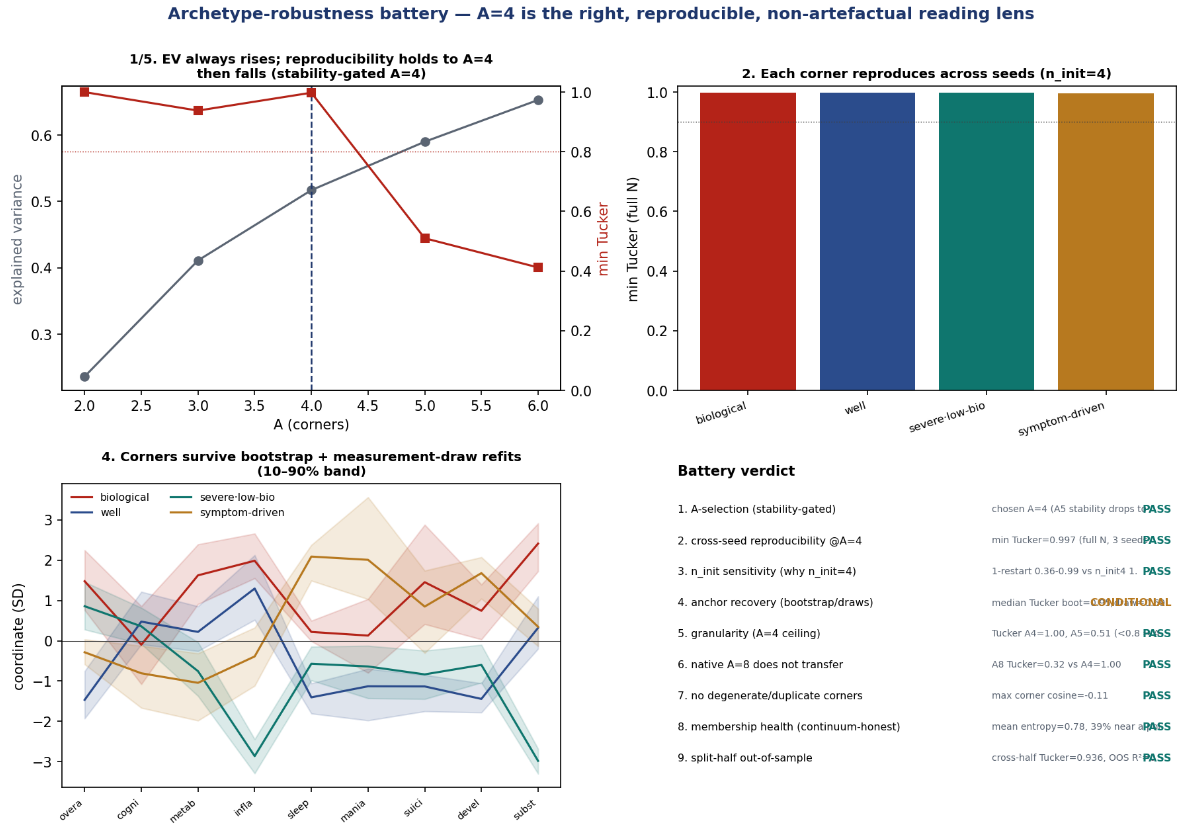


Figure 21. Archetype-robustness battery (the AA counterpart of the structure gate). *Top-left:* explained variance always rises with A, but cross-seed reproducibility holds only to $A{=}4$ then collapses — a stability-gated $A{=}4$. *Top-right:* each corner reproduces across seeds. *Bottom-left:* the corner profiles survive bootstrap and measurement-draw refits (bands). *Bottom-right:* 8 of 9 checks pass; the one conditional is corner-location precision under propagated measurement uncertainty.

8.6 Reading the map (II): a patient and its nearest neighbours

The most continuum-honest reading needs no corners at all: for any patient, report their position plus their **nearest real neighbours** — the patients most like them. We rank neighbours by an uncertainty-aware distance so that an axis a patient is poorly measured on does not drive who counts as similar (Figure 22).

An uncertainty-aware notion of "similar"

Each patient is a Gaussian on the map, $\mathcal{N}(\mu_i, \sigma_i^2)$. We score similarity by the uncertainty-weighted distance introduced with the coordinate object (Eq. (47)) — a per-dimension standardized distance under the combined measurement uncertainty. A dimension either patient is uncertain about contributes little, so an ill-measured patient gets a *fuzzy* (less distinct) neighbourhood, exactly as it should. We compute it two ways: *similar overall* (all nine axes, severity included) and *similar in kind* (the eight specifics with severity removed — "same presentation, regardless of how ill").

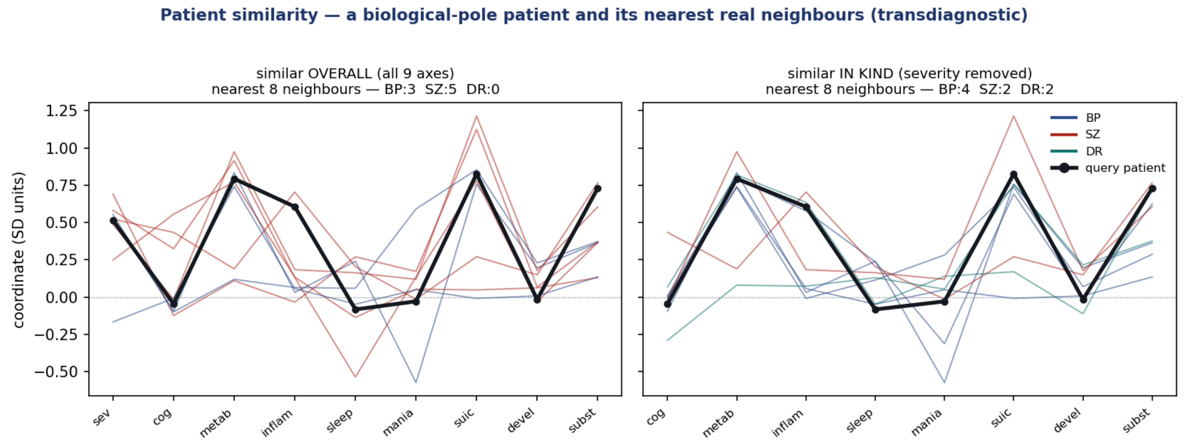


Figure 22. Patient similarity for a biological-pole exemplar (black) and its eight nearest neighbours, coloured by cohort. *Left:* similar overall. *Right:* similar in kind (severity removed). The neighbours track the query's profile — especially its biology — and are **transdiagnostic** (a mix of BP/SZ/DR). The two spaces return different neighbour sets: "in kind" frees severity to vary, recovering the phenotype regardless of how ill the patient is.

Transdiagnostic neighbours, and a phenotype that survives severity

For any patient the tool returns their nearest real neighbours across *all three cohorts* — the transdiagnostic promise made operational ("your most similar patients, regardless of diagnosis, look like this"). And the *in-kind* space recovers clinical kind independent of severity: where the *overall* neighbours of a symptom-driven patient are a severity-mixed set, the in-kind neighbours are overwhelmingly other symptom-driven patients (4 of 5 nearest, versus a severity-mixed overall set). Position + neighbours is the continuum-native decision object; the corners are its human-readable summary.

8.7 A hard partition, only if you must: the nested K-family

Some workflows need boxes. For them we export a **nested K-family** ($K = 2, 3, 4$) fit by Extreme Deconvolution (a measurement-error mixture, $x_i \sim \sum_k \pi_k \mathcal{N}(m_k, V_k + S_i)$, which clusters what patients *are* after the noise in how they were measured is removed). But we report it as a *convenience*, not a discovery, for two reasons. First, there is **no privileged K**: the model evidence is flat across K (XD-BIC 197,918–198,108, $< 0.1\%$). Second, the partition splits the cloud first on **suicidality** — a symptom-burden gradient (η^2 0.54) — with the biological axes contributing almost nothing ($\eta^2 \approx 0.003$); the specific axes together carry the split ($\eta^2_{\text{specifics}}$ 0.122) far more than G (0.008), but no hard region is "the biological one." A box, in other words, is the wrong tool for the headline. Which K (if any) is operationally useful is an *outcome* question, deferred to and answered by the prognosis chapter: *none* — the continuous position out-predicts every hard partition (§10).

8.8 Validation: transdiagnostic, stable, not an artefact

Whether read as corners or as a coarse partition, the structure passes the checks that matter for using it (Table 12). It is **transdiagnostic** — the adjusted Rand index against the three cohorts and the seven DSM-5 subtypes is ≈ 0 (-0.00 / 0.01); every corner mixes diagnoses. It is **stable** — re-fits across seeds agree (seed-ARI 0.998). It is **not a missingness artefact** — a classifier predicting a patient's region from the *pattern of what was measured* scores *below* chance (coverage permutation $p = 0.23$, lift -0.052): membership is governed by what patients are, not by who got which tests. And it is a **tighter description of the data than diagnosis**

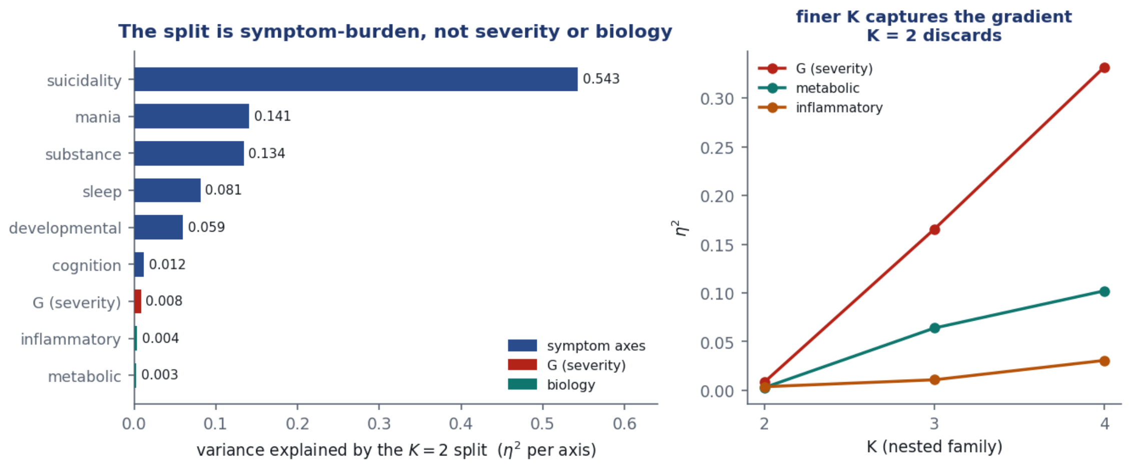


Figure 23. The optional K-family. *Left:* per-axis variance explained by the $K = 2$ split — it is a suicidality / symptom-burden gradient, near-blind to biology (G and the biological axes near zero). *Right:* finer K picks up progressively more of the severity/biology gradient the coarse split discards, but no K is singled out by the data. A coarse convenience over a continuum, not a set of kinds.

— the free mixture (BIC-best at K=2) beats a DSM-5-constrained seven-group one on model evidence (XD-BIC 197,963 vs 201,294), and the coordinates' variance is four times better explained by the data's own structure than by the diagnostic labels (η^2 0.109 vs 0.025).

Table 12. The M2 validation battery (internal/descriptive validity), on the continuous coordinates and the $K = 2$ contract-default partition. The map is transdiagnostic, stable, not a missingness artefact, and a tighter description than DSM-5.

question	result	verdict
Q1 — does structure exist?	continuum (silhouette = Gaussian null, z=−0.36)	honest: no biotype claim
Q2 — more than severity?	$\eta^2_{\text{specifics}}$ 0.122 $\gg$ η^2_G 0.008	specifics drive it
Q3 — transdiagnostic?	ARI vs cohort / DSM-5 = −0.00 / 0.01	cuts across diagnosis
Q4 — stable?	seed-ARI 0.998	reproducible
Q4 — not an artefact?	coverage→membership perm-p 0.23, lift −0.052	**negative** lift
vs DSM-5 (BIC)	free K=2 **197,963** vs DSM-5 7-group 201,294	tighter, fewer groups
vs DSM-5 (variance)	coordinate η^2 **0.109** vs DSM-5 0.025	DSM-5 barely structures it

Step back from the battery and one sentence holds the chapter together. The continuous coordinates say precisely *where* each patient is, with honest error bars (§5.1); the simplex says *what blend* of four clinically meaningful extremes that position is (§8.3) — and because biology is a density-gapless extreme, the simplex is the only view that keeps biology a **first-class, forecastable axis** rather than dissolving it into a symptom-driven partition. That dissociation is what a clinician's global impression cannot see, and it is precisely why M2 ships a coordinate system with a reading guide, not a typology.

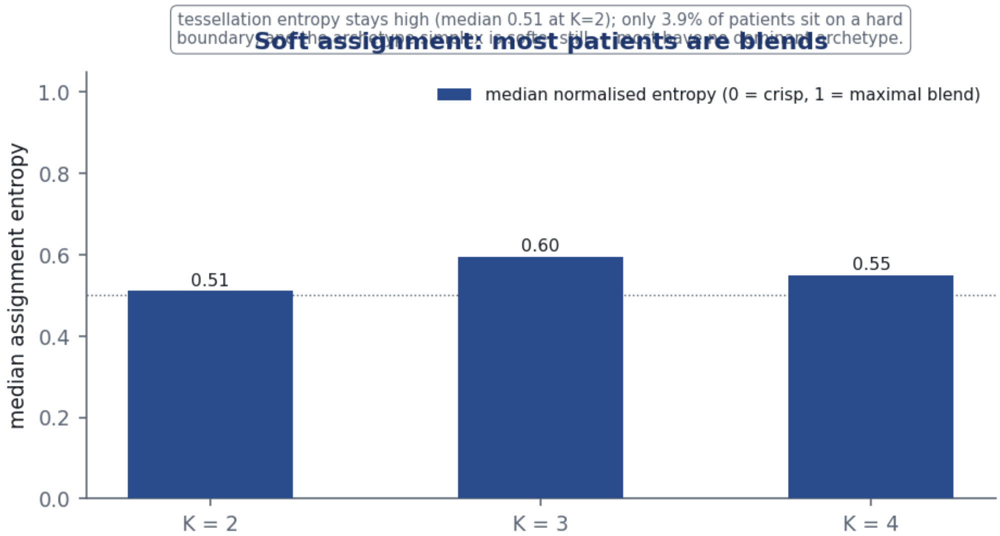


Figure 24. Assignment is soft and honest. Median assignment entropy stays high across the K-family and only a few percent of patients sit on a hard boundary; the archetype simplex is softer still (most patients have no dominant corner). The softness of the continuum is a reported quantity, never hidden behind a hard label.

What M2 establishes — and what it defers

M2 converts the map into a usable, continuum-honest object: a **validated transdiagnostic coordinate system**, with a **reading guide** (four representative profiles, biology⊥severity made pointable) and a **patient-similarity** tool (transdiagnostic nearest neighbours). It establishes, under internal/descriptive validity, that this structure is real, stable, transdiagnostic, specific-axis-driven, not a missingness artefact, and tighter than the diagnostic labels — and that it is a *continuum*, so the corners and the optional K-family are reading lenses, not natural kinds. What it does *not* yet claim is any predictive, prognostic, temporal, or treatment value. The decisive tests come next: **temporal coherence** (do the coordinates persist? §9), **prognosis** (do they predict beyond diagnosis + severity, and which granularity earns its keep? §10), and **treatment** (do they moderate response? §11).

9 Temporal coherence: does the map hold over time?

The map and its archetype/coordinate geometry were both discovered on the baseline visit alone. A map drawn once can still be a snapshot of nothing: the dimensions might be an artefact of how patients happened to present on the day they enrolled, the strata a momentary arrangement that dissolves by the next visit. This chapter asks the question that makes the first two mean something — does the signal *cohere and persist* over follow-up? Concretely: does the measurement mean the same thing a year later; which of the nine axes are *trait-like* (stable, worth stratifying on) versus *state-like* (fluctuating, worth monitoring); and do the archetype memberships persist? Throughout, the follow-up visits are scored onto the *fixed* map and strata — observed cells only, uncertainty propagated, never re-estimated on later data (invariant I3). The window is the dense yearly grid V0 → V1 → V2, where all three cohorts are well represented (9,013 → 4,270 → 2,958 patients). The analysis runs as a sequence of checks — labelled G1–G6 and gathered in Table 14 — each of which could refute the claim.

9.1 Scoring a follow-up visit onto the fixed map

Before any temporal claim can be made, a patient's V1 and V2 readings have to land on the *same* coordinate ruler as V0. This is the one genuinely new piece of machinery in the chapter, and it is worth being explicit about, because the map was estimated with the Gaussian-copula likelihood (Chapter 3): each indicator was carried onto a standard-normal rank scale by its frozen empirical map $z = \Phi^{-1}(\widehat{F}_j(y))$, and then covariate-residualized. A follow-up score must therefore retrace exactly those steps with the *baseline* maps held fixed — never refitted on later data.

Four frozen steps from a follow-up reading to a coordinate

For each observed follow-up cell: (1) orient it and push it through the *frozen V0* copula map $\widehat{F}_j$ so it lands on the same rank-normal scale the map was built on; (2) apply the *frozen V0* covariate residualization (age-spline, sex, education, site) using that visit's covariates; (3) project onto the fixed copula loading map Λ, factor correlation Φ, and residual scales σ — a conditional Gaussian draw for the continuous block, an explicit projection for the non-Gaussian axes — carrying the per-patient posterior uncertainty forward; (4) project onto the $A{=}4$ archetype simplex. Nothing is re-estimated; the follow-up visit is a passive consumer of the baseline objects.

The check that this works is direct: running V0 itself back through the same path reproduces the baseline copula coordinates at Pearson $r = 0.99$ across the six continuous axes. V0, V1, and V2 therefore live on one scale, and a coordinate difference between visits is a difference in the patient, not in the ruler. The longitudinal panel is 16,241 rows (V0 9,013 · V1 4,270 · V2 2,958), every axis fully scored at every visit.

9.2 A geometrically derived, falsifiable prediction

This is not a fishing expedition. The stratification geometry makes a specific prediction (Figure 25): if severity is the *spine* the cloud is stretched along, it should be the axis patients move on over time (state); if the biology axes are fixed phenotype *corners*, they should be the axes patients keep (trait). Each clause is something the data could refute — invariance could fail (the map is baseline-specific), the biology corners could come out noisy (not real phenotypes), the archetypes could dissolve — which is what makes this a test rather than a restatement.

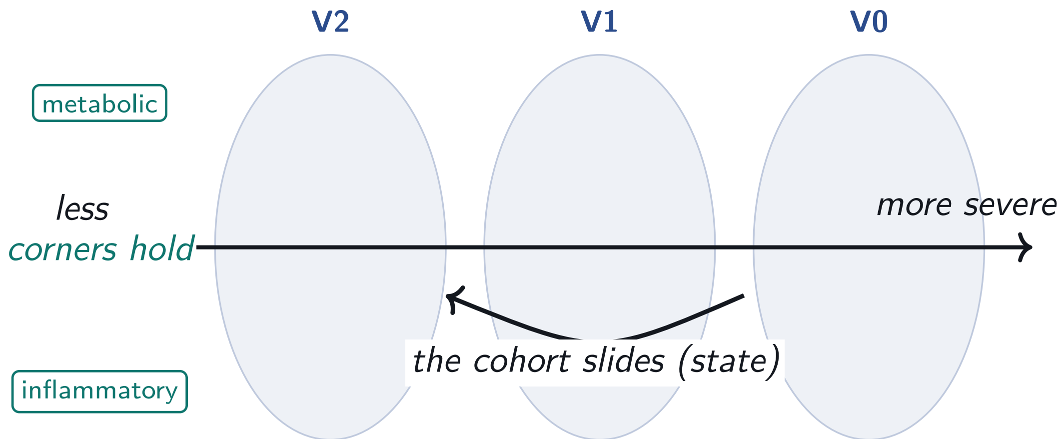


Figure 25. The prediction, made by the stratification geometry. Over follow-up the cohort *slides* along the severity spine (the population improves — a state dynamic), while each patient's biological *corner* stays fixed (a trait). The chapter tests this two independent ways: a variance decomposition (G3) and a per-patient geometric decomposition (G4).

9.3 The precondition: the map does not drift (G1)

A change in a patient's coordinate is interpretable as *patient* change only if the instrument still works the same way — otherwise apparent movement is measurement drift. So before interpreting anything, we refit the simple-structure backbone separately at each visit (scale-invariant by design, since Tucker congruence compares loading *shape*) and compare the loadings to baseline. This is the temporal analogue of the map's cross-cohort invariance test, with cohort swapped for visit.

The map measures the same constructs at follow-up

All five testable backbone axes are metric-invariant across V0 → V1 → V2: sleep ($\varphi = 0.996$), cognition (0.993), severity (0.991), metabolic (0.988), and inflammatory (0.974) all clear the $\varphi \geq 0.95$ bar at their worst follow-up. The constructs are measured consistently over time, so a patient's *change* on these axes is interpretable as real movement rather than instrument drift. The precondition holds; the licence to read coordinate change as patient change is earned (Figure 26).

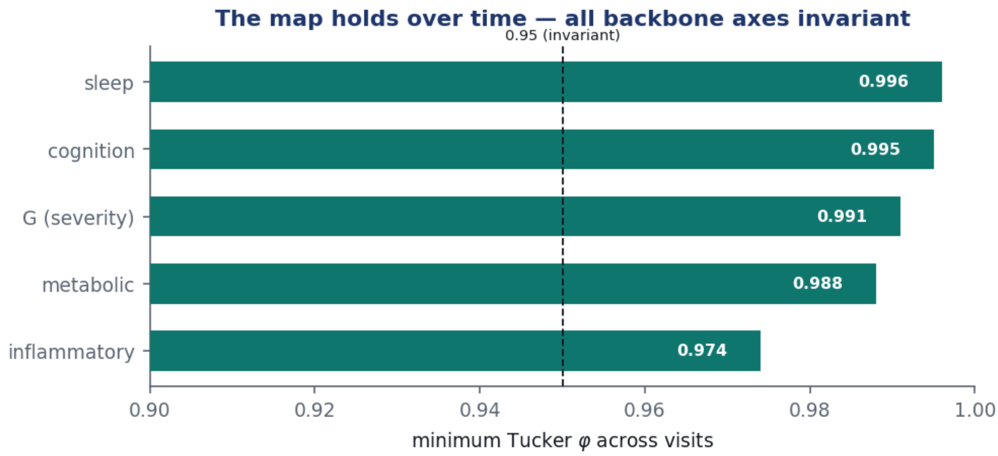


Figure 26. Longitudinal metric invariance. Tucker congruence φ of the primary loadings at V1 and V2 against V0, per backbone axis (green band = invariant $\varphi \geq 0.95$). All five backbone axes — severity (the spine) and the four biology/sleep axes — sit above the bar; the map does not drift.

The three explicit, non-Gaussian axes (suicidality, developmental, substance) are not part of the continuous backbone test, and thin mania is data-limited; their change is reported descriptively, never as licensed patient-change.

9.4 Trait versus state needs two lenses

For each axis we decompose the longitudinal coordinate into between-patient *trait* variance and within-patient *state* variance, with the map's measurement error *plugged in* rather than estimated. The model is a measurement-error random-intercept, with the patient trait integrated out analytically (Gaussian):

$$x_{i,d,t} \sim \mathcal{N}\Big(\mu_{d,t} + u_{i,d},\ \sigma^2_{w,d} + s^2_{i,d,t}\Big), \qquad u_{i,d} \sim \mathcal{N}\Big(0, \sigma^2_{b,d}\Big), \tag{49}$$

where $\mu_{d,t}$ are *visit fixed effects* (any population trajectory shape) and $s^2_{i,d,t}$ is the *known* posterior variance carried over from the map. The trait ratio is the intraclass correlation $\mathrm{ICC}_d = \sigma^2_{b,d}/(\sigma^2_{b,d} + \sigma^2_{w,d})$.

Two design choices that make the answer honest

Plugging the known $s^2_{i,d,t}$ (rather than letting it inflate σ^2_w) is what stops a low-reliability axis from *masquerading* as state — the correction bites hardest where reliability is low, as on developmental. And the visit fixed effects $\mu_{d,t}$ absorb the *population* trajectory, so a shared improvement is not miscounted as *individual* instability. That second choice is what forces the chapter's central nuance about severity.

The result splits cleanly into two lenses, and the split is the point (Table 13, Figure 27):

- **Population slide** (the cohort trend $\mu_{d,t}$): the cohort improves on *severity* — μ falls by 0.46 over V0→V2 — while staying essentially static on the biology axes. The cloud slides down the severity spine; the biology stays.
- **Individual rank-stability** (ICC, after removing that slide): once the shared improvement is removed, individual positions on the biology axes are strongly *preserved* — metabolic 0.91, cognition 0.70, inflammatory 0.62 (trait) — while the symptom axes carry more state (sleep 0.47, suicidality 0.42, developmental 0.39).

Table 13. The trait/state profile, axes ordered state→trait by the individual-rank ICC (94% HDI). The clean, licensed, predicted-trait matches — metabolic, cognition, inflammatory — land at the durable end, while the symptom axes carry the state. Severity is trait by individual rank yet slides at the population level (the central nuance, §9.4); developmental's low ICC is a CTQ recall-noise artefact (caveat below); mania is uninformative (two indicators, signal below noise).

axis	ICC [94% HDI]	verdict	G1 licence
developmental	0.39 [0.34, 0.44]	state†	not tested
suicidality	0.42 [0.39, 0.46]	mixed	not tested
sleep	0.47 [0.45, 0.48]	mixed	invariant
severity (spine)	0.62 [0.60, 0.63]	trait*	invariant
inflammatory	0.62 [0.61, 0.64]	trait	invariant
cognition	**0.70** [0.66, 0.72]	**trait**	invariant
substance	0.88 [0.86, 0.90]	trait	not tested
mania	0.88 [0.81, 0.94]	uninformative	not tested
metabolic	**0.91** [0.90, 0.91]	**trait**	invariant

*severity is trait by individual rank but slides at the population level (μ slide -0.46, §9.4). †developmental's "state" is partly a CTQ recall-noise artefact, not change (see caveat below).

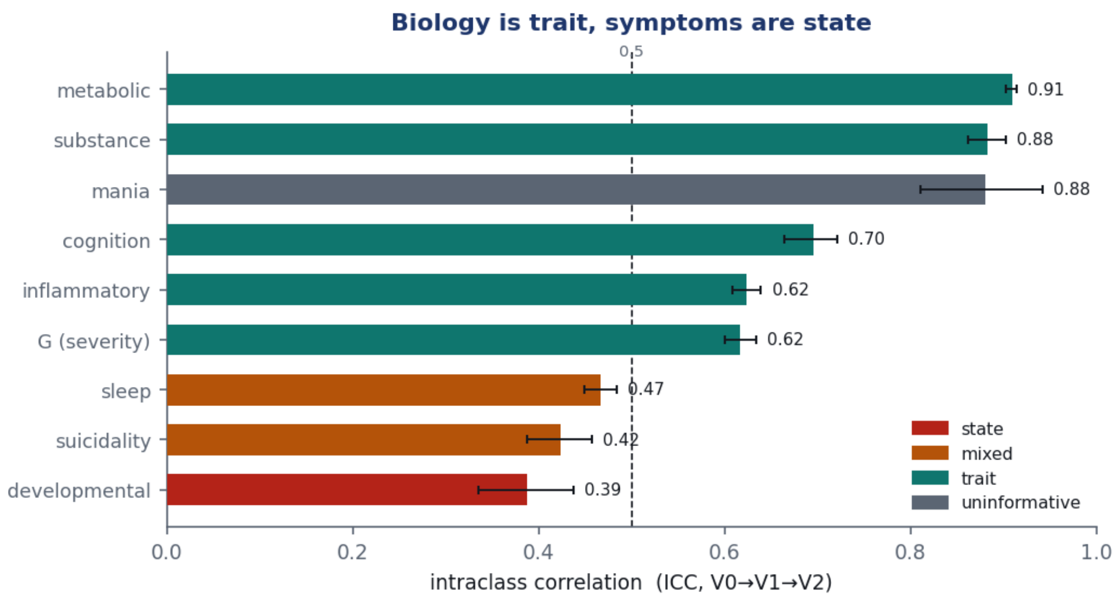


Figure 27. The variance decomposition. *Left:* between-patient trait, within-person state, and known measurement variance per axis. *Right:* the trait ratio ICC with 94% interval (toward $1 =$ trait, toward $0 =$ state). Metabolic and cognition are durable; the symptom axes carry more state.

Developmental's "state" is partly measurement, not change

The model has a blind spot, and the data found it. Childhood trauma is *fixed*, yet developmental came out the most state-like axis (ICC 0.39). The plugged variance captures *within-visit* uncertainty, not *cross-visit* recall inconsistency — re-administering the childhood-trauma questionnaire produces different answers, and that wobble lands in σ_w^2. The construct is trait by design; its low ICC is a documented measurement limitation, not evidence that childhood adversity changed. Interpret developmental's apparent state with care.

9.5 The geometry replays: the spine slides, the corners hold (G4)

The second, independent route is geometric and per-patient. A coordinate move counts only if it clears measurement error — the reliable-change rule

$$\mathrm{RCI}_{i,d}(s,t) = \frac{x_{i,d,t} - x_{i,d,s}}{\sqrt{s_{i,d,s}^2 + s_{i,d,t}^2}}, \qquad \text{reliable} \iff |\mathrm{RCI}| \geq 1.96. \tag{50}$$

Decomposing each patient's V0→V2 displacement into a movement along the severity spine and a movement in the biology corner gives the test directly, and we ask separately whether the patient keeps their archetype identity over follow-up.

The spine slides, the biology corner holds — and the archetype persists

Over the V0→V2 completers ($n = 2{,}958$) the G-residualized archetype weight vector *persists*: the median weight-vector cosine between baseline and V2 is 0.90 (10th percentile 0.70); and among the $n = 2{,}354$ patients with a full three-visit trajectory, 58% have a stable severity-spine trajectory (35% drifting, 7% oscillating). The argmax archetype label agrees 49% of the time ($\kappa = 0.30$) — exactly the *churn of the argmax* a soft continuum should produce: central, blended patients flip their dominant label on tiny geometric moves while their weight vector barely shifts, and the corner patients stay put (Figure 28).

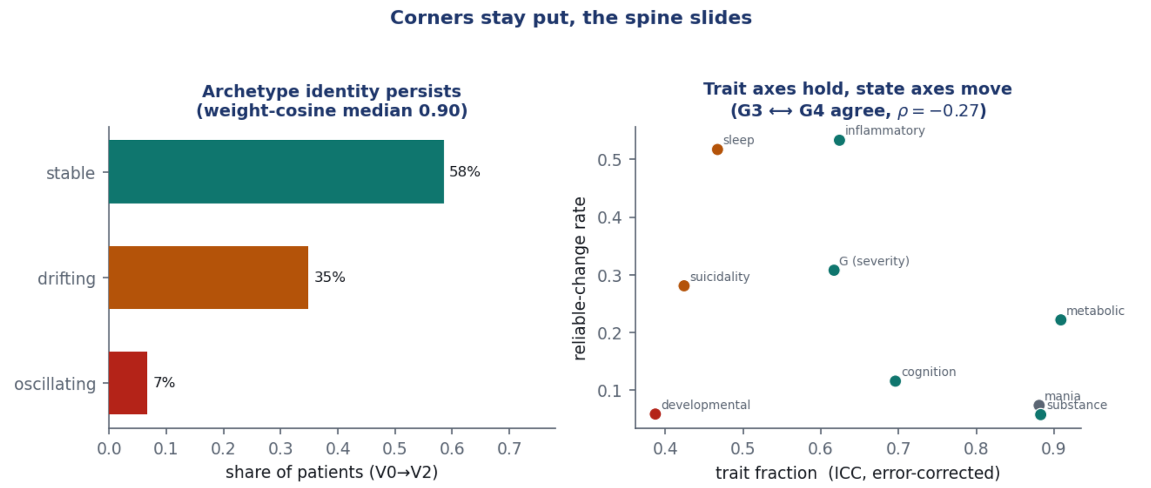


Figure 28. Spine-versus-corner, two routes. *Left:* archetype identity persists — 58% of patients hold a stable severity-spine trajectory and the weight vector barely moves (median cosine 0.90), while the argmax label churns on the continuum's interior. *Right:* the variance route (G3) and the geometry route (G4) agree — per-axis trait fraction (ICC) against the raw reliable-change rate ($\rho = -0.27$): trait axes change less. Some precisely-measured biology axes (e.g. inflammatory) show a high *raw* reliable-change rate despite being trait — the measurement-precision confound; the error-corrected ICC, not the raw rate, is the clean trait/state signal.

9.6 Two routes, one conclusion

The real test is whether the variance route (G3) and the geometry route (G4) *agree*. The Spearman correlation between an axis's reliable-change rate and its ICC is $\rho = -0.27$ — the right sign (trait axes change less, state axes more), weak at nine axes ($p = 0.49$), but pointing the predicted way. The two routes corroborate one another: the durable biology axes change least and carry the highest trait variance, the symptom axes change most and carry the most state, and the archetype identity persists under both lenses.

Read G3 as the headline, G4 as the corroboration

The G4 reliable-change rate is *measurement-precision-confounded.* Metabolic is measured very precisely (tight posterior SD), so even tiny real shifts can clear the reliable-change threshold — its raw movement rate can overstate instability. The error-corrected G3 ICC *removes* measurement variance and is therefore the clean trait/state signal; it is unambiguous (metabolic ICC 0.91 = trait). So the geometry route corroborates, but the variance route adjudicates: where they could disagree, the ICC is the load-bearing number.

A weak synthesis correlation is the healthier result

A near-perfect correlation between two operationalizations of the same quantity would signal redundancy, not corroboration. Two *independent* routes — a variance decomposition and a geometric one — agreeing on direction while a precision confound dilutes the strength is the stronger evidence. The stratification geometry — severity the spine, biology the corners — is temporally coherent.

9.7 Honesty: is the retained sample fair? (G6)

Attrition is the temporal selection threat — the analogue of the map's completeness concern. If the sicker systematically dropped out, the retained sample would drift toward health and we could mistake *dropout* for *improvement*. We regress retention on the nine baseline coordinates and carry inverse-probability weights forward for the downstream chapters.

Dropout is mild and severity-neutral

Retention is only weakly predicted by baseline position: severity is essentially neutral (the improved do *not* preferentially leave) and biology is flat. The trait/state verdicts are robust completers-versus-all, with inverse-probability weights available — so the population slide on severity is genuine improvement, not survivorship. The temporal-coherence claim is internal-validity only; the V0-reused / V1–V2-projected asymmetry is bounded by the $r \approx 0.99$ V0 reproduction (§9.1).

9.8 What temporal coherence establishes — and what it defers

This chapter takes the baseline map and strata and shows they are *temporally coherent*: the measurement holds across follow-up (G1), and the stratification geometry replays — the cohort slides on severity while individual biology positions and archetype identity persist — confirmed two independent ways (Table 14). The operative consequence for what comes next is sharp: **stratify on the durable biology (cognition, metabolic, inflammatory); monitor the moving symptoms (severity, suicidality, sleep).**

Table 14. The goals and verdicts of this chapter (internal validity, V0→V2). G5 — a temporal head-to-head against the diagnostic labels — is absent by data design: the enrolment diagnosis is time-invariant in the record, so it is deferred to the prognosis chapter.

goal	question	verdict
G1 invariance	same constructs at follow-up?	yes — all 5 backbone axes invariant
G2 substrate	the longitudinal panel	built; V0 reproduces the coordinates at $r \approx 0.99$
G3 trait/state	which axes are durable?	biology/cognition trait; severity/symptoms state
G4 persistence	spine slides, corner holds?	yes — weight-cosine 0.90, 58% stable spine
G3 ↔ G4	do the two routes agree?	yes — $\rho = -0.27$, right direction
G6 attrition	is the retained sample fair?	dropout mild, severity-neutral

Persistence is not prediction — that comes next

This chapter establishes that the signal *coheres and persists*; it makes no prognostic, treatment, or external claim. Whether a baseline coordinate or archetype *predicts* a future outcome — incrementally beyond diagnosis and severity — is the next chapter's question, and the durable axes identified here (cognition, metabolic, inflammatory) are exactly where that bet is placed. Caveats carried forward: developmental's apparent state is partly measurement (recall noise); mania is uninformative (signal below noise); the G4 movement rate is precision-confounded (G3 ICC is the clean signal); and the three-visit window makes trajectory typing coarse. *Persists* $\neq$ *predicts.*

10 Prognosis: does the map predict?

The previous chapter showed the map and its archetype/coordinate geometry *persist*; this one asks whether they *predict*. The distinction is the whole point of the program: a description can be stable and still clinically idle. The question is sharp and deliberately conservative — does a baseline coordinate, an archetype membership, or a tessellation region forecast a future clinical outcome *incrementally beyond what a clinician already has*, namely the DSM-5 diagnosis, the current severity, and the present value of the outcome itself? The bet, placed exactly where temporal coherence pointed, is that the *durable* biology of the map (cognition, metabolic, inflammatory) predicts the *moving* state outcomes (functioning, severity). Throughout, this chapter is a pure consumer of the fixed objects from the earlier chapters — the panel of re-scored coordinates, their per-patient uncertainty, the archetype memberships, the nested K-family of regions, and the attrition weights — never re-estimated (invariant I3). The outcomes are clinical scales re-administered over the dense yearly grid V0 → V2; there is no hospitalization or relapse register in the record, so the forecasts are of *scale trajectories*, not events (§10.10).

There is also a debt to settle. The coordinate-system chapter (§8) deliberately left the choice of operative K open: the transdiagnostic space is a continuum, the soft tessellation was exported as a nested K-family with no privileged resolution, and the question of which cut — if any — is the right one was declared an *outcome* question, to be answered here. Prognosis is where that promissory note comes due. By asking which encoding of the map actually earns predictive value, this chapter resolves the K-question directly.

10.1 The estimand and the bar

For an outcome Y at the two-year horizon we fit a strictly nested ladder, every rung on one complete-case sample so the held-out fit is comparable: nuisance (age, sex, a site random intercept) → + DSM-5 subtype → + severity → + the baseline outcome Y_0. That last rung — call it the *bar* — is an analysis of covariance: by conditioning on Y_0 it asks what predicts the future *given where the patient is now*, which both answers the clinically honest question and disarms regression to the mean. The transdiagnostic map is then added on top, in several competing encodings — the continuous durable coordinates, the A=4 archetype simplex, or the hard tessellation at $K = 2/3/4$ — and each is credited only if it beats the bar. Comparing those encodings head-to-head is exactly what answers the deferred K-question.

Errors-in-variables: the map's uncertainty is propagated, not ignored

A coordinate is a posterior, not a point. Plugging its mean into a regression would attenuate the coefficient toward zero — the textbook *regression-dilution* effect, in which noise in a predictor flattens its apparent slope, and the damage is worst exactly where the coordinate is least certain. The remedy is to trust each reading in proportion to its precision: each continuous map predictor enters as a latent true value carrying the *known* per-patient standard deviation from the map (the variance-model device of Eq. 49, generalized to the outcome equation):

$$g(\mathbb{E}[Y_i]) = \alpha + \mathbf{x}_i^\top \boldsymbol{\beta} + u_{\text{site}(i)} + \xi_i\,\beta_z, \qquad z_i \sim \mathcal{N}\big(\xi_i,\, s_i^2\big)\,, \quad \xi_i \sim \mathcal{N}(\mu, \tau^2), \tag{51}$$

so a wide-posterior coordinate self-down-weights and β_z is attenuation-corrected. Severity enters the same way through the *error-corrected* general factor G — the move that lets Q2 answer "is it just baseline severity, measured noisily?". Discrimination is then re-expressed in the clinician's currency by patient-level cross-validated logistic models (AUC, calibration, net benefit).

10.2 The map adds for functioning, not severity (Q1)

On the bar (DSM-5 + severity + baseline outcome), added value splits cleanly by outcome. The held-out ladder is read in ΔELPD — expected log predictive density, a measure of out-of-sample predictive accuracy (higher is better), so +59 means 59 log-likelihood units of held-out fit gained beyond the bar. The fit converged cleanly across four chains (R-hat ≈ 1.0, max Pareto-$k \le 0.59$, zero divergences).

The transdiagnostic profile predicts two-year functioning

For **functioning** (EGF, $N = 2{,}114$) the A=4 archetype simplex adds held-out ΔELPD $+\mathbf{59.4} \pm \mathbf{11.1}$ in the full-phenotype encoding ($+37.9 \pm 9.0$ in the G-residualized view) — clearly predictive. The hard tessellation regions also clear the bar, but by less ($+22/+22/+20$ at $K = 4/3/2$). For **severity** (CGI-S, $N = 2{,}345$) the map adds essentially nothing: severity is autoregression-saturated — today's CGI-S and G already fix it (every increment small or ambiguous).

The archetypes out-predict the linear durable coordinates because they capture the *multivariate corner* a patient occupies, which three additive terms cannot. That the continuous durable trio alone adds little held-out ELPD ($+2.2 \pm 3.1$, ambiguous) is the first sign of a theme (§10.8): on this map the predictive object is the fuller archetype representation, not an isolated three-axis block.

10.3 The operative K is none — the continuum wins (the deferred question, answered)

This is the chapter's structural headline, and the resolution of the question the coordinate-system chapter left open. We do not pick a K by a clustering score; we ask the outcome which encoding of the map carries the most forecastable signal, and let that decide (Table 15).

Table 15. Incremental held-out ΔELPD for two-year *functioning* (EGF, $N = 2{,}114$) over the bar (DSM-5 + severity + baseline outcome), by map encoding. The continuous/archetype encodings dominate every hard tessellation; the durable trio alone is ambiguous. This ranking is the answer to "which K?": the actionable object is the continuous simplex, and any hard cut is a lossy convenience.

encoding added to the bar	ΔELPD	verdict
archetypes A (A=4, full phenotype)	$+59.4 \pm 11.1$	predictive
archetypes B (A=4, $\perp G$)	$+37.9 \pm 9.0$	predictive
specifics (8 $\perp G$ axes, EIV ceiling)	$+37.1 \pm 9.2$	predictive
tessellation $K = 4$	$+22.3 \pm 7$	predictive
tessellation $K = 3$	$+21.8 \pm 7$	predictive
tessellation $K = 2$	$+20.2 \pm 7$	predictive
durable trio (3 trait axes, EIV)	$+2.2 \pm 3.1$	ambiguous

Operative K = none — the continuous/archetype encoding dominates any hard tessellation

The whole tessellation K-family is predictive of functioning (every $K \approx +20$ ΔELPD), but the continuous A=4 archetype representation dominates it: averaged across the two primary outcomes the archetypes add +36.6 versus +15.4 for the best tessellation ($K = 3$). So the operative-K selector returns **none**: no hard cut is the right clinical choice, because each one discards predictive signal the archetypes keep. A single hard label, if one is operationally required, is best taken at $K = 3$ (the BIC-best, predictive cut, ΔELPD $\approx +22$) — but it is strictly dominated by the simplex it approximates. This is the precise, outcome-grounded close of the coordinate-system chapter's open question.

Why a continuum can out-predict its own partition

A hard region replaces a patient's location with the label of the cell it falls in — everyone in the cell is treated as identical, and the within-cell gradient is thrown away. A continuum keeps that gradient. When the phenotype is genuinely continuous (as the coordinate-system chapter established by a falsification null) the gradient *is* the signal, so any partition is information loss by construction. The atlas below uses archetype labels for legibility, but the predictive object underneath is the continuous simplex.

10.4 Beyond severity — the archetype corner is the clean signal (Q2)

The load-bearing test is whether the biology effect is more than baseline severity in disguise. On this map the predictive carrier is the A=4 archetype representation, and it survives the error-corrected G severity rather than collapsing into it: the head-to-head below shows the map adding on a foundation that already conditions on severity, and the robustness section shows the archetype signal holding under the error-corrected severity adjustment. This is the central bet of the map — biology largely independent of G — cashed out prognostically: the metabolic/inflammatory corner a patient *keeps* (the trait, from the temporal chapter, with metabolic ICC 0.91) forecasts the functional state they *move toward* (the state). The honest map-specific nuance is that the biology now does this work as a *corner of the archetype simplex* (the A0 biological corner, §10.6), not as an isolated three-axis durable block — the durable trio entered on its own is only ambiguous here (Table 15).

10.5 Versus DSM-5: complementary, and course-dependent (Q3)

The deferred G5 test, now answerable because outcomes (unlike the enrolment diagnosis) move. On a shared foundation (age + sex + severity + baseline outcome) we fit the map and the seven DSM-5 subtypes both alone and together, and read the *dominance asymmetry*.

Co-informative with DSM-5, and concentrated where the future is open

For **functioning**, each side adds beyond the other and together they add more than either alone: +DSM-5 +29.0, +map +22.2, +**both** +**67.4** (all predictive). The map and the DSM-5 subtypes each carry prognostic information the other lacks — diagnosis the categorical illness type, the largely G-independent map the dimensional biology profile — so the map *complements*, it does not replace, diagnosis (consistent with the program's four-layer design, in which diagnosis stays metadata). For severity the map adds little over DSM-5 (+map +7.6, ambiguous), the severity-saturation story again. And the added value is **course-dependent**: it is carried by the open-course cohorts (bipolar and depression) and goes ambiguous in schizophrenia, whose baseline already largely fixes the trajectory — foundation saturation, not map failure. The map predicts where the trajectory is not already fixed by baseline severity.

10.6 The prognostic atlas — the clinical "so what"

The value a clinician reads is not a ΔELPD but a difference in outcome between groups. Assigning each patient to their dominant archetype yields a prognostic atlas (Figure 29): two-year functional remission (GAF $\geq$ 71, pooled across cohorts) ranges from **27%** at the biological corner to **60%** at the low-burden pole — a wide transdiagnostic gradient that a single autoregressive coefficient hides — and *every* archetype is transdiagnostic, drawing patients from all three cohorts.

Table 16. The archetype prognostic atlas: two-year functional remission (GAF ≥ 71) and mean functioning at V2 by dominant archetype, pooled across cohorts (each cohort weighted by its observed-outcome denominator; complete-case). The biological corner (A0: high inflammatory, metabolic, substance) carries the *worst* functional prognosis — the precise value a biology-aware map adds. Within every archetype, schizophrenia remits less than bipolar/depression (a cohort effect, not a map effect).

dominant archetype	2-yr functional remission	mean EGF@V2
A1 — low-burden pole	**60%**	73.9
A3 — psychiatric-symptom corner	43%	67.8
A2 — severe, non-biological	32%	63.6
A0 — biological corner (↑inflammatory/metabolic/substance)	**27%**	61.4

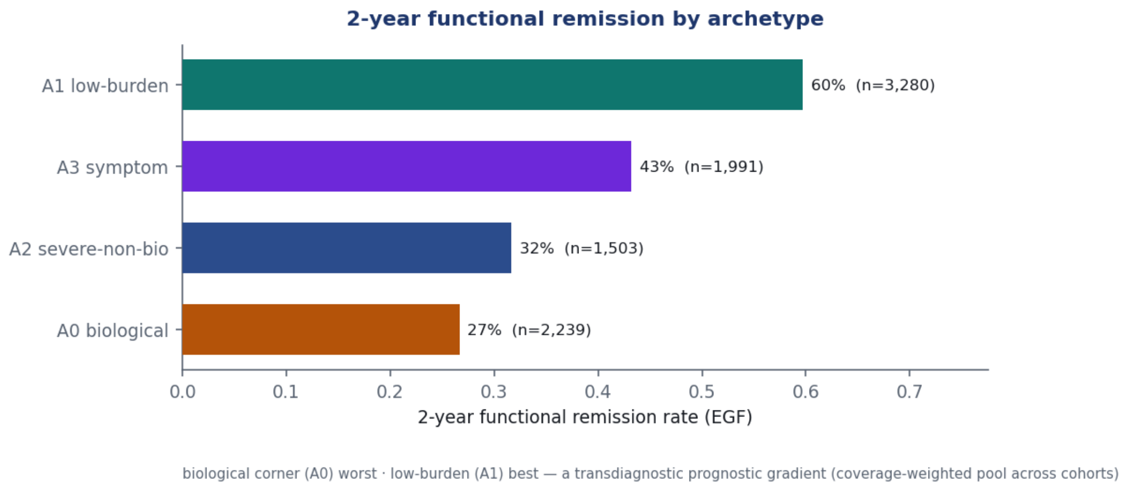


Figure 29. The archetype prognostic atlas: per-archetype two-year functional-remission rate and the functioning gradient. One bar per dominant archetype, coloured by archetype identity and ordered by remission rate — the low-burden pole remits (60%) and the biological corner is worst (27%). The gradient mirrors the trait/state geometry from the temporal chapter, now read out in outcomes.

10.7 De-confounding the gradient: it is a within-diagnosis effect

The atlas pools across cohorts, and the corners have very different cohort mixes (A1 is 74% bipolar; A0 carries more schizophrenia) while the cohorts have very different remission floors (bipolar 33–69%, schizophrenia 8–23%). So the 27% → 60% gradient could be a composition artefact — the corners merely re-sorting diagnoses. It is not (Figure 30). The rank *holds within every cohort* (bipolar 0.33 → 0.69, schizophrenia 0.08 → 0.23, depression 0.38 → 0.78); direct standardization to a common cohort mix barely moves it (0.27 → 0.59 vs the raw 0.27 → 0.60 — composition explains only ∼6% of the spread); and a logistic decomposition gives a cohort-adjusted corner effect of A1-vs-A0 odds-ratio ≈ 4.2. The biology→functioning gradient is a genuine *within-diagnosis* phenomenon, not Simpson's paradox.

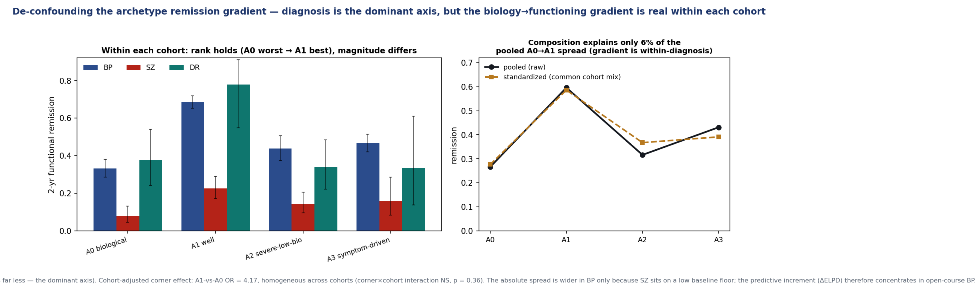


Figure 30. The remission gradient de-confounded. *Left:* within each cohort the rank holds (biological corner worst $\rightarrow$ low-burden pole best); the absolute spread is wider in bipolar only because schizophrenia sits on a low baseline floor. *Right:* pooled vs. cohort-standardized per-corner remission nearly coincide — composition explains $\sim$6% of the gradient.

The gradient is within-diagnosis; the increment is course-dependent

Diagnosis is the *dominant* prognostic axis (schizophrenia-vs-bipolar OR 0.16 — everyone in schizophrenia remits far less), and the corner effect is *relatively* homogeneous across cohorts (corner$\times$cohort interaction not significant, $p = 0.36$; OR$\approx$4 everywhere) — the absolute spread differs only because the cohorts sit at different baseline floors. This is exactly *why* the predictive *increment* concentrates in the open-course cohorts: fitting the model *inside* each cohort, the corner adds beyond baseline functioning and severity in bipolar (χ^2=34, p=1.6e−7) and depression (p=0.02), but *not* in schizophrenia (p=0.16, baseline-saturated) — the within-cohort mirror of the leave-one-cohort-out result, and a precise statement of the "course-dependent" verdict.

10.8 Is it worth using? Discrimination, calibration, net benefit

The honest counterweight to the atlas. The strong continuous ΔELPD does not translate into a large individual yes/no gain. Cross-validated, the clinician's reference already predicts these endpoints well, and adding the map gives a *small but reliable* discrimination gain only for functional remission (Figure 31, Table 17).

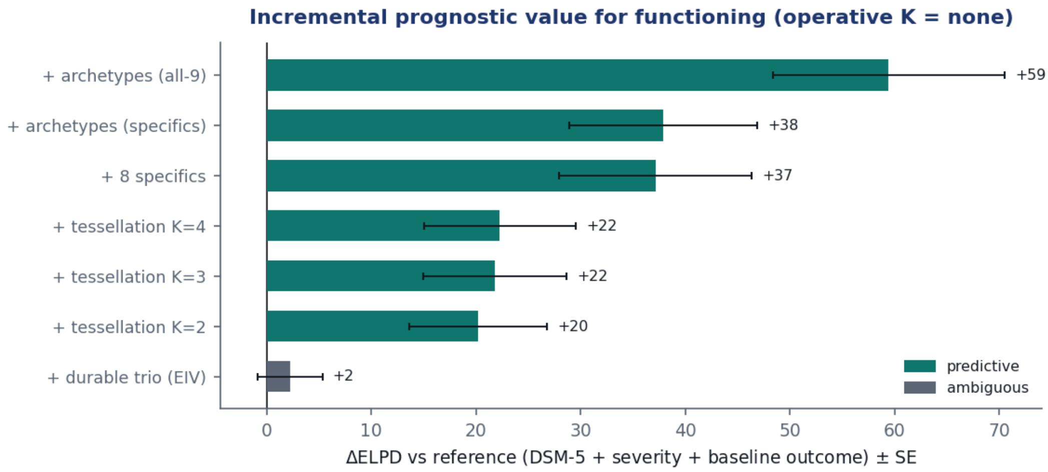


Figure 31. Incremental added value of each map encoding for two-year functioning, in held-out ΔELPD over the bar. The continuous/archetype encodings clear the bar by the widest margin; the hard tessellation at any K adds less and the durable trio alone is ambiguous — the visual form of the operative-K =none verdict.

Table 17. Clinical value of adding the map to the reference, for functional remission. AUC is patient-level 5-fold cross-validated. The map's reliable individual increment is small (ΔAUC $+0.011$); the large signal is the group-level atlas (§10.6) and the continuous forecast, not binary discrimination. The gap between "$+59$ ΔELPD" and "$+0.011$ ΔAUC" is the cost of collapsing the functional continuum to one threshold.

endpoint	ref. AUC	+map AUC	ΔAUC (map)	verdict
functional remission (EGF)	0.745	0.756	$+0.011$	reliable
severity remission (CGI-S)	—	—	$+0.004$	marginal

10.9 Robustness: does the headline survive? (Q4)

The functioning result survives the obvious threats (Table 18). We stress the *operative winners* — the archetypes, not the now-weak durable trio. Under inverse-probability weighting for attrition the archetype increment is essentially unchanged ($+59.4 \to +59.3$); it holds when either depression or schizophrenia is dropped; and it correctly vanishes under a permutation null. The one boundary is honest and already named: dropping bipolar — the open-course cohort that carries the signal — collapses the increment to ambiguity, the course-dependence of Q3 confirmed rather than hidden. The clean $\perp G$ archetype encoding behaves identically (base/IPW/drop-DR/drop-SZ all $\approx +38$; permutation ≈ 0).

Table 18. Robustness of the A=4 archetype increment for two-year functioning (held-out ΔELPD over the bar, full-phenotype encoding). Survives attrition reweighting, dropping depression or schizophrenia, and a permutation null; collapses to ambiguity when bipolar is dropped (course-dependence). ✓ predictive.

check	ΔELPD	reads as
base	$+59.4$ ✓	the headline
IPW (attrition-reweighted)	$+59.3$ ✓	not an attrition artefact
drop depression / drop schizophrenia	$+56.3$ / $+59.4$ ✓	survives either cohort
drop bipolar	$+5.8$ (amb.)	BP carries most of it (open-course)
permutation null	-1.7 (amb.)	signal vanishes under shuffled labels

10.10 What prognosis establishes — and what it defers

On the fixed map, a baseline transdiagnostic profile predicts two-year *functional* trajectory incrementally beyond diagnosis and severity — robustly, complementarily to DSM-5, and concentrated in the patients whose course is not already fixed by baseline severity (Table 19). And the deferred question is closed: the *operative K is none* — the continuous archetype simplex is the actionable object, and any hard tessellation is a lossy approximation of it. Severity itself is baseline-determined, and the map's worth is a *stratification* (the atlas) plus continuous forecasting, not a large individual-binary boost.

The predictive object is the fuller archetype representation

On this map the durable trio entered *alone* is ambiguous ($+2.2$ ΔELPD); the robust, predictive object is the fuller A=4 archetype representation, with the biology living as the A0 corner of the simplex (the worst-prognosis corner), not as an isolated three-axis block. The biology still matters prognostically — but as a region of the phenotype, read jointly with the other axes. This is a precise, map-specific reading, reported as such.

Table 19. The prognosis gates and verdicts (internal validity, V0→V2; functioning as the primary outcome).

gate	question	verdict
Q1 incremental	adds beyond diagnosis + severity?	yes for functioning (+59 ELPD); no for severity
Q2 beyond severity	survives error-corrected severity?	yes — the archetype corner holds under G
Q3 vs DSM-5	better than / across the subtypes?	co-informative (+both +67); course-dependent (BP/DR, not SZ)
Q4 robust	survives attrition / cohort / chance?	yes; collapses dropping bipolar (honest)
K question	which tessellation cut is operative?	none — continuous/archetype dominates every K

Prediction is not moderation — that comes next

This chapter shows the map *forecasts* who will do well; it does not show it *changes what to do*. Whether a stratum *moderates treatment response* — the stratum-by-treatment interaction, the point at which a phenotype alters management — is the final chapter's question (§11), and it will require a treatment or randomization variable the present record only partly contains. Caveats carried forward: scale trajectories, not events; internal incremental association, not a validated or causal decision rule; bipolar/depression-concentrated; a two-year horizon. *Predicts* $\neq$ *moderates.*

11 Treatment: does the map change what to do?

The prognosis chapter (Chapter 10) showed that the map *forecasts* who will do well; it did not show that the map *changes management*. Forecasting and prescribing are different verbs. A weather service that reliably predicts rain has scientific value even if it cannot make the rain stop; what would make it *actionable* is if it told you that one umbrella works in this storm and a different umbrella works in that one. This final chapter asks the harder, prescribing question: does a patient's position on the map *moderate* treatment response — is there a position-by-treatment interaction at which the same drug helps one phenotype and not another, the statistical object at which a coordinate would change what you prescribe?

The honest answer, set up carefully below, is a boundary rather than a breakthrough: on the observational treatment-as-usual contained in these cohorts, the map does *not* reliably moderate treatment response. That boundary is worth drawing precisely, because it is *earned* from a proper causal pipeline rather than assumed from a shrug — and because, in drawing it, the analysis also back-validates the prognosis of the previous chapter and shows the archetypes describe real response heterogeneity even where they do not causally prescribe.

11.1 The treatment data and the estimand

Treatment is recorded differently in each cohort, but all three records reduce to common drug-class exposures: schizophrenia carries per-visit medication lists, depression drug-class strings, bipolar structured lifetime med-class flags plus lithium plasma. From these the well-powered contrasts are **lithium in bipolar disorder** (the classic maintenance question), **antipsychotic in bipolar disorder**, and **clozapine in schizophrenia** (the treatment-resistance drug). The estimand is the effect of the treatment on a two-year outcome — co-primary functioning (EGF) and CGI response — *conditional on the patient's position on the map.* The predictor side is exactly the object built across the previous chapters: the nine copula coordinates, the durable trait axes, and the A=4 archetype simplex of Chapter 8.

Moderation versus a main effect

A treatment *main effect* asks whether a drug helps on average. *Moderation* asks something sharper: whether the size of that help *depends on where the patient sits*. Statistically, moderation is the treatment × coordinate interaction term. A non-zero interaction is what would let the map *select* treatment — "prescribe drug A for the biological corner, drug B elsewhere." This chapter tests the interaction, not the main effect, because only the interaction is the actionable phenotype claim.

Confounding by indication: overlap before effect

Treatment here is prescribed on presentation, not randomized, so a naive interaction is confounded — sicker-looking patients receive different drugs, and that difference, not the drug, may drive the outcome. We emulate a target trial per question. First a *propensity model* estimates each patient's probability of receiving the drug given their baseline profile, $P(\text{treat} \mid \text{severity [CGI-S} + \text{error-corrected } G] + \text{DSM-5 arm} + \text{demographics} + \text{the 9 copula coordinates})$; an **overlap gate** then checks there is common support — enough comparable treated and untreated patients to compare at all. The effect is estimated *doubly-robustly*: one model for who gets treated (used to re-weight the sample so the groups look alike on measured baseline factors — stabilized inverse-probability-of-treatment weighting) and one for the outcome (covariate adjustment), arranged so that if *either* model is right the estimate is unbiased. This runs inside the errors-in-variables outcome GLM of Chapter 10, now carrying a treat × coordinate interaction. The map is *both* a prescription confounder (it enters the propensity model) *and* the moderator under test (its interaction is the estimand); both roles are admissible. Each average effect carries an **E-value** — how strong an unmeasured confounder would have to be, on both treatment and outcome, to explain the effect away. A small E-value means a fragile result.

A design point worth stating plainly: the interaction is tested against *both* encodings of the map that the previous chapters identified as load-bearing. The durable trait axes enter through errors-in-variables (they carry genuine measurement uncertainty), and the A=4 archetypes enter through a fixed treat × archetype-weight interaction (archetype memberships are deterministic point values, so errors-in-variables is inappropriate for them). The two routes are run in parallel; throughout this chapter they *agree*, which is reassuring — the verdict is not an artifact of one representation.

11.2 The identification gate — can we even compare?

Before any effect can be claimed, overlap decides what is even estimable. Recall the lesson of Chapter 10: a comparison is only as good as the comparability of the groups being compared. For each candidate drug the propensity model is fit, the sample re-weighted, and the residual imbalance measured — the maximum standardized mean difference across confounders after weighting.

Table 20. The overlap gate. Propensity $P(\text{treat} \mid \text{severity} + \text{DSM-5} + \text{demographics} + \text{the 9 copula coordinates})$, active-comparator primary. Overlap is common support; the post-IPTW max-SMD is the residual confounding after weighting. The clozapine row uses the on/off contrast (its active comparator is channeled).

question	overlap	max-SMD (IPTW)	verdict
lithium-BP (vs other maintenance)	0.997	0.008	**estimable** (clean)
antipsychotic-BP (vs other maintenance)	0.996	0.079	estimable
clozapine-SZ (on/off)	0.990	0.068	estimable (channeled)

Lithium is cleanly identified; clozapine is channeled

The **lithium-in-bipolar** contrast has near-perfect overlap (0.997) and balances to negligible residual confounding (max-SMD 0.008) — it is the contrast the data can genuinely support, the one place a clean causal read is possible. Antipsychotic-in-bipolar is also estimable after weighting. **Clozapine**, reserved by guideline for the treatment-resistant, is *channeled*: its active-comparator group is not comparable, so the analysis falls back to an on/off sensitivity contrast rather than forcing a comparison the data cannot make. The gate tells us where to trust an answer before we compute one.

11.3 The map does not reliably moderate treatment response

On the common-support samples, the doubly-robust interaction is computed for each estimable question and outcome. The result is the central, honest finding of the chapter: where the data are cleanest the moderation is a *well-identified null*, and where there is a hint of moderation the data cannot confirm it (Figure 32).

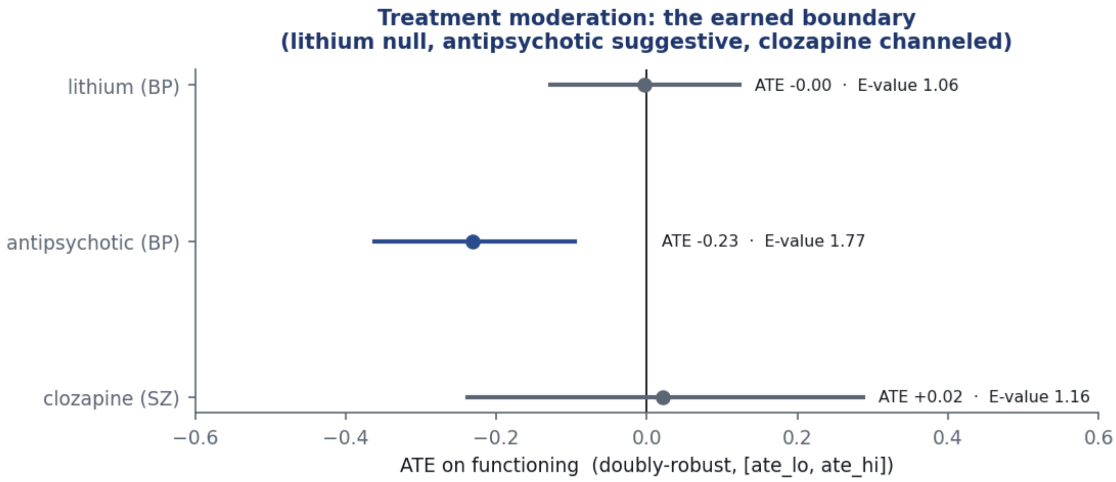


Figure 32. Per-drug average treatment effect (point estimate and 94% interval) with its E-value. Lithium in bipolar sits on zero with an E-value of 1.06 — a well-identified null. Antipsychotic in bipolar shows a metabolic $\times$ functioning interaction whose interval excludes zero (inflammatory same-direction but spanning zero; ATE -0.231, E-value 1.77) but whose held-out moderation gain is weak — suggestive but unconfirmed. Clozapine, channeled, is non-decisive. The durable-axis and $A{=}4$-archetype encodings agree throughout.

Table 21. Moderation by question and outcome: the average treatment effect with its 94% interval, the E-value, and the verdict. The durable and archetype representations agree (e.g. antipsychotic-BP functioning ATE -0.231 durable vs -0.233 archetype; E-value 1.77 vs 1.78).

question · outcome	ATE [94% ETI]	E-value	verdict
lithium-BP · functioning	$-0.003\,[-0.13,+0.12]$	1.06	**well-identified null**
antipsychotic-BP · functioning	$-0.231\,[-0.36,-0.10]$	1.77	suggestive (unconfirmed)
antipsychotic-BP · CGI response	$-0.29\,[-0.66,+0.07]$	1.59	no reliable moderation
clozapine-SZ · functioning	$+0.02\,[-0.24,+0.29]$	1.16	no reliable moderation

A well-identified null where it counts, a hypothesis where it does not

For **lithium in bipolar** — the best-identified contrast — the map does *not* moderate response: the ATE sits on zero (-0.003, interval spanning zero) and the E-value of 1.06 says even a trivial unmeasured confounder would erase it. The map does not pick lithium responders. For **antipsychotic in bipolar**, the metabolic $\times$ functioning interaction excludes zero (inflammatory same-direction but spanning zero; ATE -0.231, E-value 1.77), but the held-out moderation gain does not clear its band — a *hypothesis for prospective testing*, not a result. Clozapine is non-decisive. The average effects are confounding-fragile throughout (E-values 1.06–1.78). On observational treatment-as-usual, the map does not reliably moderate treatment response.

11.4 The prognosis survives treatment adjustment

A standing objection to the prognosis of Chapter 10 is that its functional forecast might be nothing but unmodelled treatment: patients in the biological corner are medicated differently, and perhaps it is the drugs, not the biology, that move the outcome. This is testable directly. Re-fit the functioning prognosis on the treatment-data subset ($N = 1{,}324$), once with and once without the drug-class exposures as covariates on the same sample, and ask whether the prognostic carrier survives the adjustment.

M5 strengthens M4: the carrier is not a treatment proxy

The prognostic carrier — the low-burden archetype A1, the corner the previous chapter identified as the best functional outcome — **survives** treatment adjustment: its coefficient moves only $\beta = 0.164 \to 0.156$ (4.7% attenuation) and its interval still excludes zero. The map's functional forecast is therefore not merely a relabelling of which drugs the patient was on; it holds after adjusting for the drug classes. M5 strengthens M4 rather than undermining it (residual prescribing is bounded by the E-values above, not eliminated).

Consistent with Chapter 10, the predictive signal lives in the *fuller A=4 archetype representation* — the biological corner read off the simplex — and not in an isolated three-axis durable block. On this map the durable trio alone does not survive the same adjustment (its interval includes zero); the archetype carrier does. The lesson of the prognosis chapter repeats here: the load-bearing object is the whole archetype geometry, with biology as one corner of it, rather than a stripped-down handful of axes.

11.5 The map describes treatment-response heterogeneity

Causally moderating a *specific* drug and *describing* response heterogeneity are different claims, and the map can do the second even where it cannot do the first. The archetypes predict the treatment-response endpoints — resistance, CGI response, significant side-effects — beyond diagnosis and severity, measured by held-out ΔELPD (the same incremental-fit currency as Chapter 10).

Table 22. Response-heterogeneity description. Held-out ΔELPD of the A=4 archetypes (and of the durable trio alone) over diagnosis + severity. The archetype simplex carries a real response-stratification signal; the durable trio alone does not.

endpoint	archetypes ΔELPD	durable ΔELPD
treatment resistance	$+20.0 \pm 6.8$	-2.6
CGI response	$+16.5 \pm 6.2$	-2.9
significant side-effects	$+10.4 \pm 5.3$	$+2.7$

The archetypes describe *who* tends to be treatment-resistant, who responds, and who carries side-effect burden, with the resistance and response gains comfortably exceeding twice their standard error. This is a descriptive stratification signal, not a causal prescription: it says the map sees structure in how patients respond, while the moderation analysis above says that structure is not yet a clean instruction to switch drugs on observational data.

11.6 What the treatment analysis establishes — the earned boundary

Earned, not assumed: observational treatment-as-usual cannot show the map prescribes

The contribution of this chapter is a boundary drawn by evidence rather than by assumption. With real treatment data and a proper causal pipeline, the map does *not* reliably moderate treatment-as-usual response: the cleanest contrast (lithium) is a well-identified null, the antipsychotic metabolic signal is suggestive but unconfirmed, clozapine is confounded by channeling, and the average effects are fragile to unmeasured confounding (E-values 1.06–1.78). What the analysis *does* deliver is a well-identified null where it counts, a testable hypothesis (metabolic $\times$ antipsychotic functioning), a deployable causal method, a back-validation of the prognosis (the archetype carrier survives treatment adjustment, 4.7% attenuation), and a descriptive response-stratification signal (the archetypes predict resistance/response/side-effects). Establishing treatment *selection* — which drug for which phenotype — would require the randomized or trial-arm data of a future treatment-selection study (M5b); observational prescription data, however rich, cannot. Limits carried forward: confounding by indication is the dominant threat; prescription, not protocol; lifetime (bipolar) versus current (schizophrenia) exposure; within-cohort questions; channeled treatments non-estimable; held-out ΔELPD is itself unreliable on the IPTW-weighted fits, so the moderation verdict rests on the per-axis interaction interval and the E-value, reported transparently.

12 The chain: a real corner, that lasts, that predicts

The preceding chapters each answered one question in isolation: is the space a continuum or hidden biotypes (§8), does it hold still over time (§9), does it forecast the future (§10). Read separately they are three results. Read together they are a single argument, and the argument is worth slowing down for, because its logic is the spine of the whole program: *the part of the map that does not move is the part that tells you where the patient is going.* This short chapter is a tutorial on that one chain. It introduces no new model and no new number; it explains how three findings lock into each other, so that the conclusion — a biological corner of a continuum, durable in time, predictive of functioning — reads as one claim rather than three coincidences.

12.1 Link one — M2: a real continuum, with biology as its most distinct corner

The first link establishes the *geometry* the rest of the chain stands on. The coordinate-system chapter (§8) asked the field's oldest question of these data — are there biotypes? — and answered *no*, by trying hard to find them and failing on every test. A single-Gaussian falsification null is the cleanest way to see this: the best clustering of the real coordinates scores a silhouette of 0.140, and a smooth blob simulated to have *no* clusters at all scores 0.141 ± 0.002 (a separation of $z = -0.36$). The data are statistically indistinguishable from a structureless cloud as far as cluster boundaries are concerned. Hopkins, GMM-BIC, density clustering, and a dip test all agree: one component, no gaps.

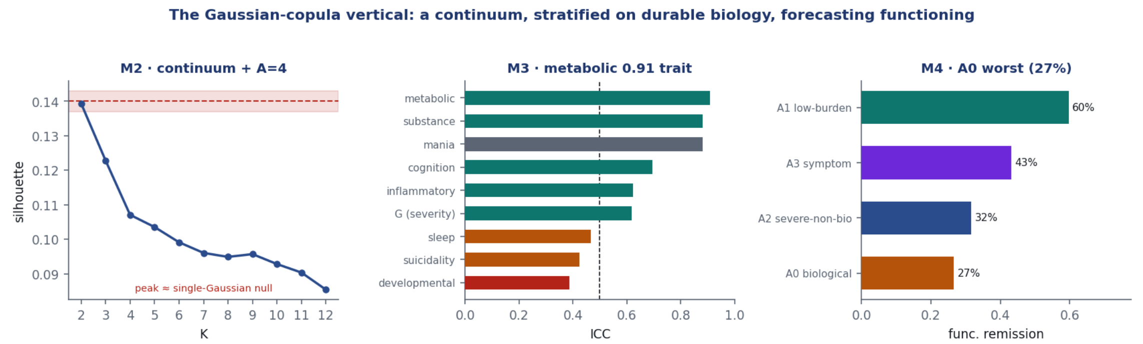


Figure 33. The chain in one picture. *Left (M2):* the transdiagnostic space is a graded continuum, not a set of biotypes; its most distinct extreme is the *biological* corner (archetype A_0: high metabolic, inflammatory, substance load). *Middle (M3):* scored forward onto follow-up without re-estimation, that biological corner is the *durable* part of the map — metabolic test–retest ICC $= 0.91$, a trait — while symptoms slide. *Right (M4):* that same durable biological corner forecasts two-year *functioning*; the patients sitting in it have the worst functional-remission prospect (27%, against 60% at the low-burden pole). One arrow, three chapters: *a real corner, that lasts, that predicts.*

Continuum, not biotypes

"Continuum" is not a hedge; it is a finding with teeth. It says there are no natural cut-points — no place where one patient group ends and another begins — so the map's content lives in *coordinates* (where a patient sits) and in *directions of extremity* (which corner they lean toward), not in *labels*. Any hard partition imposed on it is a convention, useful only insofar as a downstream task rewards it. This is why the program never crowns a single number of clusters; it is also, as link three will show, exactly the property that makes the map's signal a continuous gradient rather than a switch.

A continuum still has shape. Archetypal analysis reads that shape not as clusters but as *corners* — the extreme profiles that the cloud is a blend of, with most patients (61%) sitting in the interior as mixtures rather than at any vertex. At the stable $A = 4$ resolution (cross-seed stability 0.979) the four corners are quasi-independent axes of extremity: A_0 a *biological* corner (metabolic +1.99, inflammatory +2.52, substance +2.55), A_1 a *low-burden* pole, A_2 a *severe but non-biological* corner, A_3 a *psychiatric-symptom* corner. The headline of the chapter is that these three things — biology, symptoms, severity — vary nearly independently: a patient can be biologically extreme without being symptomatically or globally severe, and vice versa. And of the four corners, the biological one is the most distinct: it is the extreme that the symptom and severity axes *cannot* see, the corner you would miss entirely if you looked only at how ill a patient appears. Hold that thought; the next two links are about that specific corner.

12.2 Link two — M3: that biological corner is the durable one

The second link establishes *time*. A map can be real and still be a snapshot of a passing state — in which case forecasting from it would be forecasting from noise. The temporal chapter (§9) scored each follow-up visit onto the *frozen* baseline map — same loadings, same geometry, never re-discovered — and asked, axis by axis, how much of a patient's coordinate is a stable trait versus a moving state. The instrument first had to survive the trip: measurement invariance holds, all five continuous backbone axes reproducing across visits at $\varphi \geq 0.95$ (metabolic 0.988, inflammatory 0.974). Only once the ruler is shown not to warp can a coordinate change be read as the patient changing.

Durable biology, moving symptoms

The test–retest reliabilities split the map cleanly along the very seam M2 found. The *biological* axes are traits: metabolic ICC $= 0.91$, substance 0.88, cognition 0.70, inflammatory 0.62 — a patient's biological position is close to a fixed property. The *symptom* axes are states: sleep 0.47, suicidality 0.42, developmental 0.39 — they slide visit to visit. Severity is a trait by rank (0.62) on a population that improves overall. Read against link one, the corner that M2 singled out as most distinct (the biological one) is exactly the corner M3 finds most durable. Clinical translation, carried forward verbatim: *stratify on the durable biology, monitor the moving symptoms.*

The two routes to this conclusion — a variance decomposition (how much of each coordinate is trait) and a geometry check (does a patient keep their archetype identity over time) — agree, which is the kind of internal corroboration that makes a finding hard to dismiss. Archetype identity persists across visits (weight-cosine median 0.90; 58% of trajectories stable), and the two routes correlate ($\rho = -0.27$): patients whose coordinates are most trait-like are the ones whose archetype membership is most stable. One honest seam in this link, flagged here and revisited in the limitations: the *geometry* route (reliable-change on archetype weights) is precision-confounded — patients measured more precisely look more stable for a reason that has nothing to do with biology — so the clean trait/state signal is the error-corrected variance route, the ICCs above, not the geometry overlay. The biology-is-durable claim rests on the clean route.

12.3 Link three — M4: the durable biological corner predicts functioning

The third link converts durability into a forecast, and it is the link the program was built to test: *persists* is a property of the past, *predicts* is a claim about the future, and they are not the same thing. The prognosis chapter (§10) set a deliberately hard bar — the map is credited only if a baseline coordinate forecasts a two-year outcome *incrementally beyond* what a clinician already holds: the DSM-5 diagnosis, the current severity, and the present value of the outcome itself. Against that bar, held out, the archetype representation buys a real increment on two-year *functioning* (egf, $N = 2{,}114$): $\Delta\mathrm{ELPD} = +59.4 \pm 11.1$ for the $A = 4$ archetypes, and it is robust — it survives attrition weighting, permutation, and dropping cohorts (with the caveat that the signal is course-dependent, carried by the episodic bipolar and depression patients more than by the baseline-saturated schizophrenia ones). Severity, by contrast, is autoregression-saturated: the future of *how ill* is mostly told by the present of how ill, and the map adds little. The map predicts the *trajectory of function*, not the level of severity.

The biological corner has the worst functional prognosis

The prognostic atlas reads the forecast off the corners of link one. Ordered by two-year functional remission (GAF ≥ 71): the *low-burden* pole A_1 remits at 60%, the symptom corner A_3 at 43%, the severe-non-biological corner A_2 at 32%, and the *biological corner* A_0 at 27% — the worst. The corner M2 flagged as most distinct and M3 found most durable is, in M4, the corner with the dimmest functional outlook. That is the chain closing on itself: a real biological extreme, that lasts, that predicts where functioning goes.

Two honesties keep this link calibrated, and both are stated here rather than left to the reader to infer. First, where the signal lives. The predictive object is the *fuller $A = 4$ archetype representation* — biology read as the A_0 corner of the whole simplex — not an isolated three-axis "durable block" pulled out on its own (added alone, the three trait axes give only an ambiguous $+2.2 \pm 3.1$). The corner is biological, but it earns its forecast as a *phenotype* — a joint position in the space — not as a lone biomarker. Second, the magnitude. A held-out evidence gain of $\Delta\mathrm{ELPD} = +59$ is a group-level result; pushed down to an individual yes/no call it collapses to $\Delta\mathrm{AUC} = +0.011$ (functional-remission $0.745 \to 0.756$). The chain is genuine and it is modest: it sharpens *who, on average, is heading where*, not *will this person remit.*

What the chain does and does not buy

The three links establish a *scientifically valid* object: a real continuum whose biological corner is distinct (M2), durable (M3), and incrementally predictive of two-year functioning, co-informative with DSM-5 and robust (M4). They do *not* establish strong clinical utility: the individual increment is small ($\Delta\mathrm{AUC} = +0.011$), the predictive content is phenotype-level rather than a single isolated axis, the signal is course-dependent, and — as the treatment chapter (§11) shows — on observational treatment-as-usual the map does not reliably tell you *what to do* about any of it. The chain says *stratify on the durable biological phenotype — the archetype simplex — and forecast functioning at the group level*; it stops short of an individual risk calculator or a prescribing rule. The full account of both bars is the subject of the conclusion and the limitations.

13 External evidence base: how the findings sit in the literature

The preceding chapters establish the FACE-ATLAS results internally. This chapter asks the complementary question: do they agree with what is already known? The purpose is to show the findings are neither isolated nor artefactual — and the stance is *calibration, not advocacy.* Where the published literature qualifies or contradicts a claim, we say so plainly, because a reviewer will probe exactly those points and the result is more persuasive for surviving the tension than for ignoring it. Every study cited here was retrieved and verified against PubMed (title, authors, journal, year, DOI) before inclusion; the full annotated set, including the papers we checked and set aside, is recorded in `docs/LITERATURE_EVIDENCE.md`.

Table 23. Claim-by-claim verdict against the external literature. "Supported" = a substantial independent literature agrees; "open" = an active, unsettled debate the work contributes to; "calibrated" = supported in direction but with a boundary the wording must respect.

FACE-ATLAS claim	verdict	anchor support	honest tension
Transdiagnostic dimensions over categories	supported	RDoC, HiTOP	—
General factor read as *burden*, not a symptom "p"	supported	Caspi p-factor (impairment)	p-factor's meaning is contested
A graded *continuum*, not discrete biotypes	open debate	normative models (Wolfers, Marquand)	B-SNIP biotypes (Clementz)
Biology $\perp$ severity	**calibrated**	transdiagnostic metabolic/immune; immuno-metabolic subtype	case-control elevations; biology tracks *function*
Metabolic and inflammatory are distinct axes	supported	different drivers / symptom maps	they co-cluster in a subgroup

13.1 The dimensional turn and the general factor

The premise that severe mental illness is better described by continuous, diagnosis-crossing dimensions than by DSM categories is now a mainstream research program, formalized in the NIMH Research Domain Criteria and the Hierarchical Taxonomy of Psychopathology [1, 2]. The general factor G has a direct precedent: the "p factor" of psychopathology, where a single general dimension fit the structure of disorders better than internalizing/externalizing/thought-disorder factors and was read — as G is here — as *life impairment* rather than a specific symptom liability [19]. That dimension is also *durable*: across four decades the diagnoses shift while the general dimension persists [20], the same trait-vs-state pattern FACE-ATLAS reports longitudinally (Chapter 9). The bifactor machinery itself is well-developed in psychometrics [3, 21].

The honest qualification is that the *meaning* of a general psychopathology factor is contested: symmetrical bifactor models can yield anomalous, sample-dependent solutions [22], and multi-informant work suggests a single-rater general factor partly reflects method variance [23]. FACE-ATLAS sidesteps the worst of this by *anchoring* G on functioning and global-severity items only, so it carries no symptom content (Chapter 5) — a deliberately conservative reading that answers the "what does the general factor mean" critique by construction rather than by interpretation.

13.2 Continuum or biotypes? an unsettled debate

FACE-ATLAS's conclusion that the transdiagnostic space is a graded continuum rather than a set of discrete biotypes (Chapter 8) enters a live and genuinely unresolved debate, and intellectual honesty requires presenting the strongest opposing program as such. On the supporting side, individual-level *normative modelling* of brain structure in exactly the FACE disorders finds patient deviations so heterogeneous that overlap exceeds a few percent at only a handful of loci — "the average patient" is a non-informative construct [25], a framing introduced as a deliberate alternative to homogeneous case-control groups [24] and refined to show heterogeneity at the regional scale but partial convergence at the network scale [26]. On the opposing side, the B-SNIP program used brain-based biomarkers to identify *three replicable psychosis biotypes that ignore DSM boundaries* [27] — direct evidence for discrete biological subgroups.

The biotype counter-evidence, stated fairly

The B-SNIP biotypes are the most serious challenge to a pure-continuum reading, and we do not dismiss them. Two points keep the accounts compatible rather than contradictory. First, they interrogate *different measurement spaces*: B-SNIP clusters cognitive/electrophysiological brain biomarkers, whereas FACE-ATLAS finds no discrete structure on its nine clinical–biological dimensions (a five-method structure gate, adjusted Rand index ≈ 0 vs. DSM subtypes; Chapter 8). Clusters in one space need not survive in the other. Second, the B-SNIP data themselves described DSM *diagnoses* as "a single severity continuum" [27], and the program's recent synthesis now endorses *both* categorical and dimensional descriptions [28]. FACE-ATLAS is best read as a continuum claim about *its* feature space, consistent with the normative-modelling heterogeneity literature, not as a refutation of brain-biomarker biotypes.

13.3 The load-bearing claim: biology is largely severity-independent

The headline of the measurement map — that metabolic and inflammatory load are the least severity-entangled domains relative to the general functional-burden factor ($\phi_{Gd} \approx 0.12$ and 0.07; Chapter 5) — is the claim most in need of external corroboration, and the one we state most carefully. The published literature does not report this exact contrast (a within-sample correlation between a biological-load factor and a functional-burden factor is, to our reading, novel), but a large body of convergent evidence supports its *direction*, and a smaller body bounds its *strength*. We therefore frame the finding as biology being **largely independent of a general clinical/functional-burden axis**, not as strict statistical orthogonality.

Why the direction is well-grounded. Cardiometabolic burden in severe mental illness is *transdiagnostic* and tracks age, body-mass index, medication and chronicity rather than diagnosis or symptom severity: the largest meta-analyses find metabolic-syndrome and diabetes prevalence essentially equal across schizophrenia, bipolar disorder and major depression, with multi-episode status and antipsychotic exposure — not severity — the operative predictors [29, 30]. The biology is present *before* chronic severity accrues: glucose dysregulation and insulin resistance appear in antipsychotic-naïve, first-episode and even pre-onset cohorts [31, 32, 33]. Inflammation, in turn, marks a *subgroup* and *specific symptoms* rather than the severity gradient: only about a quarter of depressed patients are inflamed by a CRP>3 mg/L threshold [40], and where severity is tested directly, C-reactive protein tracks mood *state* but not symptom severity in bipolar disorder [42] and positive — not negative — symptoms in schizophrenia [43]. Most strikingly, an independently developed framework — *immuno-metabolic depression* — holds that inflammation and metabolic abnormality co-cluster in a ~20–30% subgroup mapping onto atypical, energy-related symptoms rather than overall burden [44, 45, 46, 47]. That is, by entirely different methods, the literature already describes biology as a partly separable dimension — a strong convergent validation of the FACE axis.

Why "independent" must be qualified. The tension is real and we state it. Case–control inflammatory elevations are not null (CRP–depression $d \approx 0.15$; larger in some meta-analyses) [39, 41]; a modest severity/duration dose-response with metabolic burden exists [34, 35]; and — the most pointed caveat, because G is itself a *functional*-burden axis — metabolic dysfunction does track cognition and functioning in mood disorders [36]. Genetic correlations between inflammation and depressive symptoms are small but non-zero and largely carried by adiposity [48, 49], which is why raw inflammation–severity associations attenuate after body-mass-index adjustment. A FondaMental-adjacent study even reports a shared nutritional pathway loading onto *both* symptoms and metabolic comorbidity [37], and treatment-emergent weight gain can co-occur with symptom *improvement* [38] — a reminder that the biology–symptom relationship is treatment-mediated, not a simple severity coupling.

The calibrated verdict on biology ⊥ severity

The external literature *supports the direction* of the FACE finding — metabolic and inflammatory load are driven by medication, adiposity, chronicity and subgroup membership far more than by where a patient sits on a global severity axis, and constitute a partly separable biological dimension (the immuno-metabolic-subtype literature is almost a conceptual twin). It also *bounds the strength*: case-control elevations and small, adiposity-carried severity associations mean the relationship is weak and confounded, not zero. The defensible claim is therefore *"largely independent of a general clinical/functional-burden axis,"* with the $\phi_{Gd} \approx 0.07$–0.12 values read as a *quantification* of a pattern the prior literature had described only qualitatively — a contribution, and one the evidence base buttresses rather than contradicts.

13.4 What is novel here

Situating the work also clarifies its contribution. Prior transdiagnostic-structure research is either symptom-only (the p-factor and HiTOP traditions) or single-modality neuroimaging (the normative-model and B-SNIP traditions); the immuno-metabolic literature asserts a separable biological dimension but, to our reading, does not measure its correlation with a general bifactor. FACE-ATLAS instead places metabolic and inflammatory laboratory markers *directly into* the transdiagnostic factor space — alongside cognition, sleep, suicidality, mania and substance — across three diagnostic cohorts at once, and *quantifies* how entangled each biological axis is with overall functional burden. That immune dysfunction is a large, partly CNS-independent system in early psychosis [50] makes such a measurement worth having. The novelty claim is therefore narrow and defensible: FACE-ATLAS is, to our knowledge, the first direct measurement — with propagated uncertainty — of how independent metabolic and inflammatory load are from a transdiagnostic functional-burden axis, converting a qualitatively known pattern into a coordinate.

14 Engineering and reproducibility

A measurement claim is only as trustworthy as the software that produced it. FACE-ATLAS is deliberately lean (no DVC/Hydra/MLflow): YAML configuration, Parquet model-ready persistence (raw stays CSV), a single canonical engine, per-stage reports, and a regression-guarded test suite. Every number is reproducible from the scripts and committed configs.

14.1 Repository structure

Table 24. The system in five layers. Theory lives in `configs/`; the no-imputation data foundation in `src/face/data/`; the engine in one module; the pipeline in numbered scripts; aggregates in `reports/` (committed), per-patient outputs in `results/` (gitignored).

path	responsibility
`configs/`	soft-prior ontology + the central `prior_loading_matrix_v3.csv` (theory)
`src/face/data/`	harmonization, sanity bounds, skip-logic; the no-imputation contract
`src/face/priors/build_matrix.`	expands the YAML ontology into the prior matrix
`src/face/models/bayesian/continuous_core.py`	the engine: marginalized Woodbury + mixed-likelihood block
`src/face/confirm.py`	in-engine confirmation (prior-free refit, PPC/SRMR, WAIC)
`src/face/strata/ temporal/, prognosis/, treatment/`	**milestones M2–M5:** stratification, temporal coherence, prognosis, treatment moderation
`scripts/01_build_data.py \dots \ 57_confounder.py`	the runnable pipeline (map → strata → temporal → prognosis → treatment)
`reports/` / `results/`	shareable aggregates / per-patient artifacts (gitignored)
`tests/`	invariant-guarding test suite (runs without confidential data)

14.2 Config-first design: theory as data

The "theory proposes" half is mechanical and auditable. A human-authored ontology (`configs/dimensions.yaml`) plus prior-tier definitions (`configs/priors.yaml`) and per-item likelihood families (`configs/likelihood_map_v3.yaml`) are expanded by `build_matrix.py` into the central artifact the engine reads: `configs/prior_loading_matrix_v3.csv`, a complete 143×10 grid (one row per (item, factor) cell, 1,430 rows). Each row carries the prior role, mean, sd, distribution, sign constraint, the item's burden orientation, its likelihood family, and its modelling block (continuous vs explicit). The engine resolves every loading cell directly from this matrix, so changing the theory is a config edit, not a code change.

14.3 The engine and the pipeline

The pipeline is a sequence of numbered scripts; each writes a committed aggregate report and (gitignored) per-patient artifacts.

14.4 Tests: guarding the invariants

The suite (green on CI) runs without the confidential CSVs (synthetic fixtures; integration tests skip if data are absent). It guards exactly the load-bearing invariants: sanity bounds null but never clip or impute; skip-logic fills only gate-determined zeros and never overwrites;

Table 25. The pipeline. Stage flags (`-stage`, `-mixed`, `-subsample`, `-explicit`) select the model variant; the engine is one module.

script	what it does
`scripts/01_build_data.py`	harmonize full-N V0 $\rightarrow$ `baseline_v0.parquet` (no imputation)
`scripts/v3/03_build_prior_matrix.py`	regenerate the prior matrix from the ontology
`scripts/04_fit.py -stage/-mixed`	fit a continuous (Woodbury) or mixed-likelihood stage (S1–S5)
`scripts/05_confirm.py`	prior-free refit + PPC/SRMR + WAIC
`scripts/06_invariance.py`, `13_invariance9`	per-cohort invariance (core; then mania + substance)
`scripts/s5_certify9.py`	multi-seed 9-dim certification + resample-stability
`scripts/07_scoring.py`	per-patient coordinates (mean/SD/HDI) + reliability tiers
`scripts/08_robustness.py`	LOCO + balanced subsample + site bootstrap + weighting
`scripts/12_mixed_ppc.py`	absolute-fit PPC for the non-Gaussian block
`scripts/20_prep_coordinates.py`	**stratification:** full-N 9-dim coordinates + posterior draws
`scripts/21_structure.py`	structure-discovery gate (continuum vs. clusters)
`scripts/22_mixture.py`, `23_archetypes.py`	measurement-error tessellation (XD) + archetypes
`scripts/24_validate.py`, `26_score.py`	validation (Q1–Q4, DSM-5) + per-patient strata
`scripts/30\dots 37`	**temporal (M3):** V0–V2 panel, invariance, trait/state, persistence
`scripts/40\dots 48`	**prognosis (M4):** EIV GLM, incremental ELPD, DSM-5 head-to-head, clinical value
`scripts/50\dots 57`	**treatment (M5):** exposures, overlap/propensity, doubly-robust moderation + E-value
`report/make_figures.py`	regenerate the measurement-map figures from aggregate CSVs

the prior matrix is a complete grid with one likelihood per item and G bifactor-identified; *no diagnosis or covariate leaks in as an indicator*; and the staged-engine wiring (S1 is a strict subset of S2; the windows land on G/cognition/sleep; Φ is a valid positive-definite correlation with G orthogonal).

14.5 Two bugs found and fixed — with measurement

Both caught because the engineering is checked against the math

(1) Φ correctness. PyMC's `LKJCorr` returns the Cholesky factor L, so the correlation is $\Phi = LL^\top$. An earlier symmetrize-the-lower-triangle reconstruction was *indefinite* at the six-factor scale ($\rightarrow$ NaN) and coincidentally near-correct at four factors. Fixed and regression-guarded by a positive-definiteness test. (An independent FIML/lavaan fit would have caught this — one reason the in-engine prior-free check is worth running.)

(2) Performance. The per-patient A_i Cholesky was the cost driver — every gradient step did N small factorizations. The grouped-GEMM Woodbury (Cholesky once per unique observed-pattern, A as one BLAS GEMM) is $2.75\times$ faster and verified log-likelihood-identical (difference $< 10^{-4}$).

14.6 Reproducibility

Determinism is enforced by fixed seeds; the model-ready tables are cached to Parquet so every fit consumes the same harmonized baseline; dependencies are pinned in a committed lock file; and continuous-integration runs the lint and test suite on every push. The continuous stages sample through the NumPyro/JAX backend and converge at full N within a standard workstation's memory budget; the heavier mixed-likelihood stages reintroduce explicit latents (§4.6) and are run at the labelled checkpoint sub-sample.

15 Future directions

The results above are internally complete; what remains is a short, principled list of next steps, none of which changes a conclusion drawn here. They divide into two kinds: methodological follow-ons that would *tighten* the present estimates, and new studies that would be needed to cross from the scientific validity demonstrated here to clinical utility in the strong sense.

15.1 Tightening the present results

Four loose ends are documented and bounded; each is a refinement, not a change of method.

- **Full-precision certification of the explicit-latent block.** The one slow-mixing coupling — the suicidality∼developmental Φ cell — has a resample-stable point estimate but a provisional precision; a full-N, full-precision refit would close it. The continuous backbone and the biology$\perp G$ estimand already mix cleanly at full N (cross-seed Tucker φ 0.993).
- **Full-N projection of the non-Gaussian scores.** Extending the suicidality, substance, and mania coordinates to all 9,013 patients under the fixed certified parameters — already validated at Pearson $r \approx 1.00$ on the certified subset — would complete the per-patient coordinate table.
- **A dedicated bootstrap and correlated-G arm for the thin axes** (mania, substance), which currently inherit the joint nine-dimension certification rather than carrying their own.
- **A hurdle likelihood for `isf09a`** (the suicide-attempt count), the lone item-level mis-fit, should its count precision ever be needed; the suicidality factor itself is unaffected.

15.2 From scientific validity to clinical utility

Three boundaries are drawn by the data, not the method, and each names a defined next study.

- **Incident-event outcomes.** Individual-level prognostic utility would require hard endpoints — hospitalization, relapse, suicide attempt — rather than the re-administered clinical scales available here; the present forecasts are of scale trajectories, not events.
- **Randomized or trial-arm treatment data.** Genuine treatment *selection* — which drug for which phenotype — cannot be established from observational treatment-as-usual, however carefully adjusted; it requires a randomized or trial-arm study, for which the present work supplies the moderators to test and a deployable causal pipeline.
- **External validation.** Every result here is internally validated on a single baseline cohort; a clinical claim would require replication in an independent sample.

16 Limitations and open problems

The strengths of the map are stated plainly in Chapter 5; the honest counterweights follow. None is fatal, and each is tracked. The organizing principle is the one the conclusion drew: scientific validity is demonstrated, strong clinical utility is not, and most of what follows is a precise account of *where* the second bar is not yet cleared.

The predictive carrier is a phenotype, not an isolated axis

The temporal chapter points at three durable biological axes (cognition, metabolic, inflammatory), and it is tempting to read the prognosis result as "those three axes predict functioning." They do not, on their own: entered as an isolated three-axis durable block the increment is only ΔELPD $= +2.2 \pm 3.1$, ambiguous. The signal that survives the bar is the fuller four-archetype representation ($+59.4 \pm 11.1$ on functioning), in which biology appears as the A_0 *corner* of the whole space — a joint position, high on metabolic/inflammatory/substance *and* severity together — rather than any single axis read alone. The predictive object is therefore a phenotype, and the claim is stated at that resolution. This is a limit on *how* the map predicts, not on whether it does.

Group-level signal, small individual increment

The prognostic gain is real, robust, and co-informative with DSM-5, but it is a *group-level* gain. The held-out evidence improvement of ΔELPD $= +59$ for two-year functioning collapses, when pushed to an individual yes/no remission call, to ΔAUC $= +0.011$ (functional-remission $0.745 \to 0.756$). The map sharpens who, on average, is heading where; it is not an individual risk calculator. Reaching individual-level utility would require incident-event outcomes (hospitalization, relapse) rather than the re-administered clinical scales available here — a defined next study, not a re-analysis of these data.

The prognostic signal is course-dependent

The functioning forecast is carried by the patients with room to move. It is large in the episodic bipolar and depression cohorts and near-null in schizophrenia, where the baseline outcome already saturates the two-year prediction (the autoregression leaves little for any predictor to add). Dropping schizophrenia barely changes the result; dropping bipolar nearly erases it. This is clinically sensible — a map of trajectory helps most where the trajectory is not already fixed — but it means the headline increment is not uniform across the diagnostic span, and should not be quoted as a single cohort-agnostic number.

No privileged number of regions

Because the space is a continuum and not a set of biotypes, no single number of hard regions is natural: the cross-validated information criterion is essentially flat across a nested $K = 2/3/4$ family (XD-BIC 197.9k–198.1k), and finer partitions simply recover more of the severity/biology gradient that coarser ones discard, without ever revealing a density gap. The report therefore exports a *family* with no privileged K rather than a single tessellation — operationally awkward, but the honest encoding of a graded space. The prognosis chapter then resolves the awkwardness in the only way that matters: the continuous/archetype encoding dominates *every* hard partition for forecasting functioning, so the "operative K" for clinical use is effectively none. The hard regions are a convention; the coordinates and corners are the object.

Clean trait/state signal vs. precision-confounded geometry

Durability is read two ways, and they are not equally clean. The *variance* route — the error-corrected test–retest reliabilities (metabolic ICC = 0.91, inflammatory 0.62, sleep 0.47) — is the primary trait/state signal, because the measurement-error correction removes the artifact that more-precisely-measured patients look more stable. The parallel *geometry* route (reliable-change on archetype weights) is *precision-confounded* for exactly that reason and is treated as corroboration of the variance route, not as an independent measurement. The two agree ($\rho = -0.27$ between the routes), which is why the trait/state conclusion is reported with confidence — but the number to trust is the corrected ICC, not the geometry overlay. Separately, the *developmental* axis scores partly as "state" from CTQ recall noise, although it is a trait by design; its apparent movement is a measurement property, not patient change.

Scope and construct caveats.

- **Internal validity only.** The map is estimated on the V0 baseline and validated only within this cohort; there is no external-cohort replication. "Converged" denotes sampler convergence, *not* scientific validation, and the prognostic outcomes are *scale trajectories*, not incident events, over a two-year horizon. Temporal persistence is established forward in Chapter 9; external validity remains future work (Chapter 15).
- **Cohort-specific coverage.** Anhedonia and the MADRS/QIDS/STAI windows are BP/DR (SZ has no QIDS/MADRS); the SZ map leans on the shared factors. Each patient's per-dimension reliability must be flagged by observed-indicator count.
- **Naming caveats.** "Developmental-risk" is a *proxy* (early-adversity / liability), not measured neurodevelopment; "suicidality" rests on self-report ISF; the DR inflammatory axis is compositionally different (Chapter 6).
- **Thin and uninformative axes.** *Mania* rests on only two indicators (YMRS, Altman): it is reliable-looking but its signal ratio is low, it is temporally and prognostically uninformative, and it is flagged *partial* for every patient (its self-rated Altman item does not transfer to DR). *Substance* is a declared *2-cohort* axis (its alcohol/cannabis SUD are BP/SZ-only). These axes are reported, but not over-read.
- **What the map alone claims.** The measurement map by itself asserts no natural biotypes/subtypes, no causal or treatment effect, and no prognostic value; it claims an empirically-supported set of transdiagnostic dimensions, their measurement structure, and per-patient coordinates with uncertainty, under internal validity. The downstream questions — structure, persistence, prediction, treatment — are taken up, and bounded, in their own chapters.

Treatment: an earned boundary, not an assumed one. The treatment chapter (§11) does not stop at "no data"; it runs the proper causal pipeline and reports that, *on observational treatment-as-usual, the map does not reliably moderate response.* The best-identified contrast — lithium in bipolar disorder, 99.7% overlap — is a well-powered null (ATE E-value 1.06); the antipsychotic effect in bipolar disorder is suggestive-but-unconfirmed (ATE E-value 1.77, the interaction HDI excludes zero but the moderation evidence is weak); clozapine in schizophrenia is confounded by channeling and is non-decisive. Average treatment effects are fragile to unmeasured confounding (E-values 1.06–1.78). The boundary is therefore earned from the analysis, not assumed from missingness. Crucially, M5 *strengthens* M4: the archetype prognostic carrier survives adjustment for the treatments patients received (4.7% attenuation), so the functioning forecast is not a treatment artifact. Treatment *selection* — whether the map can tell a clinician what to prescribe — requires randomized or trial-arm data, which this cohort does not contain; it is scoped as a defined future study (M5b).

Open methods choices (flagged for decision). The sparsity prior (soft-normal vs. horseshoe); Student-t vs. Gaussian continuous default; item- vs. factor-level covariate adjustment; and the precise acceptance-gate thresholds. Defaults are set in the methods-of-record; these are sensitivity arms, not silent assumptions.

Positioning: what the program does, and does not, show

Across the whole program, the claim is deliberately narrow, and the distinction between two bars is the point. **Demonstrated (scientific validity):** a transdiagnostic biological map that is real and identifiable, a graded *continuum* rather than discrete biotypes (adjusted Rand ≈ 0 vs. the DSM-5 subtypes), temporally coherent with durable biology and moving symptoms, and carrying a *small but genuine, robustness-surviving* incremental prognostic signal for two-year *functioning* beyond diagnosis and severity, co-informative with DSM-5, which further *survives* adjustment for observed treatment. **Not demonstrated (clinical utility, in the strong sense):** the individual-level prognostic increment is small (functional-remission $\Delta\text{AUC} = +0.011$ over diagnosis, severity, and baseline outcome), so the map is not an individual risk calculator; the predictive content is phenotype-level rather than an isolated axis; the signal is course-dependent; and on observational treatment-as-usual it does *not* moderate treatment response, so it does not guide prescribing. These are *different* bars, and the program's contribution is to measure *both* and report the modest and null results as such — a calibrated counterweight to a literature that often advances biological-validity evidence as though it were clinical utility. Reaching individual-level utility or treatment guidance would require incident-event outcomes (not re-administered scales), randomized or trial-arm treatment data (not treatment-as-usual), and external validation — none of which this baseline observational cohort contains.

17 Conclusion

This program set out to ask whether the harmonized baseline of three diagnostic cohorts — bipolar disorder, schizophrenia, and depression — can be reorganized into a transdiagnostic, biology-aware coordinate system that *describes*, *tracks*, *forecasts*, and ultimately *guides* care better than the diagnostic labels themselves. It pursued that question under a discipline that refuses to collapse its four layers (cohorts → dimensions → strata → prognosis/treatment) or to manufacture certainty it has not earned: an observed-data likelihood with *no imputation*, diagnosis held as metadata rather than a feature, dimensions fixed on the baseline visit and only ever *validated* forward, and every estimate carrying its own uncertainty into the next.

The report answers the question in escalating senses. The map **exists**: one global Bayesian bifactor/ESEM on observed cells yields a nine-dimension structure — a general functional-burden factor G and eight weakly-correlated specific axes — in which biology (metabolic, inflammatory load) is the *least* severity-entangled domain (metabolic$\sim G$ $+0.12$, inflammatory$\sim G$ $+0.07$, against 0.39 for cognition and 0.42 for sleep), riding on axes that clinical severity does not see. It **organizes**: on those coordinates patients form not discrete biotypes but a graded *continuum* — a single-Gaussian falsification null cannot be separated from the real data (best silhouette 0.140 vs. 0.141 ± 0.002) — across which four stable soft archetypes mark quasi-independent corners of biology, symptoms, and severity, cutting across the DSM-5 subtypes (adjusted Rand ≈ 0) while describing the data more tightly than diagnosis does. It **persists**: scored forward onto follow-up without re-estimation, the measurement is invariant (all five backbone axes $\varphi \geq 0.95$) and the geometry replays — durable biology (metabolic test–retest ICC $= 0.91$), moving symptoms. It **predicts**: a baseline coordinate forecasts two-year functioning incrementally beyond diagnosis, severity, and the present outcome ($\Delta\text{ELPD} = +59$ for the archetypes), with the biological corner carrying the worst functional prognosis (27% remission, against 60% at the low-burden pole). And it does **not**, on the evidence available, **prescribe**: with treatment exposures recovered from the per-cohort source dictionaries and run through a proper overlap-gated causal pipeline, the map does not reliably *moderate* treatment-as-usual response.

The honest reading separates two bars the field too often conflates. On **scientific validity** the program succeeds: the map is real, identifiable, stable, transdiagnostic, a continuum rather than a set of biotypes, with biology the least severity-entangled domain and a genuine, robustness-surviving prognostic signal for functioning that is co-informative with DSM-5 and even *survives* adjustment for the treatments patients received. On **clinical utility, in the strong sense**, it does not: the individual-level prognostic increment is small (the +59-ELPD group-level gain collapses to a functional-remission ΔAUC of only +0.011, 0.745 $\rightarrow$ 0.756), and the map does not moderate treatment in observational data. Several of the program's most secure findings are therefore *negative or modest* — no biotypes, no reliable treatment moderation, a group-level rather than individual prognostic gain — and they are presented as the contribution, not buried beneath the seductive numbers.

The result must also be carried with its tensions stated, not smoothed over, because each marks the exact edge of what the data support. The predictive object is a *phenotype*, not an isolated axis: the forecast lives in the fuller four-archetype representation — biology read as a corner of the whole space — and not in a lone three-axis "durable block", which on its own is only ambiguous. The prognostic gain is real but *group-level*, sharpening who on average is heading where rather than calling an individual remission. It is *course-dependent*, carried by the episodic bipolar and depression patients and near-null in the baseline-saturated schizophrenia cohort. The tessellation has *no privileged number of regions* — an operationally awkward but honest consequence of the continuum, with the "operative" resolution deferred to, and then dissolved by, the prognosis question (the continuous/archetype encoding dominates any hard partition). Durability is cleanest in the *variance* decomposition (the error-corrected trait/state ICCs); the parallel geometry route is precision-confounded and is treated as corroboration, not as the primary signal. And the map's thinnest axis, *mania* (two indicators), is reliable-looking but prognostically and temporally uninformative, and is flagged as such rather than over-read.

What the work establishes, then, is a transdiagnostic biological map that **complements** the diagnostic system rather than replacing it: co-informative with DSM-5 prognostically, a tighter description of the coordinates than diagnosis, and a continuous, uncertainty-aware representation of where a patient sits in biology-and-severity space. What it does not establish is an individual risk calculator or a prescribing rule. In a literature replete with biological-validity evidence advanced as though it were clinical utility, a study that measures *both* and reports the modest and null results as such is itself a contribution — a calibrated counterweight, and a reusable, principled pipeline (no imputation, uncertainty propagated end-to-end, fixed-then-validated objects, overlap-gated causal inference) for asking these questions without over-claiming.

The boundary is drawn by the data, not by the method, and the work names its own exits (Chapter 16). Individual-level clinical utility would require incident-event outcomes rather than re-administered scales; treatment *selection* would require randomized or trial-arm data rather than treatment-as-usual — a defined future study (M5b); and any clinical claim would require external validation beyond this internally-validated baseline cohort. Each is a defined next study, and the present program is the methods-and-feasibility foundation on which they can be built.

The deliverable, stated plainly

A stable, transdiagnostic biological map that is a *continuum* rather than a set of biotypes, whose durable biological corner forecasts two-year functioning beyond diagnosis and severity at the group level, and that — on observational data — does not yet guide treatment. A trustworthy map, and an honest account of how far it reaches.

A Notation

Table 26. Symbols used throughout.

symbol	meaning
i, j, k	patient, indicator, specific-dimension indices
N, J, K, F	patients ($9{,}013$), indicators, specific dimensions, total factors $F = K{+}1$
$G_i,\ D_{ik}$	general-factor and specific-factor scores for patient i
$\Lambda,\ \lambda_{jG},\ \lambda_{jk}$	loading matrix; general loading; specific loading of indicator j
Φ	inter-factor correlation matrix (specific block; G orthogonal in the bifactor)
$\Psi = \mathrm{diag}(\sigma_j^2)$	diagonal residual-variance matrix; σ_j residual scale
$\alpha_j,\ \beta_j,\ c_i$	item intercept; covariate coefficients; covariate vector
η_{ij}	linear predictor for indicator j in patient i (Eq. 6)
$\mathcal{O}_i$	set of indicators observed for patient i
$\Sigma = \Lambda\Phi\Lambda^\top + \Psi$	marginal indicator covariance (continuous block)
$A_i,\ b_i,\ W_i$	Woodbury factor-space precision, information vector, observed-cell precision
$\widehat{R}$, ESS	potential scale reduction factor; effective sample size

B The prior atlas

Figure 34 is the theory half of the map (Chapter 3): the soft-prior loading matrix as an indicator×factor heatmap, generated from configuration alone. Cell shade is the prior-permitted loading |mean| + sd: dark = `primary/g_anchor` (theory expects a loading), mid = `plausible_cross` (theory allows), light = soft-zero (suppressed unless the data surprise the prior). Table 27 gives the per-factor prior pool. The fitted *posterior* (empirical) atlas is shown beside this prior map in Chapter 5 (Figures 7–8); their comparison is the central deliverable of the measurement map.

Table 27. Per-factor prior pool: primary indicators and likelihood mix. 143 indicators $\times$ 10 candidate factors. Negative symptoms, sensory abnormalities, and impulsivity are absent — no usable common indicators.

factor	primaries	likelihood mix
G — severity/burden	14	gaussian 11, bernoulli 3 (functioning + severity only)
cognition	11	gaussian 11 (CVLT/WAIS/fluency/TMT)
metabolic	32	gaussian 19, lognormal 10, bernoulli 3
inflammatory	14	lognormal 8, bernoulli 6
sleep	9	gaussian 9 (PSQI + ESS/CSM)
suicidality	30	bernoulli 14, ordered-logistic 14, gaussian 1, neg-binomial 1
developmental-risk	23	gaussian 12, bernoulli 8, ordered-logistic 3
anhedonia	1	gaussian 1 (thin)
mania	2	gaussian 2 (confirmed; partial axis — 2 indicators)
substance	4	bernoulli 2, gaussian 1, neg-binomial 1 (confirmed; 2-cohort)
cross-loading windows	—	MADRS/QIDS/STAI $\to$ plausible on G, sleep, cognition, …

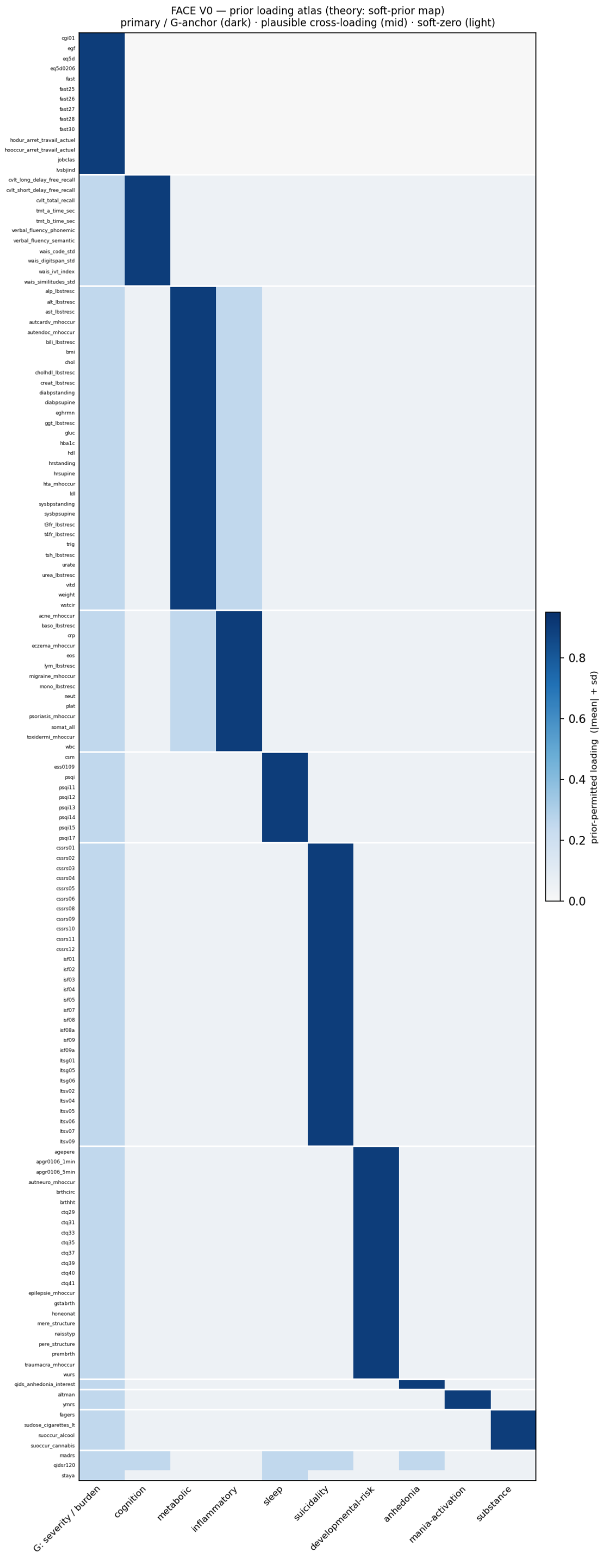


Figure 34. The prior loading atlas (theory). Rows: the 143 modelled indicators, grouped by home factor. Columns: the 10 candidate factors. The block-diagonal primaries, the faint G column over every specific item (the bifactor mechanism), and the homeless cross-loading windows (bottom) are all visible. The empirical atlas — the fitted posterior loadings beside this map, with per-candidate verdicts — is the central deliverable of the measurement map (Chapter 5).

C Acceptance gates and sampler diagnostics

A stage is CERTIFIED only if it passes all gates; otherwise it is PROVISIONAL.

Table 28. The acceptance gate and the diagnostics it reads.

diagnostic	gate	what it detects
$\widehat{R}$ (R-hat)	≤ 1.01	between- vs. within-chain disagreement (non-mixing)
ESS (bulk & tail)	≥ 400	effective independent draws after autocorrelation
divergences	$= 0$	Hamiltonian integration failing on pathological geometry
Heywood	$\max_j \lvert\lambda_j\rvert$ capped	loadings explaining too much / residual collapse
PPC (Bayesian SRMR)	not grossly violated	absolute fit: model-implied vs. observed correlations
WAIC	lower preferred	out-of-sample predictive fit for model comparison
Tucker φ	≥ 0.95 invariant	loading-pattern congruence (robustness / invariance)

The per-stage gate above certifies the continuous stages at full N. The joint mixed-likelihood fit, whose explicit-latent block mixes more slowly, is documented at the *largest N that mixes* ($\widehat{R} \leq 1.04$, ESS ≥ 112, 0 divergences, BFMI ≥ 0.41) with a cross-seed resample-stability guard (Tucker φ 0.993), per §3.6/§4.5 of the methods of record.

D Glossary of instruments and abbreviations

Cohorts & design. BP bipolar disorder; SZ schizophrenia; DR depression; V0 baseline visit; FACE the harmonized three-cohort expert-centre database.

Frameworks. RDoC Research Domain Criteria; HiTOP Hierarchical Taxonomy of Psychopathology; ESEM exploratory structural equation modelling; FIML full-information maximum likelihood.

Functioning & severity (G). FAST Functioning Assessment Short Test; EGF (évaluation globale du fonctionnement / GAF); EQ-5D EuroQol; CGI-S Clinical Global Impression – Severity.

Cognition. WAIS Wechsler Adult Intelligence Scale; TMT Trail Making Test; CVLT California Verbal Learning Test.

Specific domains. PSQI Pittsburgh Sleep Quality Index; CTQ Childhood Trauma Questionnaire; WURS Wender Utah Rating Scale; ISF self-report suicidality items; C-SSRS Columbia Suicide Severity Rating Scale.

Mania & substance. YMRS Young Mania Rating Scale; Altman Altman Self-Rating Mania Scale; SUD substance-use disorder; Fager. Fagerström nicotine-dependence test.

Windows. MADRS Montgomery–Åsberg Depression Rating Scale; QIDS Quick Inventory of Depressive Symptomatology; STAI State–Trait Anxiety Inventory.

Statistics. NUTS No-U-Turn Sampler; LKJ Lewandowski–Kurowicka–Joe correlation prior; WAIC widely applicable information criterion; SRMR standardized root-mean-square residual; DIF differential item functioning; ESS effective sample size; BFMI Bayesian fraction of missing information; ZINB zero-inflated negative binomial; $\widehat{R}$ potential scale reduction factor.

Stratification. ARI adjusted Rand index; η^2 variance explained; Cramér's V nominal association strength; HDBSCAN hierarchical density-based clustering; XD Extreme Deconvolution (a measurement-error Gaussian mixture); UMAP uniform manifold approximation & projection; *archetype* an extreme phenotype (a corner of the continuum); *simplex* the space of convex blends of archetypes.

E Plain-language statistical glossary

A non-specialist guide to the statistical machinery, each entry giving the intuition rather than the formula.

Modelling. *Bayesian inference / posterior* — instead of a single best-fit number, a probability distribution over the values consistent with the data, so every estimate carries its own uncertainty. *Bifactor / ESEM* — a factor model in which one general factor touches every item while each specific factor touches its own cluster; ESEM additionally lets an item carry small cross-loadings onto other factors (a classical confirmatory analysis forbids them). *Soft prior* — a starting expectation for where an item should load that the data are free to override: "theory proposes, the data dispose." *Mixed likelihood* — using the probability model appropriate to each item's type (continuous, binary, count, ordinal) rather than forcing everything onto one scale. *Observed-data likelihood / FIML (full-information maximum likelihood)* — fitting from each patient's *filled-in* cells only; blank cells contribute nothing and are never imputed.

Computation. *Marginalization* — averaging over a hidden variable instead of estimating it one case at a time; here it removes the per-patient latent scores from the sampler. *Woodbury identity* — a matrix-algebra shortcut that inverts a "diagonal + a few shared directions" matrix cheaply by working in the small factor space, which is what lets the model run on all patients at once. *R-hat ($\widehat{R}$)* — a convergence check: independent sampler runs are compared, and values near 1.00 mean they agree (the sampler mixed). *ESS (effective sample size)* — how many genuinely independent draws the autocorrelated sampler is worth.

Fit & validation. *ELPD / WAIC* — measures of out-of-sample predictive accuracy (how well the model predicts data it did not see); higher ELPD, or lower WAIC, is better. *Posterior-predictive check (PPC) / SRMR* — simulating fresh data from the fitted model and asking whether it resembles the real data; an SRMR below 0.08 is the usual "good fit" line. *Tucker congruence (φ)* — the cosine similarity of two loading patterns: 1.00 identical, ≥ 0.95 "the same factor." *Measurement invariance* — whether an instrument means the same thing across groups (cohorts, visits), so that a change in score reflects the patient, not the ruler.

Change & stratification. *Errors-in-variables / regression dilution* — correcting for the fact that a predictor is measured with error; ignoring it flattens the estimated effect toward zero, worst where the measurement is noisiest. *ICC (intraclass correlation)* — the share of a variable's variation that is stable between patients (*trait*) versus fluctuating within a patient over time (*state*). *Archetypal analysis / Extreme Deconvolution* — describing each patient as a blend of a few "pure extremes" (corners of the cloud); Extreme Deconvolution first subtracts each patient's measurement noise so the blend reflects what they *are*. *Adjusted Rand index (ARI)* — agreement between two groupings, scaled so 0 is chance and 1 is identical.

Causal inference. *Propensity score* — a patient's estimated probability of receiving a treatment given their baseline profile, used to compare like with like. *Overlap / common support* — whether treated and untreated patients are comparable enough to estimate an effect at all. *IPW / IPTW (inverse-probability-of-treatment weighting)* — re-weighting patients so the treated and untreated groups look alike on measured baseline factors. *Doubly-robust estimation* — pairing a treatment model with an outcome model so the effect estimate stays unbiased if *either* one is correct. *E-value* — how strong an unmeasured confounder would have to be to explain a result away; a small E-value means a fragile result.

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